



Programme : Bachelor in Data Science

Subject : BAIT3003 Data Warehouse Technology (202505)

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BAIT3003 Data Warehouse Technology**Assignment Assessment Form**

Tas k No.	Task Descriptions	Weightage	Criteria	Ratings	Marks	CLO
1	Design of Data warehouse (logical design)	5%	• Include the relevant dimensions. • Include the correct measures in the fact table.	• Excellent (5) • Good (4) • Moderate (2-3) • Poor (0-1)		1
	Design of Data warehouse (physical design)	10%	• Create TABLE statements • Appropriate data types and size of attributes • Proper Integrity constraints	• Excellent (9-10) • Good (7-8) • Moderate (4-6) • Poor (0-3)		1
2	ETL (initial loading)	15%	• VIEWS, SELECT, INSERT, PROCEDURES for each of the dimensions and fact table. • Variety of techniques necessary to achieve the correct data loading	• Excellent (13-15) • Good (10-12) • Moderate (6-9) • Poor (0-5)		1
	ETL (subsequent loading)	15%	• VIEWS, SELECT, INSERT, PROCEDURES for each of the dimensions and fact table. • Logic to scrub dirty data	• Excellent (13-15) • Good (10-12) • Moderate (6-9) • Poor (0-5)		1
3	*Business Analytic queries design (Individual marks awarded))	45%	• Clear and proper identification of information needs • Flexible query to cater for variety of inputs, use of multiple dimensions • Meaningful visualisation • Data values formatted accordingly • Relevant decision making based on facts presented	• Excellent (35-45) • Good (25-34) • Moderate(13-24) • Poor (0-12)		3

4	Assignment Report	10%	• Comprehensive coverage • Quality of report presented • All tasks numbered, header / footer used, proper formatting	• Excellent (9-10) • Good (7-8) • Moderate (4-6) • Poor (0-3)		1
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Group Member:

1. Jesmine Tey Khai Jing

Task 3 marks

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Total marks

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2. Gan Khai Li

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3. Kit Chin Jie Ying

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4. Ethel Ng Yi Yan

()

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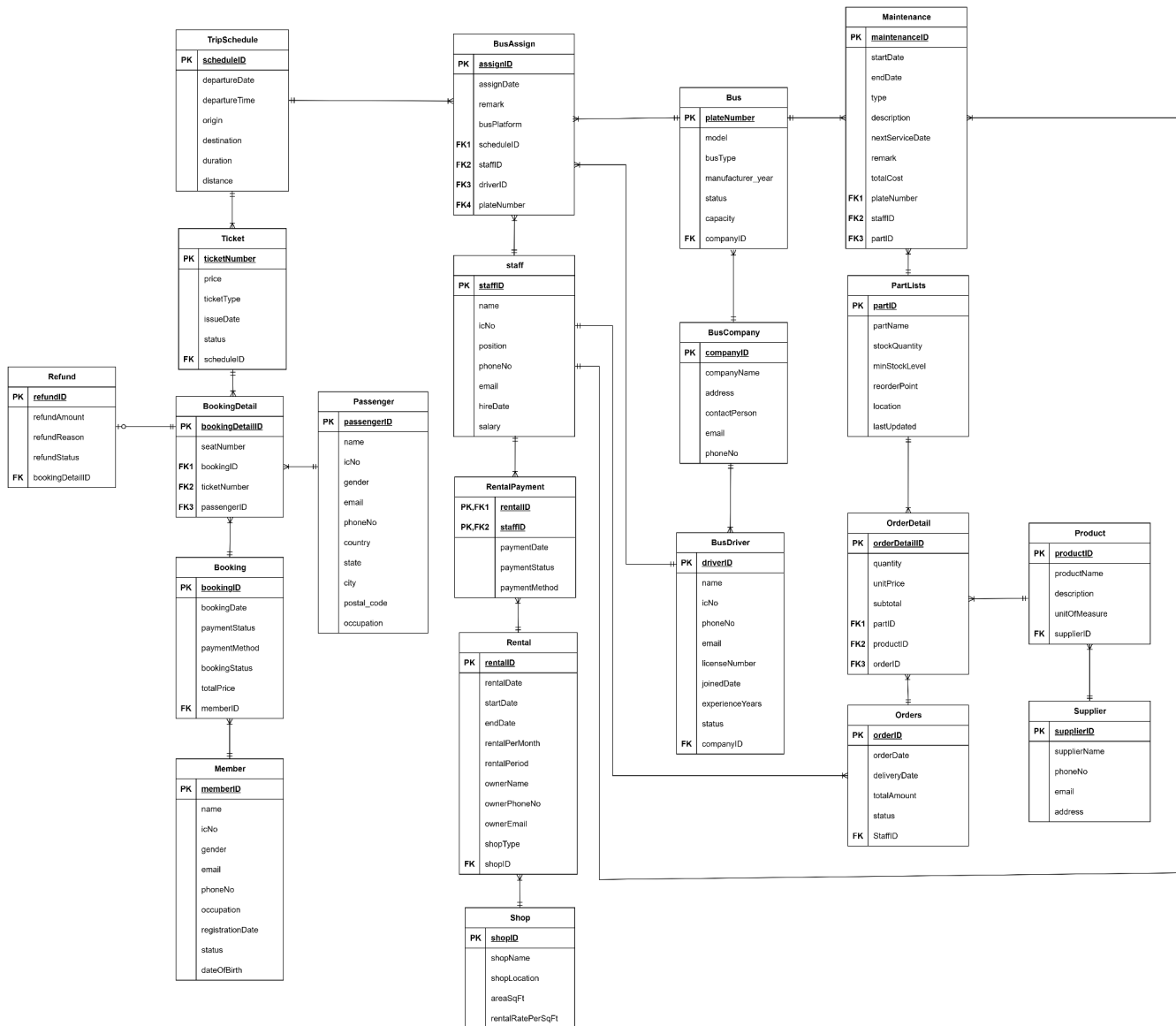
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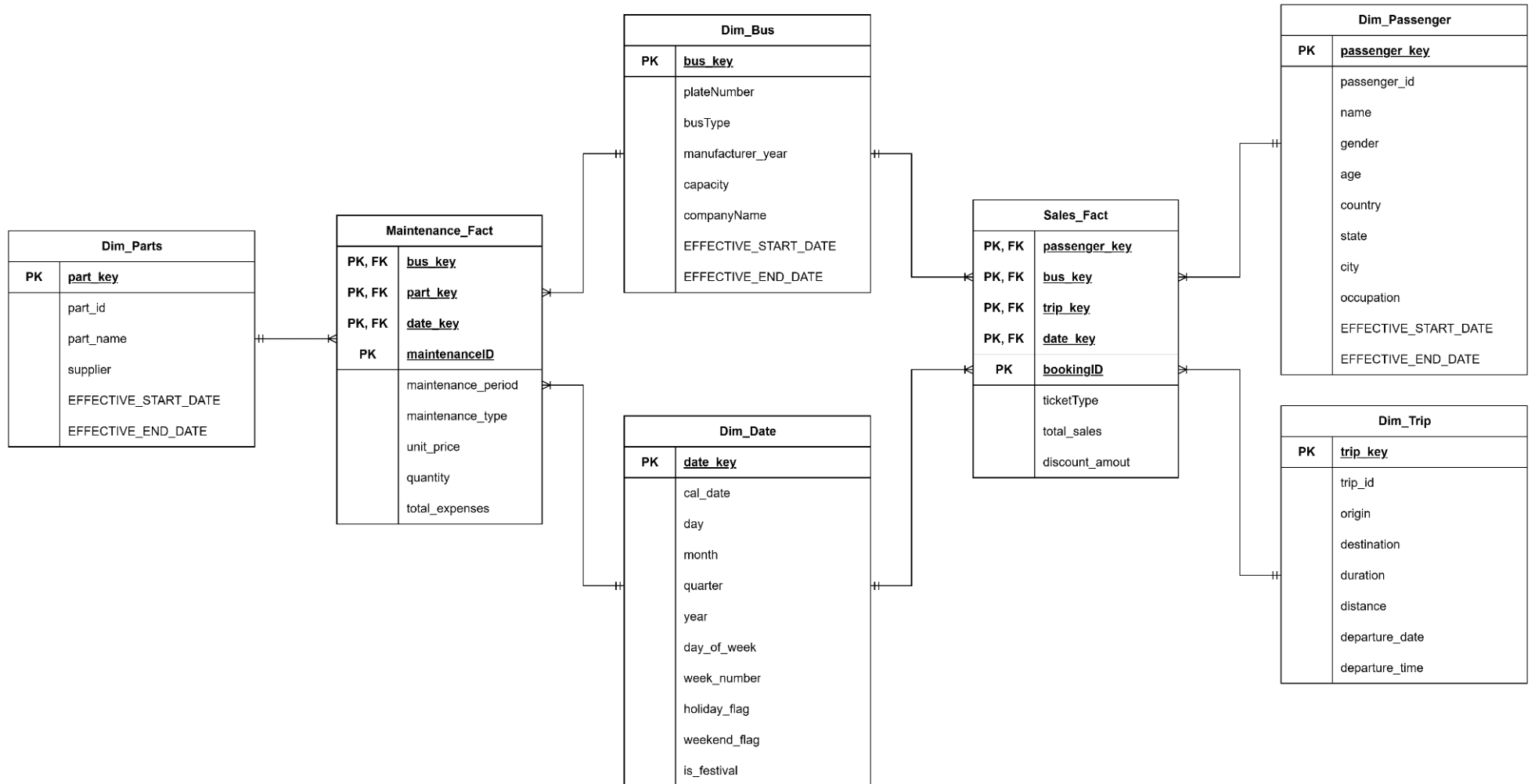
Chapter 1 Design of Data Warehouse

1.1 Logical Design

1.1.1 Original Database (Entity Relationship Diagram)



1.1.2 Star Schema Dimension and Fact Tables



1.2 Physical Design

1.2.1 Dimension Tables

Date Dimension Table

```
create table Dim_Date(  
    date_key      number      not null,  
    cal_date      date        not null,  
    day           number      not null,  
    month         number      not null,  
    quarter       varchar(2)  not null,  
    year          number(4)    not null,  
    day_of_week   varchar(9)  not null,  
    week_number   varchar(3)  not null,  
    holiday_flag  varchar(1)  not null,  
    weekend_flag   varchar(1)  not null,  
    is_festival   varchar(50) not null,  
    constraint Dim_Date_PK primary key(date_key)  
);
```

Bus Dimension Table

```
create table Dim_Bus(  
    bus_key      number      not null,  
    plateNumber   varchar(8)  not null,  
    busType       varchar(50) not null,  
    manufacturer_year number    not null,  
    capacity      number      not null,  
    companyName   varchar(100) not null,  
    EFFECTIVE_START_DATE date    not null,  
    EFFECTIVE_END_DATE date     not null,  
    constraint Dim_Bus_PK primary key(bus_key)  
);
```

Passenger Dimension Table

```
create table Dim_Passenger(  
    passenger_key          number          not null,  
    passenger_id           varchar(6)       not null,  
    name                   varchar(100)    not null,  
    gender                 varchar(1)      not null,  
    age                   varchar(3)       not null,  
    country               varchar(50)     not null,  
    state                 varchar(50)     not null,  
    city                 varchar(50)     not null,  
    occupation            varchar(50)     not null,  
    EFFECTIVE_START_DATE  date           not null,  
    EFFECTIVE_END_DATE    date           not null,  
    constraint Dim_Passenger_PK primary key(passenger_key)  
);
```

Trip Dimension Table

```
create table Dim_Trip(  
    trip_key              number          not null,  
    trip_id              varchar(6)       not null,  
    origin               varchar(50)     not null,  
    destination          varchar(50)     not null,  
    duration             number          not null,  
    distance             number(3)       not null,  
    departure_date       date           not null,  
    departure_time       date           not null,  
    constraint Dim_Trip_PK primary key(trip_key)  
);
```

Parts Dimension Table

```
create table Dim_Parts(  
    part_key             number          not null,  
    part_id              varchar(6)       not null,  
    part_name            varchar(50)     not null,
```

```

supplier          varchar(50)  not null,
EFFECTIVE_START_DATE  date          not null,
EFFECTIVE_END_DATE    date          not null,
constraint Dim_Parts_PK primary key(part_key)
);

```

1.2.2 Fact Tables

Sales Fact

```

create table Sales_Fact(
    passenger_key    number          not null,
    bus_key          number          not null,
    trip_key         number          not null,
    date_key         number          not null,
    bookingID       varchar(6)       not null,
    ticketType      varchar(12)      not null,
    total_sales     number(10,2)     not null,
    discount_amount  number(4,2)     not null,
    constraint SF_PK primary key(passenger_key, bus_key, trip_key,
date_key, bookingID),
    constraint SF_Dim_Passenger_FK foreign key (passenger_key)
references Dim_Passenger (passenger_key),
    constraint SF_Dim_Bus_FK foreign key (bus_key) references
Dim_Bus (bus_key),
    constraint SF_Dim_Trip_FK foreign key (trip_key) references
Dim_Trip (trip_key),
    constraint SF_Dim_Date_FK foreign key (date_key) references
Dim_Date (date_key)
);

```

Maintenance Table

```

create table Maintenance_Fact(
    bus_key          number          not null,
    part_key         number          not null,
    date_key         number          not null,

```

```

maintenanceID      varchar(6)      not null,
maintenance_period  number          not null,
maintenance_type    varchar(50)     not null,
unit_price          number(5,2)      not null,
quantity            number(3)        not null,
total_expenses      number(7,2)      not null,
constraint MF_PK primary key (bus_key, part_key, date_key,
maintenanceID),
constraint MF_Dim_Bus_FK foreign key (bus_key) references
Dim_Bus (bus_key),
constraint MF_Dim_Part_FK foreign key (part_key) references
Dim_Parts (part_key),
constraint MF_Dim_Date_FK foreign key (date_key) references
Dim_Date (date_key)
);

```

Chapter 2 Extract, Transform, Load Process

2.1 Script for initial loading

```
alter session set nls_date_format = 'DD-MM-YYYY';
```

Date Dimension Table

```
create sequence Dim_Date_Sequence
start with 100001
increment by 1;
```

```
----- Date Dimension
```

```
declare
```

```
    startDate      date := to_date('01-01-2020', 'dd-mm-yyyy');
    endDate        date := to_date('31-12-2026', 'dd-mm-yyyy');
```

```
    v_date_key      number;
    v_cal_date      date;
    v_day           number(2);
    v_month         number(2);
    v_quarter       varchar(2);
    v_year          number(4);
    v_day_of_week   varchar(9);
    v_week_number   varchar(3);
    v_holiday_flag  varchar(1);
    v_weekend_flag  varchar(1);
    v_is_festival   varchar(50);
```

```
begin
```

```
    while (startDate <= endDate) loop
        v_date_key := Dim_Date_Sequence.nextval;
        v_cal_date := startDate;
        v_day := to_char(startDate, 'DD');
        v_month := to_char(startDate, 'MM');
        v_quarter := 'Q' || to_char(startDate, 'Q');
        v_year := to_number(to_char(startDate, 'YYYY'));
        v_day_of_week := to_char(startDate, 'DAY');
```

```

        v_week_number := 'W' || to_char(startDate, 'IW');
        v_holiday_flag := 'N';
        if (rtrim(v_day_of_week) in ('SATURDAY', 'SUNDAY'))
then
            v_weekend_flag := 'Y';
        else
            v_weekend_flag := 'N';
        end if;
        v_is_festival := 'N';

        insert into Dim_Date values(
            v_date_key,
            v_cal_date,
            v_day,
            v_month,
            v_quarter,
            v_year,
            v_day_of_week,
            v_week_number,
            v_holiday_flag,
            v_weekend_flag,
            v_is_festival
        );

        startDate := startDate + 1;
    end loop;
end;
/

```

Bus Dimension Table

```

create sequence Dim_Bus_Sequence
start with 100001
increment by 1;

insert into Dim_Bus

```

```

select Dim_Bus_Sequence.nextval,
       upper(b.plateNumber),
       upper(b.busType),
       b.manufacture_year,
       b.capacity,
       upper(c.companyName),
       sysdate,
       '31-12-9999'
from Bus b
join BusCompany c on b.companyID=c.companyID;

```

Passenger Dimension Table

```

create sequence Dim_Passenger_Sequence
start with 100001
increment by 1;

insert into Dim_Passenger
select Dim_Passenger_Sequence.nextval,
       passengerID,
       upper(name),
       upper(gender),
       extract(year from sysdate) -
       (case
          when to_number(substr(icNo,1,2)) <=
to_number(to_char(sysdate,'YY'))
          then 2000 + to_number(substr(icNo,1,2))
          else 1900 + to_number(substr(icNo,1,2))
       end),
       upper(country),
       upper(state),
       upper(city),
       upper(occupation),
       sysdate,
       to_date('31-12-9999','dd-mm-yyyy')
from Passenger;

```

Trip Dimension Table

```
create sequence Dim_Trip_Sequence
start with 100001
increment by 1;
```

```
insert into Dim_Trip
select Dim_Trip_Sequence.nextval,
       scheduleID,
       upper(origin),
       upper(destination),
       duration,
       distance,
       departureDate,
       departureTime
from TripSchedule;
```

Parts Dimension Table

```
create sequence Dim_Parts_Sequence
start with 100001
increment by 1;
```

```
insert into Dim_Parts
select Dim_Parts_Sequence.nextval,
       p.partID,
       upper(p.partName),
       upper(s.supplierName),
       sysdate,
       '31-12-9999'
from PartLists p
join orderDetail o on p.partID=o.partID
join Product prod on o.productID=prod.productID
join Supplier s on prod.supplierID=s.supplierID;
```

Sales Fact

```
insert into Sales_Fact
select dp.passenger_key,
       db.bus_key,
       dt.trip_key,
       dd.date_key,
       b.bookingID,
       upper(tk.ticketType),
       b.totalPrice,
       nvl(r.refundAmount, 0)
from bookingDetail bd
join booking b on b.bookingID=bd.bookingID
join ticket tk on tk.ticketNumber=bd.TicketNumber
join dim_passenger dp on dp.passenger_id=bd.passengerID
join dim_trip dt on dt.trip_id=tk.scheduleID
join busAssign ba on ba.scheduleID=dt.trip_id
join dim_bus db on db.plateNumber=ba.plateNumber
join dim_date dd on dd.cal_date=b.bookingDate
left join refund r on r.bookingDetailID=bd.bookingDetailID;
```

Maintenance Table

```
insert into Maintenance_Fact
select db.bus_key,
       dp.part_key,
       dd.date_key,
       m.maintenanceID,
       nvl(m.endDate, m.startDate) - m.startDate,
       upper(m.type),
       od.unitPrice,
       od.quantity,
       od.unitPrice * od.quantity
from maintenance m
join dim_bus db on db.plateNumber=m.plateNumber
join dim_parts dp on dp.part_id=m.partID
join orderDetail od on od.partID=dp.part_id
```

```
join orders o on o.orderID=od.orderID  
join dim_date dd on dd.cal_date=o.orderDate;
```

2.2 Script for subsequent loading (Insert Value)

```
alter session set nls_date_format = 'DD-MM-YYYY';
```

Date Dimension Table

```
CREATE OR REPLACE PROCEDURE P_UPDATE_HOLIDAYS AS
BEGIN
  /*=====
    FEDERAL / NATIONAL HOLIDAYS (every year)
    =====*/
  -- New Year's Day
  UPDATE Dim_Date SET holiday_flag='Y', is_festival='New Year's
Day'
  WHERE TO_CHAR(TRUNC(cal_date), 'MMDD')='0101';

  -- Labour Day
  UPDATE Dim_Date SET holiday_flag='Y', is_festival='Labour Day'
  WHERE TO_CHAR(TRUNC(cal_date), 'MMDD')='0501';

  -- Merdeka Day
  UPDATE Dim_Date SET holiday_flag='Y', is_festival='Merdeka Day'
  WHERE TO_CHAR(TRUNC(cal_date), 'MMDD')='0831';

  -- Malaysia Day
  UPDATE Dim_Date SET holiday_flag='Y', is_festival='Malaysia Day'
  WHERE TO_CHAR(TRUNC(cal_date), 'MMDD')='0916';

  -- Christmas Day
  UPDATE Dim_Date SET holiday_flag='Y', is_festival='Christmas Day'
  WHERE TO_CHAR(TRUNC(cal_date), 'MMDD')='1225';

  /*=====
    MOVEABLE FEDERAL FESTIVALS
    =====*/
  -- Chinese New Year (Day 1 & Day 2)
  UPDATE Dim_Date SET holiday_flag='Y', is_festival='Chinese New
Year (Day 1)'
```

```

WHERE TRUNC(cal_date) IN (DATE '2020-01-25', DATE '2021-02-12',
DATE '2022-02-01',
DATE '2023-01-22', DATE '2024-02-10',
DATE '2025-01-29');
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Chinese New
Year (Day 2)'
WHERE TRUNC(cal_date) IN (DATE '2020-01-26', DATE '2021-02-13',
DATE '2022-02-02',
DATE '2023-01-23', DATE '2024-02-11',
DATE '2025-01-30');

-- Hari Raya Aidilfitri (Day 1 & 2)
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Hari Raya
Aidilfitri (Day 1)'
WHERE TRUNC(cal_date) IN (DATE '2020-05-24', DATE '2021-05-13',
DATE '2022-05-02',
DATE '2023-04-22', DATE '2024-04-10',
DATE '2025-03-31');
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Hari Raya
Aidilfitri (Day 2)'
WHERE TRUNC(cal_date) IN (DATE '2020-05-25', DATE '2021-05-14',
DATE '2022-05-03',
DATE '2023-04-23', DATE '2024-04-11',
DATE '2025-04-01');

-- Hari Raya Haji (Aidiladha)
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Hari Raya Haji'
WHERE TRUNC(cal_date) IN (DATE '2020-07-31', DATE '2021-07-20',
DATE '2022-07-10',
DATE '2023-06-29', DATE '2024-06-17',
DATE '2025-06-07');

-- Awal Muharram (Islamic New Year)
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Awal Muharram'

```

```

WHERE TRUNC(cal_date) IN (DATE '2020-08-20', DATE '2021-08-10',
DATE '2022-07-30',
DATE '2023-07-19', DATE '2024-07-07',
DATE '2025-06-27');

```

```

-- Prophet Muhammad's Birthday (Maulidur Rasul)
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Prophet
Muhammad's Birthday'
WHERE TRUNC(cal_date) IN (DATE '2020-10-29', DATE '2021-10-19',
DATE '2022-09-09',
DATE '2023-09-28', DATE '2024-09-16',
DATE '2025-09-05');

```

```

-- Wesak Day
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Wesak Day'
WHERE TRUNC(cal_date) IN (DATE '2020-05-07', DATE '2021-05-26',
DATE '2022-05-15',
DATE '2023-05-04', DATE '2024-05-22',
DATE '2025-05-12');

```

```

-- Deepavali
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Deepavali'
WHERE TRUNC(cal_date) IN (DATE '2020-11-14', DATE '2021-11-04',
DATE '2022-10-24',
DATE '2023-11-12', DATE '2024-10-31',
DATE '2025-10-20');

```

```

UPDATE Dim_Date SET holiday_flag='Y', is_festival='Federal
Territory Day'
WHERE TRUNC(cal_date) IN (DATE '2020-02-01', DATE '2021-02-01',
DATE '2022-02-01',
DATE '2023-02-01', DATE '2024-02-01',
DATE '2025-02-01');

```

```

/*=====

```

SCHOOL HOLI

=====*/

-- 2020

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Mid  
Term 1)'
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2020-03-13' AND DATE  
'2020-03-22';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Mid  
Term 2)'
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2020-07-22' AND DATE  
'2020-07-27';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Term  
2)'
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2020-08-14' AND DATE  
'2020-08-23';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Group  
A - Year End)'
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2020-12-18' AND DATE  
'2021-01-03';
```

-- 2021

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Mid  
Term 1)'
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2021-03-26' AND DATE  
'2021-04-04';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Mid  
Term 2)'
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2021-07-16' AND DATE  
'2021-07-25';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Term  
2)'
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2021-08-13' AND DATE  
'2021-08-22';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Year  
End) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2021-12-10' AND DATE  
'2022-01-02';
```

```
-- 2022
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Mid  
Term) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2022-02-26' AND DATE  
'2022-03-20';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Term  
1) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2022-06-03' AND DATE  
'2022-06-12';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Term  
2) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2022-09-02' AND DATE  
'2022-09-10';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Year  
End) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2022-12-09' AND DATE  
'2023-01-01';
```

```
-- 2023
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Mid  
Term 1) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2023-04-20' AND DATE  
'2023-04-30';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Term  
1) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2023-05-26' AND DATE  
'2023-06-04';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Term
2) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2023-08-25' AND DATE
'2023-09-03';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Year
End) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2023-12-15' AND DATE
'2024-01-01';
```

```
-- Group A
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Mid
Term) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2024-03-08' AND DATE
'2024-03-17';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Term
1) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2024-05-24' AND DATE
'2024-06-02';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Term
2) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2024-08-23' AND DATE
'2024-09-01';
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Year
End) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2024-12-13' AND DATE
'2025-01-05';
```

```
-- Group A: term breaks + short festival breaks
```

```
UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi
(Raya) '
```

```
WHERE TRUNC(cal_date) BETWEEN DATE '2025-03-30' AND DATE
'2025-04-04';
```

```

    UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Term
1)'
    WHERE TRUNC(cal_date) BETWEEN DATE '2025-05-29' AND DATE
'2025-06-09';
    UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Term
2)'
    WHERE TRUNC(cal_date) BETWEEN DATE '2025-09-12' AND DATE
'2025-09-21';
    UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi
(Deepavali)'
    WHERE TRUNC(cal_date) BETWEEN DATE '2025-10-19' AND DATE
'2025-10-22';
    UPDATE Dim_Date SET holiday_flag='Y', is_festival='Sch Holi (Year
End)'
    WHERE TRUNC(cal_date) BETWEEN DATE '2025-12-19' AND DATE
'2026-01-11';

    COMMIT;
    DBMS_OUTPUT.PUT_LINE('Public holidays + School Holi updates
completed (2020-2025).');

EXCEPTION
    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE('Error in P_UPDATE_HOLIDAYS: '||SQLERRM);
        ROLLBACK;
END P_UPDATE_HOLIDAYS;
/

-- Execute Procedure
exec P_UPDATE_HOLIDAYS;

```

Bus Dimension Table

```

-- clean data
create or replace procedure clean_dim_bus as
begin

```

```

-- clean dirty data
update dim_bus
set plateNumber = upper(trim(plateNumber)),
    busType = upper(trim(busType)),
    companyName = upper(trim(companyName))
where plateNumber is not null;

-- remove duplicates
delete from dim_bus db
where bus_key not in(
    select bus_key
    from (
        select bus_key,
            row_number() over(
                partition by plateNumber
                order by effective_start_date desc
            ) as rn
        from dim_bus
    )
    where rn = 1
);
dbms_output.put_line('Duplicate records removed.');
```

end;

/

```
exec clean_dim_bus;
```



```
-- check duplicate data exists
select plateNumber, count(*) as count
from dim_bus
group by plateNumber
having count(*) > 1;
```



```
-- detect new record
create or replace procedure load_new_dim_bus as
begin
```

```

insert into dim_bus(
    bus_key,
    plateNumber,
    busType,
    manufacturer_year,
    capacity,
    companyName,
    effective_start_date,
    effective_end_date
)
select dim_bus_sequence.nextval,
    upper(trim(b.plateNumber)),
    upper(trim(b.busType)),
    b.manufacture_year,
    b.capacity,
    upper(trim(c.companyName)),
    sysdate,
    to_date('31-12-9999', 'DD-MM-YYYY')
from bus b
join busCompany c on b.companyID = c.companyID
where not exists(
    select 1
    from dim_bus db
    where db.plateNumber = upper(trim(b.plateNumber))
        and db.companyName = upper(trim(c.companyName))
        and db.busType = upper(trim(b.busType))
        and db.manufacturer_year = b.manufacture_year
        and db.capacity =b.capacity
        and db.effective_end_date = to_date('31-12-9999',
'DD-MM-YYYY')
    );
    dbms_output.put_line('New records inserted into Dim_Bus.');
```

end;

/

exec load_new_dim_bus;

Passenger Dimension Table

```
CREATE OR REPLACE PROCEDURE P_UPD_PASSENGER_CSC AS
    -- List of West and East Malaysia states and Federal Territories
    TYPE t_state_list IS TABLE OF VARCHAR2(50);
    v_states t_state_list := t_state_list(
        'JOHOR', 'KEDAH', 'KELANTAN', 'MELAKA', 'NEGERI SEMBILAN',
        'PAHANG', 'PERAK', 'PERLIS', 'PENANG',
        'SELANGOR', 'TERENGGANU', 'KUALA LUMPUR', 'PUTRAJAYA',
        'SABAH', 'SARAWAK'
    );

    -- City Mapping
    TYPE t_city_list IS TABLE OF VARCHAR2(50);
    TYPE t_city_map IS TABLE OF t_city_list INDEX BY VARCHAR2(50);
    v_city_map t_city_map;

    v_random_index PLS_INTEGER;
    v_random_state VARCHAR2(50);
    v_random_city VARCHAR2(50);

    -- Helper function to check if a state is valid
    FUNCTION is_valid_state(p_state VARCHAR2) RETURN BOOLEAN IS
    BEGIN
        FOR i IN 1..v_states.COUNT LOOP
            IF UPPER(TRIM(p_state)) = v_states(i) THEN
                RETURN TRUE;
            END IF;
        END LOOP;
        RETURN FALSE;
    END;

BEGIN
    -- Initialize city mapping
```

```

    v_city_map('JOHOR') := t_city_list('JOHOR BAHRU', 'MUAR',
'KLUANG');
    v_city_map('KEDAH') := t_city_list('ALOR SETAR', 'SUNGAI
PETANI', 'KULIM');
    v_city_map('KELANTAN') := t_city_list('KOTA BHARU', 'TANAH
MERAH', 'TUMPAT');
    v_city_map('MELAKA') := t_city_list('MELAKA', 'ALOR GAJAH',
'JASIN');
    v_city_map('NEGERI SEMBILAN') := t_city_list('SEREMBAN', 'PORT
DICKSON', 'NILAI');
    v_city_map('PAHANG') := t_city_list('KUANTAN', 'TEMERLOH',
'RAUB');
    v_city_map('PERAK') := t_city_list('IPOH', 'TAIPING', 'TELUK
INTAN');
    v_city_map('PERLIS') := t_city_list('KANGAR', 'ARAU');
    v_city_map('PENANG') := t_city_list('GEORGE TOWN', 'BATU
FERRINGHI', 'BUKIT MERTA JAM');
    v_city_map('SELANGOR') := t_city_list('SHAH ALAM', 'PETALING
JAYA', 'KLANG');
    v_city_map('TERENGGANU') := t_city_list('KUALA TERENGGANU',
'KEMAMAN', 'DUNGUN');
    v_city_map('KUALA LUMPUR') := t_city_list('KUALA LUMPUR');
    v_city_map('PUTRAJAYA') := t_city_list('PUTRAJAYA');
    v_city_map('SABAH') := t_city_list('KOTA KINABALU', 'SANDAKAN',
'TAWAU', 'LAHAD DATU');
    v_city_map('SARAWAK') := t_city_list('KUCHING', 'MIRI', 'SIBU',
'BINTULU');

```

```

-- Set all country values to uppercase Malaysia

```

```

UPDATE Dim_Passenger
SET country = 'MALAYSIA';

```

```

-- Fix state and city

```

```

FOR rec IN (SELECT passenger_id, state FROM Dim_Passenger) LOOP
    v_random_state := UPPER(TRIM(rec.state));

```

```

-- If state is invalid
IF NOT is_valid_state(v_random_state) THEN
    v_random_index := TRUNC(DBMS_RANDOM.VALUE(1,
v_states.COUNT + 1));
    v_random_state := v_states(v_random_index);

    UPDATE Dim_Passenger
    SET state = v_random_state
    WHERE passenger_id = rec.passenger_id;
END IF;

-- Randomly pick a city from the selected state
IF v_city_map.EXISTS(v_random_state) THEN
    v_random_index := TRUNC(DBMS_RANDOM.VALUE(1,
v_city_map(v_random_state).COUNT + 1));
    v_random_city :=
v_city_map(v_random_state)(v_random_index);
ELSE
    v_random_city := 'UNKNOWN CITY';
END IF;

UPDATE Dim_Passenger
SET city = v_random_city
WHERE passenger_id = rec.passenger_id;
END LOOP;

-- Change occupation to uppercase
UPDATE Dim_Passenger
SET occupation = 'UNEMPLOYMENT'
WHERE UPPER(TRIM(occupation)) = 'NONE';

-- Remove duplicate passenger_id (keep the one with the lowest
passenger_key)
DELETE FROM Dim_Passenger dp

```

```

WHERE dp.passenger_key NOT IN (
    SELECT MIN(passenger_key)
    FROM Dim_Passenger
    GROUP BY passenger_id
);

COMMIT;

END P_UPD_PASSENGER_CSC;
/
exec P_UPD_PASSENGER_CSC;

CREATE OR REPLACE PROCEDURE load_new_dim_passenger AS
BEGIN
    INSERT INTO Dim_Passenger (
        passenger_key,
        passenger_id,
        name,
        gender,
        age,
        country,
        state,
        city,
        occupation,
        effective_start_date,
        effective_end_date
    )
    SELECT Dim_Passenger_Sequence.NEXTVAL,
        p.passengerID,
        UPPER(p.name),
        UPPER(p.gender),
        EXTRACT(YEAR FROM SYSDATE) -
            (CASE
                WHEN TO_NUMBER(SUBSTR(p.icNo,1,2)) <=
TO_NUMBER(TO_CHAR(SYSDATE,'YY'))
                THEN 2000 + TO_NUMBER(SUBSTR(p.icNo,1,2))

```

```

        ELSE 1900 + TO_NUMBER(SUBSTR(p.icNo,1,2))
    END),
    UPPER(p.country),
    UPPER(p.state),
    UPPER(p.city),
    UPPER(p.occupation),
    SYSDATE,
    TO_DATE('31-12-9999','dd-mm-yyyy')
FROM Passenger p
WHERE NOT EXISTS (
    SELECT 1
    FROM Dim_Passenger d
    WHERE d.passenger_id = p.passengerID
);

COMMIT;

END;

/

exec load_new_dim_passenger;

```

Trip Dimension Table

```

CREATE OR REPLACE PROCEDURE ETL_Trip IS
BEGIN
    -- 1. Transform Step: Clean TripSchedule
    -- Convert to uppercase and replace NULL with 'UNKNOWN'
    UPDATE TripSchedule
    SET origin      = NVL(UPPER(origin), 'UNKNOWN'),
        destination = NVL(UPPER(destination), 'UNKNOWN');

    -- Ensure origin ≠ destination
    UPDATE TripSchedule
    SET destination = 'UNKNOWN'
    WHERE origin = destination;

    -- 2. Distance Validation

```

```

UPDATE TripSchedule
SET distance = 100
WHERE distance IS NULL OR distance < 50 OR distance > 600;

-- 3. Duration Estimation (distance ÷ 80 km/h)
UPDATE TripSchedule
SET duration = ROUND(distance / 80);

-- 5. Load Step: Insert into Dim_Trip if not exists, remove
duplicates
INSERT INTO Dim_Trip (
    trip_key, trip_id, origin, destination, duration, distance,
    departure_date, departure_time
)
SELECT Dim_Trip_Sequence.NEXTVAL,
    scheduleID,
    origin,
    destination,
    duration,
    distance,
    departureDate,
    departureTime
FROM (
    SELECT ts.*,
        ROW_NUMBER() OVER (PARTITION BY ts.scheduleID
                            ORDER BY ts.departureDate,
ts.departureTime) AS rn
    FROM TripSchedule ts
) t
WHERE t.rn = 1
    AND NOT EXISTS (
        SELECT 1 FROM Dim_Trip d
        WHERE d.trip_id = t.scheduleID
    );

```

```

COMMIT;
END ETL_Trip;
/

exec ETL_Trip;

```

Parts Dimension Table

```

CREATE OR REPLACE PROCEDURE P_SUBSEQUENT_LOAD_DIM_PARTS AS
BEGIN
    -- Step 1: Clean dirty data
    -- Example: trim spaces and ignore nulls
    DELETE FROM PartLists
    WHERE partID IS NULL OR partName IS NULL;

    DELETE FROM Supplier
    WHERE supplierName IS NULL;

    -- Step 2: Remove duplicates
    -- Keep distinct partID + supplier combination
    MERGE INTO PartLists p
    USING (
        SELECT DISTINCT partID, partName
        FROM PartLists
    ) src
    ON (p.partID = src.partID)
    WHEN NOT MATCHED THEN
        INSERT (partID, partName)
        VALUES (src.partID, src.partName);

    -- Step 3: Insert new records into Dim_Parts
    INSERT INTO Dim_Parts (
        part_key,
        part_id,
        part_name,
        supplier,

```

```

        EFFECTIVE_START_DATE,
        EFFECTIVE_END_DATE
    )
SELECT
    Dim_Parts_Sequence.nextval,
    TRIM(p.partID),
    TRIM(p.partName),
    TRIM(s.supplierName),
    SYSDATE,
    TO_DATE('31-12-9999','DD-MM-YYYY')
FROM PartLists p
JOIN orderDetail o ON p.partID = o.partID
JOIN Product prod ON o.productID = prod.productID
JOIN Supplier s ON prod.supplierID = s.supplierID
LEFT JOIN Dim_Parts dp
    ON p.partID = dp.part_id
    AND dp.EFFECTIVE_END_DATE =
TO_DATE('31-12-9999','DD-MM-YYYY')
WHERE dp.part_id IS NULL;

COMMIT;
END P_SUBSEQUENT_LOAD_DIM_PARTS;
/

exec P_SUBSEQUENT_LOAD_DIM_PARTS;

```

Sales Fact

```

-- Load New Records into Sales_Fact
CREATE OR REPLACE PROCEDURE LOAD_NEW_SALES_FACT AS
BEGIN
    INSERT INTO Sales_Fact (
        passenger_key,
        bus_key,
        trip_key,
        date_key,

```

```

        bookingID,
        ticketType,
        total_sales,
        discount_amount
    )
SELECT
    dp.passenger_key,
    db.bus_key,
    dt.trip_key,
    dd.date_key,
    b.bookingID,
    tk.ticketType,
    b.totalPrice,
    NVL(r.refundAmount, 0)
FROM bookingDetail bd
JOIN booking b ON b.bookingID = bd.bookingID
JOIN ticket tk ON tk.ticketNumber = bd.ticketNumber
JOIN dim_passenger dp ON dp.passenger_id = bd.passengerID
JOIN dim_trip dt ON dt.trip_id = tk.scheduleID
JOIN busAssign ba ON ba.scheduleID = dt.trip_id
JOIN dim_bus db ON db.plateNumber = ba.plateNumber
JOIN dim_date dd ON dd.cal_date = b.bookingDate
LEFT JOIN refund r ON r.bookingDetailID = bd.bookingDetailID
WHERE NOT EXISTS (
    SELECT 1
    FROM Sales_Fact sf
    WHERE sf.passenger_key = dp.passenger_key
        AND sf.bus_key = db.bus_key
        AND sf.trip_key = dt.trip_key
        AND sf.date_key = dd.date_key
        AND sf.bookingID = b.bookingID
);

DBMS_OUTPUT.PUT_LINE('New rows inserted into Sales_Fact.');
```

END;

/

```
exec LOAD_NEW_SALES_FACT;
```

Maintenance Table

```
CREATE OR REPLACE PROCEDURE LOAD_NEW_MAINTENANCE_FACT AS  
BEGIN
```

```
    INSERT INTO Maintenance_Fact (  
        bus_key,  
        part_key,  
        date_key,  
        maintenanceID,  
        maintenance_period,  
        maintenance_type,  
        unit_price,  
        quantity,  
        total_expenses  
    )
```

```
SELECT
```

```
    db.bus_key,  
    dp.part_key,  
    dd.date_key,  
    m.maintenanceID,  
    NVL(m.endDate, m.startDate) - m.startDate,  
    UPPER(m.type),  
    od.unitPrice,  
    od.quantity,  
    od.unitPrice * od.quantity
```

```
FROM maintenance m
```

```
JOIN dim_bus db ON db.plateNumber = m.plateNumber
```

```
JOIN dim_parts dp ON dp.part_id = m.partID
```

```
JOIN orderDetail od ON od.partID = dp.part_id
```

```
JOIN orders o ON o.orderID = od.orderID
```

```
JOIN dim_date dd ON dd.cal_date = o.orderDate
```

```
WHERE NOT EXISTS (
```

```

SELECT 1
FROM Maintenance_Fact mf
WHERE mf.bus_key = db.bus_key
      AND mf.part_key = dp.part_key
      AND mf.date_key = dd.date_key
      AND mf.maintenanceID = m.maintenanceID
);

DBMS_OUTPUT.PUT_LINE('New rows inserted into
Maintenance_Fact.');
```

END;

/

exec LOAD_NEW_MAINTENANCE_FACT;

2.3 Script for updating (Type 2 Changes)

Bus Dimension Table

```
CREATE OR REPLACE PROCEDURE LOAD_DIM_BUS_SCD2 AS
BEGIN
    -- 1. Expire old records where details changed
    UPDATE Dim_Bus db
    SET effective_end_date = SYSDATE - 1
    WHERE EXISTS (
        SELECT 1
        FROM Bus b
        JOIN BusCompany c ON b.companyID = c.companyID
        WHERE db.plateNumber = UPPER(b.plateNumber)
        AND db.effective_end_date =
TO_DATE('31-12-9999','dd-mm-yyyy')
        AND (
            db.busType          <> UPPER(b.busType)
            OR db.capacity      <> b.capacity
            OR db.companyName   <> UPPER(c.companyName)
        )
    );

    -- 2. Insert new versions for changed or new buses
    INSERT INTO Dim_Bus (
        bus_key,
        plateNumber,
        busType,
        manufacturer_year,
        capacity,
        companyName,
        effective_start_date,
        effective_end_date
    )
    SELECT Dim_Bus_Sequence.NEXTVAL,
        UPPER(b.plateNumber),
        UPPER(b.busType),
```

```

        b.manufacture_year,
        b.capacity,
        UPPER(c.companyName),
        SYSDATE,
        TO_DATE('31-12-9999','dd-mm-yyyy')
FROM Bus b
JOIN BusCompany c ON b.companyID = c.companyID
WHERE NOT EXISTS (
    SELECT 1
    FROM Dim_Bus db
    WHERE db.plateNumber = UPPER(b.plateNumber)
        AND db.effective_end_date =
TO_DATE('31-12-9999','dd-mm-yyyy')
        AND db.busType = UPPER(b.busType)
        AND db.capacity = b.capacity
        AND db.companyName = UPPER(c.companyName)
);

DBMS_OUTPUT.PUT_LINE('Dim_Bus loaded with SCD Type 2 changes.');
```

END;

/

exec LOAD_DIM_BUS_SCD2;

Passenger Dimension Table

```

CREATE OR REPLACE PROCEDURE LOAD_DIM_PASSENGER_SCD2 AS
BEGIN
    -- 1. Expire old records if passenger attributes have changed
    UPDATE Dim_Passenger dp
    SET effective_end_date = SYSDATE - 1
    WHERE dp.effective_end_date = TO_DATE('31-12-9999','dd-mm-yyyy')
        AND EXISTS (
            SELECT 1
            FROM Passenger p
            WHERE dp.passenger_id = p.passengerID

```

```

        AND (
            dp.name          <> UPPER(p.name)
            OR dp.gender     <> UPPER(p.gender)
            OR dp.age        <> (EXTRACT(YEAR FROM SYSDATE) -
                                (CASE
                                    WHEN
TO_NUMBER(SUBSTR(p.icNo,1,2)) <= TO_NUMBER(TO_CHAR(SYSDATE,'YY'))
                                        THEN 2000 +
TO_NUMBER(SUBSTR(p.icNo,1,2))
                                        ELSE 1900 +
TO_NUMBER(SUBSTR(p.icNo,1,2))
                                    END))
            OR dp.country   <> UPPER(p.country)
            OR dp.state     <> UPPER(p.state)
            OR dp.city      <> UPPER(p.city)
            OR dp.occupation <> UPPER(p.occupation)
        )
    );

```

-- 2. Insert new rows for new passengers or changed passengers

```

INSERT INTO Dim_Passenger (
    passenger_key,
    passenger_id,
    name,
    gender,
    age,
    country,
    state,
    city,
    occupation,
    effective_start_date,
    effective_end_date
)
SELECT Dim_Passenger_Sequence.NEXTVAL,
       p.passengerID,

```

```

        UPPER(p.name),
        UPPER(p.gender),
        EXTRACT(YEAR FROM SYSDATE) -
            (CASE
                WHEN TO_NUMBER(SUBSTR(p.icNo,1,2)) <=
TO_NUMBER(TO_CHAR(SYSDATE,'YY'))
                    THEN 2000 + TO_NUMBER(SUBSTR(p.icNo,1,2))
                ELSE 1900 + TO_NUMBER(SUBSTR(p.icNo,1,2))
            END),
        UPPER(p.country),
        UPPER(p.state),
        UPPER(p.city),
        UPPER(p.occupation),
        SYSDATE,
        TO_DATE('31-12-9999','dd-mm-yyyy')
FROM Passenger p
WHERE NOT EXISTS (
    SELECT 1
    FROM Dim_Passenger dp
    WHERE dp.passenger_id = p.passengerID
        AND dp.effective_end_date =
TO_DATE('31-12-9999','dd-mm-yyyy')
        AND dp.name          = UPPER(p.name)
        AND dp.gender        = UPPER(p.gender)
        AND dp.age           = (EXTRACT(YEAR FROM SYSDATE) -
                                (CASE
                                    WHEN
TO_NUMBER(SUBSTR(p.icNo,1,2)) <= TO_NUMBER(TO_CHAR(SYSDATE,'YY'))
                                        THEN 2000 +
TO_NUMBER(SUBSTR(p.icNo,1,2))
                                    ELSE 1900 +
TO_NUMBER(SUBSTR(p.icNo,1,2))
                                END))
        AND dp.country       = UPPER(p.country)
        AND dp.state         = UPPER(p.state)

```

```

        AND dp.city          = UPPER(p.city)
        AND dp.occupation = UPPER(p.occupation)
    );

    DBMS_OUTPUT.PUT_LINE('SCD Type 2 load completed for
Dim_Passenger. ');
END;
/

exec LOAD_DIM_PASSENGER_SCD2;

```

Parts Dimension Table

```

CREATE OR REPLACE PROCEDURE P_UPDATE_DIM_PARTS AS
BEGIN

    UPDATE Dim_Parts dp
    SET EFFECTIVE_END_DATE = SYSDATE
    WHERE EXISTS (
        SELECT 1
        FROM PartLists p
        JOIN orderDetail o ON p.partID = o.partID
        JOIN Product prod ON o.productID = prod.productID
        JOIN Supplier s ON prod.supplierID = s.supplierID
        WHERE p.partID = dp.part_id
        AND (p.partName <> dp.part_name OR s.supplierName <>
dp.supplier)
        AND dp.EFFECTIVE_END_DATE =
TO_DATE('31-12-9999','DD-MM-YYYY')
    );

    COMMIT;
END P_UPDATE_DIM_PARTS;
/

exec P_UPDATE_DIM_PARTS;

```

Chapter 3 Business Analytics Reports

3.1 Jesmine Tey Khai Jing

3.1.1 Sales Performance by Longest Holiday and Trip Type

SQL Code:

```
-- Sales Performance by Longest Holiday and Trip Type
set pagesize 41
set linesize 156

TTITLE CENTER 'Sales Trend Analysis by Longest Holiday and Trip Type' -
          SKIP 1 CENTER 'Yearly Report' -
          SKIP 1 RIGHT 'Page: ' SQL.PNO

BTITLE CENTER 'End of Page ' SQL.PNO

COLUMN "Year"                FORMAT 9999                HEADING 'Year'
COLUMN "Long Trip"           FORMAT 999,999,999.99        HEADING 'Long Trip'
COLUMN "Diff_Long_Trip(%)"   FORMAT 999,990.00            HEADING 'Diff_Long_Trip(%)'
COLUMN "Short Trip"          FORMAT 999,999,999.99        HEADING 'Short Trip'
COLUMN "Diff_Short_Trip(%)"  FORMAT 999,990.00            HEADING 'Diff_Short_Trip(%)'
COLUMN "Total"               FORMAT 999,999,999.99        HEADING 'Total'
COLUMN "Diff_Total(%)"       FORMAT 999,990.00            HEADING 'Diff_Total(%)'

WITH cte_holiday_marks AS (
  SELECT dd.date_key,
         dd.cal_date,
         dd.holiday_flag,
         CASE
           WHEN dd.holiday_flag = 'Y'
                AND (LAG(dd.holiday_flag) OVER (ORDER BY dd.cal_date) = 'N'
                     OR LAG(dd.holiday_flag) OVER (ORDER BY dd.cal_date) IS NULL)
           THEN 1 ELSE 0
         END AS start_flag
  FROM Dim_Date dd
),
cte_holiday_groups AS (
  SELECT date_key,
         EXTRACT(YEAR FROM cal_date) AS year,
         holiday_flag,
         SUM(start_flag) OVER (ORDER BY cal_date) AS holiday_group
  FROM cte_holiday_marks
),
cte_holiday_periods AS (
  SELECT year,
         holiday_group,
         COUNT(*) AS holiday_days
```

```

        FROM cte_holiday_groups
        WHERE holiday_flag = 'Y'
        GROUP BY year, holiday_group
    ),
    cte_rank_period AS (
        SELECT year,
               holiday_group,
               holiday_days,
               RANK() OVER (PARTITION BY year ORDER BY holiday_days DESC) AS rnk
        FROM cte_holiday_periods
    ),
    cte_longest_period AS (
        SELECT year,
               holiday_group,
               holiday_days
        FROM cte_rank_period
        WHERE rnk = 1
    ),
    cte_sales_raw AS (
        SELECT
            lp.year,
            CASE WHEN dt.duration <= 2 THEN 'Short Trip' ELSE 'Long Trip' END AS
trip_type,
            sf.total_sales
        FROM Sales_Fact sf
        JOIN Dim_Date dd ON sf.date_key = dd.date_key
        JOIN Dim_Trip dt ON sf.trip_key = dt.trip_key
        JOIN cte_holiday_groups hg ON dd.date_key = hg.date_key
        JOIN cte_longest_period lp
            ON hg.holiday_group = lp.holiday_group
        AND hg.year = lp.year
        WHERE dd.holiday_flag = 'Y'
    ),
    cte_sales_agg AS (
        SELECT
            year,
            trip_type,
            SUM(total_sales) AS total_sales
        FROM cte_sales_raw
        GROUP BY year, trip_type
    ),
    cte_pivot_sales AS (
        SELECT
            year,
            SUM(CASE WHEN trip_type = 'Long Trip' THEN total_sales ELSE 0 END) AS
long_trip,
            SUM(CASE WHEN trip_type = 'Short Trip' THEN total_sales ELSE 0 END) AS
short_trip
        FROM cte_sales_agg

```

```

GROUP BY year
),
cte_final_summary AS (
    SELECT
        year,
        long_trip,
        short_trip,
        (long_trip + short_trip) AS total_sales,
        ROUND(
            100.0 * (long_trip - LAG(long_trip) OVER (ORDER BY year))
            / NULLIF(LAG(long_trip) OVER (ORDER BY year), 0),
            2
        ) AS diff_long_trip,
        ROUND(
            100.0 * (short_trip - LAG(short_trip) OVER (ORDER BY year))
            / NULLIF(LAG(short_trip) OVER (ORDER BY year), 0),
            2
        ) AS diff_short_trip,
        ROUND(
            100.0 * ((long_trip + short_trip)
                - LAG(long_trip + short_trip) OVER (ORDER BY year))
            / NULLIF(LAG(long_trip + short_trip) OVER (ORDER BY year), 0),
            2
        ) AS diff_total
    FROM cte_pivot_sales
)
SELECT
    year AS "Year",
    long_trip AS "Long Trip",
    diff_long_trip AS "Diff_Long_Trip(%)",
    short_trip AS "Short Trip",
    diff_short_trip AS "Diff_Short_Trip(%)",
    total_sales AS "Total",
    diff_total AS "Diff_Total(%)
FROM cte_final_summary
ORDER BY year;

tttitle off;
bttitle off;

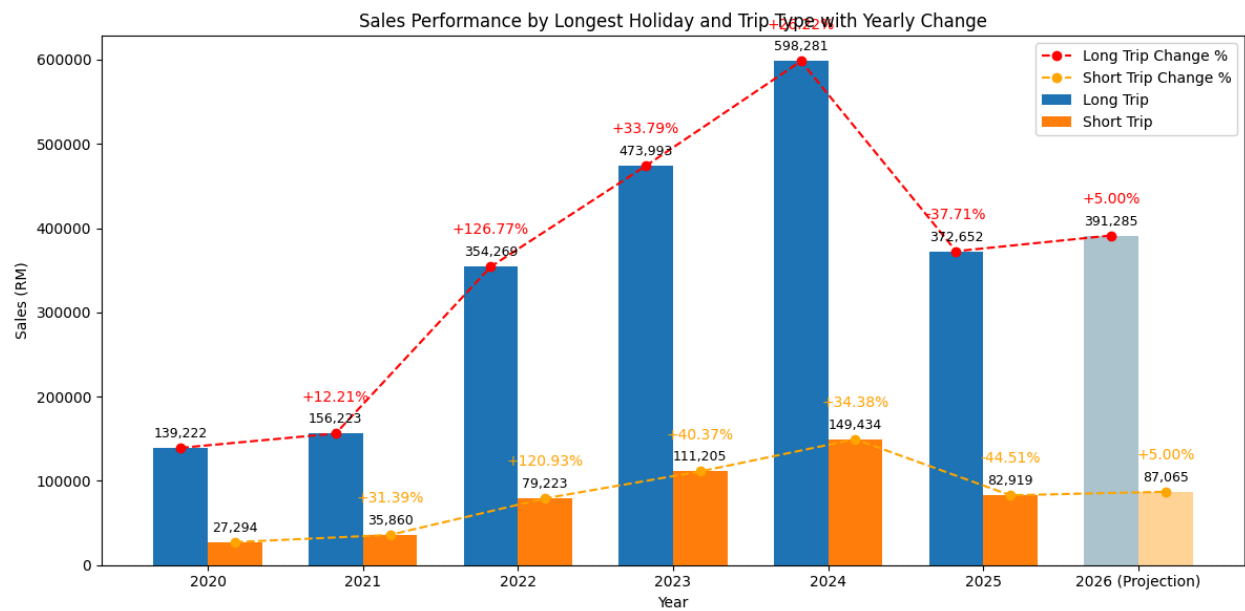
```

Output:

Sales Trend Analysis by Longest Holiday and Trip Type Yearly Report							Page:	1
Year	Long Trip	Diff_Long_Trip(%)	Short Trip	Diff_Short_Trip(%)	Total	Diff_Total(%)		
2020	139,221.67		27,293.65		166,515.32			
2021	156,223.14	12.21	35,859.77	31.39	192,082.91	15.35		
2022	354,269.25	126.77	79,223.20	120.93	433,492.45	125.68		
2023	473,992.97	33.79	111,204.58	40.37	585,197.55	35.00		
2024	598,280.77	26.22	149,434.42	34.38	747,715.19	27.77		
2025	372,652.12	-37.71	82,919.04	-44.51	455,571.16	-39.07		

End of Page 1

Visualization:



Analysis:

The sales performance during the longest holiday period between 2020 and 2025 shows a clear distinction between long trips and short trips. Short trips are trips that are within 2 hours. The long trips consistently contribute a majority of revenue. The contribution of long trips and short trips of each of the year is shown below:

Year	Long_Trip	Short_Trip	Contribution (% / %)
2020	139,221.67	27,293.65	83.6% / 16.4%
2021	156,223.14	35,859.77	81.3% / 18.7%
2022	354,269.25	79,223.20	81.7% / 18.3%
2023	473,992.97	111,204.58	81.0% / 19.0%
2024	598,280.77	149,434.42	80.0% / 20.0%
2025	372,652.12	82,919.04	81.8% / 18.2%

From the table, it can indicate the higher sales of longer trips during the longest holiday every year. In 2021, both categories have a growth which is long trips increase by 12.21% and short trips increase by 31.39%. This brings to an increase in total of 15.35. In 2022, there is a significant growth in both trip types, with long trips increasing by 126.77% and short trips increasing by 120.93%, producing a total increase of 125.68%. This sharp rebound was largely driven by the recovery from pandemic-related restrictions with pent-up travel demand fueling sales.

The upward trend continued in 2023 and 2024 although at a slower pace compared to 2022. In 2023, long trips sales increased by 33.79% while short trips increased by 40.37%. In 2024, long trips sales increased by 26.22% while short trips increased by 34.38%. This led to the total sales having a strong 27.77% increase. These gains were supported by longer holiday seasons, higher tourism demand and effective promotional campaigns.

However, in 2025, sales declined sharply. Long trips decrease by 37.71% and short trips decrease by 44.51%. The total sales decreased by 39.07% compared to the previous year. The downturn was influenced by economic slowdown and shorter holiday schedules which weakened overall demand.

The slight increase in sales from 2020 to 2021 was driven by the easing of COVID-19 restrictions, which allowed more people to travel during the longest holiday period. The significant

overall growth of 168.88% between 2022 and 2024 resulted from the full recovery of travel activities following the pandemic, supported by stronger economic conditions, longer holiday periods and effective promotional campaigns. In contrast, the sharp decline in 2025 occurred due to the rise of alternative travel modes, as a large portion of Malaysian citizens owned private vehicles or chose air travel as a faster and more convenient option. Additionally, the economic slowdown and shorter holiday schedules further reduced demand, leading to lower ticket sales.

Projection for 2026:

	Long Trip	Short Trip
Bus Assigned in 2025	716	165
Forecast Bus Assigned in 2026	$716 \times 1.05\% = 752 \text{ Bus}$ $\approx 800 \text{ Bus}$	$165 \times 1.05\% = 174 \text{ Bus}$ $\approx 200 \text{ Bus}$
Total Sales in 2025	372,652	82,919
Forecast Sales in 2026	$372,652 \times 1.05\% = 391,285$ $\approx 400,000$	$82,919 \times 1.05\% = 87,065$ $\approx 90,000$

In 2025, the analysis of the longest holiday period shows that a total of 716 buses were assigned for long trips and 165 buses for short trips. Moving forward, it is recommended to forecast bus requirements for the upcoming year to ensure sufficient capacity during peak holiday demand. Applying a 5% growth projection, the number of buses needed in 2026 is estimated to be 800 for long trips and 200 for short trips. By using this growth projection, the company can get a total sales of approximately RM 400,000 for long trips and RM 90,000 for short trips.

To meet this expected growth, the company should consider expanding routes or introducing express services for the more popular trip types, particularly long trips where demand is consistently higher. At the same time, increasing the number of buses allocated will help avoid service disruptions and overcrowding during peak travel days. In addition, launching targeted promotions and forming strategic partnerships can further stimulate demand, balance passenger loads across trip types and maximize revenue opportunities during the holiday season.

3.1.2 Top Bus Company Maintenance Cost Analysis by Year & Quarter

SQL Code:

```
-- Top Bus Company Maintenance Cost Analysis by Year & Quarter
SET PAGESIZE 41
SET LINESIZE 156
```

```
TTITLE CENTER 'Top Bus Company Maintenance Cost Analysis' -
          SKIP 1 CENTER 'Yearly and Quarterly Report' -
          SKIP 1 RIGHT 'Page: ' SQL.PNO
```

```
BTITLE CENTER 'End of Page ' SQL.PNO
```

```
COLUMN Year          FORMAT 9999
COLUMN Quarter       FORMAT A7
COLUMN Engine        FORMAT 999,999,990.00 HEADING 'Engine'
COLUMN Engine_Pct    FORMAT 999,990.00 HEADING 'Engine %'
COLUMN Brake         FORMAT 999,999,990.00 HEADING 'Brake'
COLUMN Brake_Pct     FORMAT 999,990.00 HEADING 'Brake %'
COLUMN Electrical    FORMAT 999,999,990.00 HEADING 'Electrical'
COLUMN Electrical_Pct FORMAT 999,990.00 HEADING 'Elect %'
COLUMN Tires         FORMAT 999,999,990.00 HEADING 'Tires'
COLUMN Tires_Pct     FORMAT 999,990.00 HEADING 'Tires %'
COLUMN Total         FORMAT 999,999,990.00 HEADING 'Total'
```

```
BREAK ON Quarter SKIP 2
COMPUTE SUM OF Engine ON Quarter
COMPUTE SUM OF Brake ON Quarter
COMPUTE SUM OF Electrical ON Quarter
COMPUTE SUM OF Tires ON Quarter
COMPUTE SUM OF Total ON Quarter
```

```
WITH cte_top_company AS (
    SELECT * FROM (
        SELECT db.companyName AS BusCompany,
               SUM(mf.total_expenses) AS Total_Expenses
        FROM Maintenance_Fact mf
        JOIN Dim_Bus db ON mf.bus_key = db.bus_key
        GROUP BY db.companyName
        ORDER BY Total_Expenses DESC
    )
    WHERE ROWNUM = 1
),
cte_quarterly_costs AS (
    SELECT dd.year AS Year,
           dd.quarter AS Quarter,
           mf.maintenance_type,
           SUM(mf.total_expenses) AS Total_Cost
    FROM Maintenance_Fact mf
    JOIN Dim_Date dd ON mf.date_key = dd.date_key
```

```

JOIN Dim_Bus db ON mf.bus_key = db.bus_key
WHERE db.companyName = (SELECT BusCompany FROM cte_top_company)
      AND dd.year BETWEEN 2023 AND 2025
GROUP BY dd.year, dd.quarter, mf.maintenance_type
),
cte_pivot_table AS (
  SELECT *
  FROM cte_quarterly_costs
  PIVOT(
    SUM(Total_Cost)
    FOR maintenance_type IN (
      'ENGINE'      AS Engine,
      'TRANSMISSION' AS Transmission,
      'BRAKE'       AS Brake,
      'ELECTRICAL'  AS Electrical,
      'TIRES'       AS Tires
    )
  )
),
cte_with_lag AS (
  SELECT Quarter,
         Year,
         NVL(Engine,0)      AS Engine,
         ROUND((Engine - LAG(Engine) OVER (PARTITION BY Quarter ORDER BY Year))
              / NULLIF(LAG(Engine) OVER (PARTITION BY Quarter ORDER BY Year),0)
         * 100, 2) AS Engine_Pct,
         NVL(Brake,0)      AS Brake,
         ROUND((Brake - LAG(Brake) OVER (PARTITION BY Quarter ORDER BY Year))
              / NULLIF(LAG(Brake) OVER (PARTITION BY Quarter ORDER BY Year),0)
         * 100, 2) AS Brake_Pct,
         NVL(Electrical,0) AS Electrical,
         ROUND((Electrical - LAG(Electrical) OVER (PARTITION BY Quarter ORDER BY
Year))
              / NULLIF(LAG(Electrical) OVER (PARTITION BY Quarter ORDER BY
Year),0)
         * 100, 2) AS Electrical_Pct,
         NVL(Tires,0)      AS Tires,
         ROUND((Tires - LAG(Tires) OVER (PARTITION BY Quarter ORDER BY Year))
              / NULLIF(LAG(Tires) OVER (PARTITION BY Quarter ORDER BY Year),0)
         * 100, 2) AS Tires_Pct,
         (NVL(Engine,0) + NVL(Transmission,0) + NVL(Brake,0) +
          NVL(Electrical,0) + NVL(Tires,0)) AS Total
  FROM cte_pivot_table
)
SELECT Quarter,
       Year,
       Engine, Engine_Pct,
       Brake, Brake_Pct,
       Electrical, Electrical_Pct,
       Tires, Tires_Pct,

```

Total
FROM cte_with_lag
ORDER BY Quarter, Year;

ttitle off;
btitle off;

Output:

Top Bus Company Maintenance Cost Analysis Yearly and Quarterly Report										Page:	1
QUARTER	YEAR	Engine	Engine %	Brake	Brake %	Electrical	Elect %	Tires	Tires %	Total	
Q1	2023	1,984,311.10		1,654,566.94		2,437,486.92		1,687,726.95		7,764,091.91	
	2024	2,486,830.31	25.32	3,092,699.87	86.92	892,848.00	-63.37	1,515,325.11	-10.22	7,987,703.29	
	2025	1,720,542.54	-30.81	2,119,676.22	-31.46	493,284.00	-44.75	2,016,180.13	33.05	6,349,682.89	

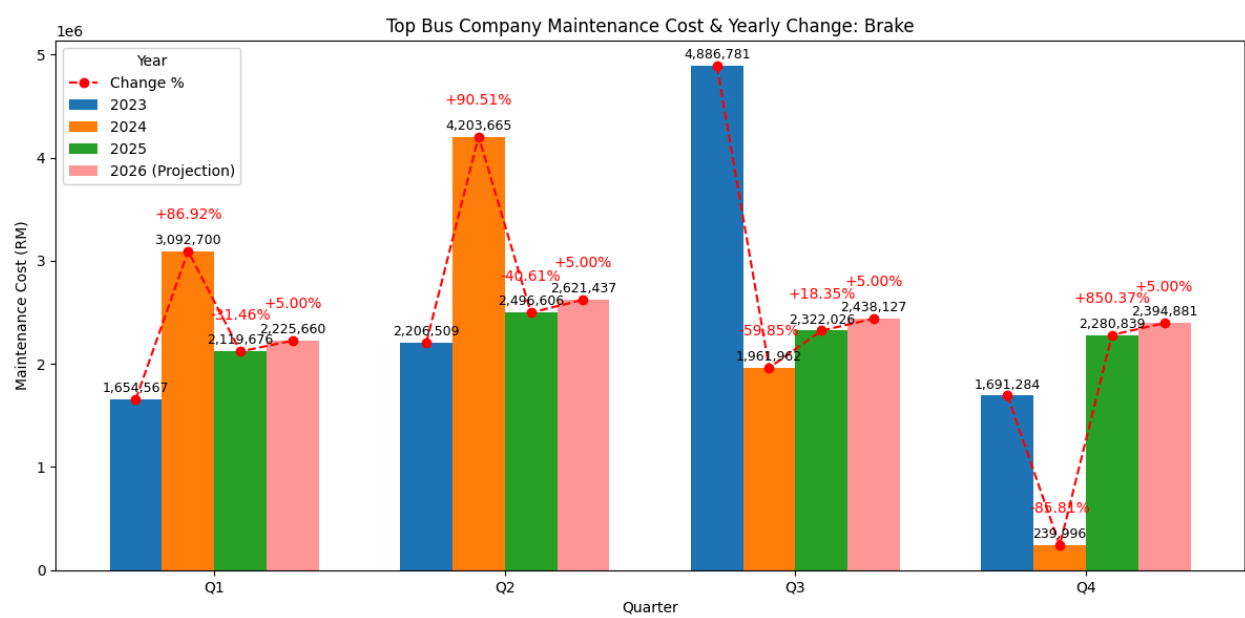
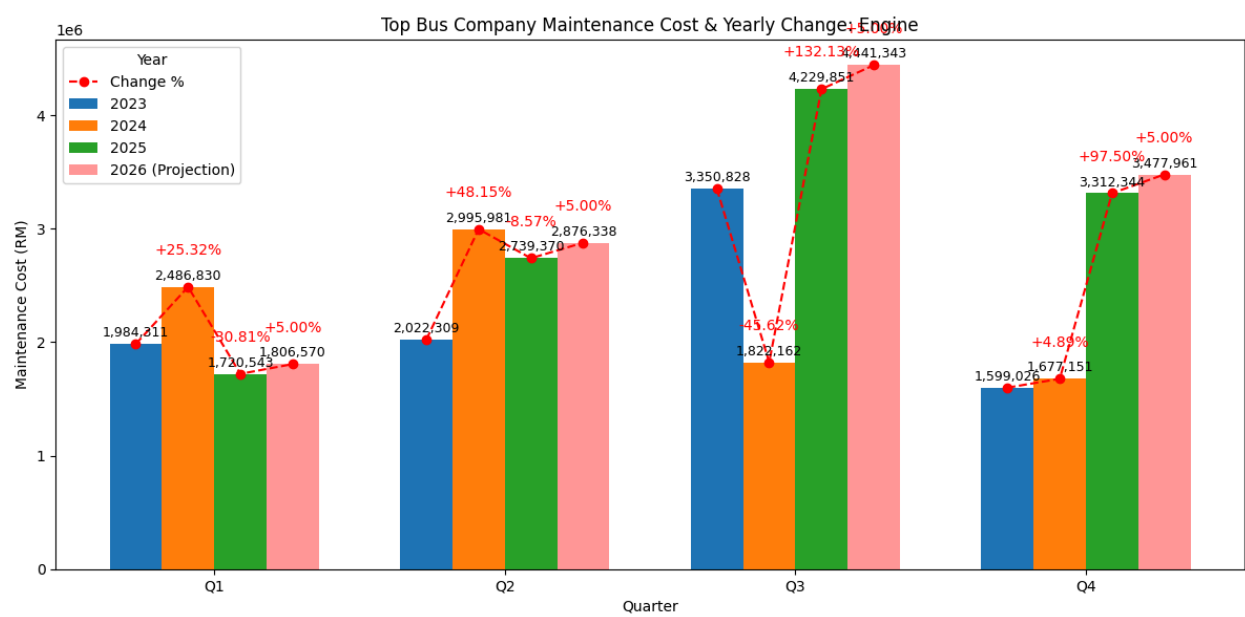
sum		6,191,683.95		6,866,943.03		3,823,618.92		5,219,232.19		22,101,478.09	
Q2	2023	2,022,309.48		2,206,509.21		796,833.18		1,025,174.09		6,050,825.96	
	2024	2,995,981.48	48.15	4,203,665.35	90.51	692,308.32	-13.12	276,503.03	-73.03	8,168,458.18	
	2025	2,739,369.94	-8.57	2,496,606.28	-40.61	1,581,095.01	128.38	285,540.09	3.27	7,102,611.32	

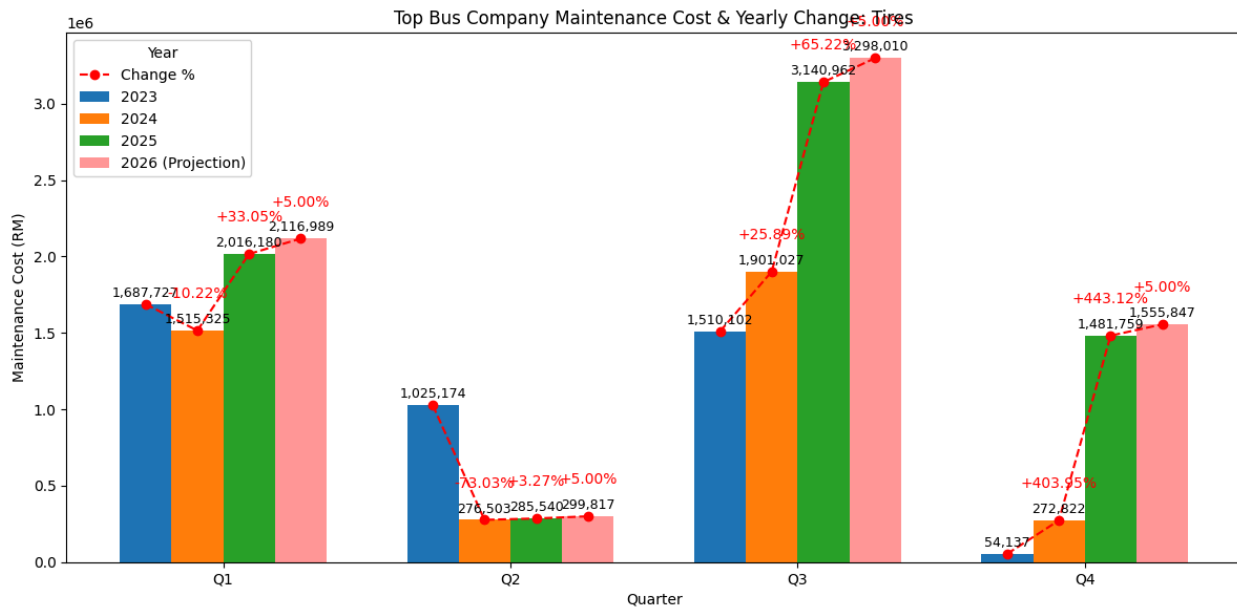
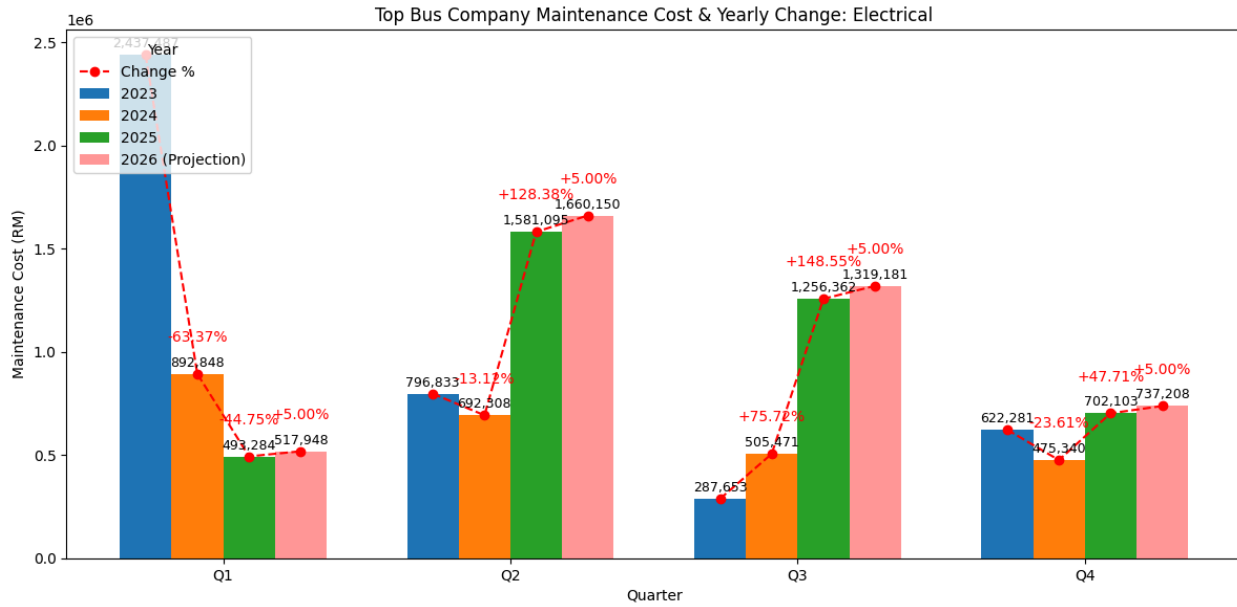
sum		7,757,660.90		8,906,780.84		3,070,236.51		1,587,217.21		21,321,895.46	
Q3	2023	3,350,827.83		4,886,781.21		287,653.08		1,510,102.44		10,035,364.56	
	2024	1,822,161.55	-45.62	1,961,961.91	-59.85	505,471.23	75.72	1,901,027.26	25.89	6,190,621.95	
	2025	4,229,850.84	132.13	2,322,025.52	18.35	1,256,362.47	148.55	3,140,962.13	65.22	10,949,200.96	

sum		9,402,840.22		9,170,768.64		2,049,486.78		6,552,091.83		27,175,187.47	
Q4	2023	1,599,025.62		1,691,283.58		622,280.67		54,137.16		3,966,727.03	
	2024	1,677,151.36	4.89	2,239,995.80	-85.81	475,339.59	-23.61	272,821.56	403.95	2,665,308.31	
	2025	3,312,344.24	97.50	2,280,838.64	850.37	702,103.05	47.71	1,481,759.40	443.12	7,777,045.33	

sum		6,588,521.22		4,212,118.02		1,799,723.31		1,808,718.12		14,409,080.67	

Visualization:





Analysis:

The maintenance cost analysis of SWIFT COACH from 2023 to 2025 has higher fluctuations in spending across the four main components which are engine, brake, electrical systems and tires. These figures reflect the operational pressures of managing an aging fleet seasonal travel demand and the impact of accidents or efficiency improvements.

Quarter 1 (Q1)

In Quarter 1 (Q1), the data shows that in 2023 the total maintenance cost was RM 7,764,091.91, with the highest spending on electrical systems (RM 2,437,486.92) and engines (RM 1,984,311.10). In 2024, the cost increased slightly to RM 7,987,703.29 because engine costs rose by 25.32% and brake expenses increased sharply by 86.92%. At the same time, electrical costs dropped significantly by -63.37% to RM 892,848.00. In 2025, the total cost decreased to RM 6,349,682.89 because engine costs fell by -30.81% and brake costs dropped by -31.46%. However, tire costs went up by 33.05% to RM 2,016,180.13. This shows that in 2024, the increase was significant because of major engine and brake repairs, while in 2025 the reduction reflected better cost control but also a shift in problems, with tire wear becoming a bigger issue. The main reasons behind these changes were aging engines and braking systems in 2024, followed by more frequent and longer trips in 2025 which increased tire usage.

Quarter 2 (Q2)

In Quarter 2 (Q2), the total cost in 2023 was RM 6,050,825.96. It surged to RM 8,168,458.18 in 2024, mainly due to a 48.15% increase in engine costs (RM 2,995,981.48) and a 90.51% jump in brake costs (RM 4,203,665.35). Tire costs, however, dropped drastically by -73.03% to RM 276,503.03, which suggests better tire management. In 2025, the total cost fell slightly to RM 7,102,611.32 because engine costs decreased by -8.57%. However, electrical costs rose sharply by 128.38% to RM 1,581,095.01. The significant rise in 2024 showed high pressure on the system, while the spike in electrical costs in 2025 highlighted a new issue that disrupted cost stability. These changes happened because of heavy mid-year travel demand and deferred maintenance in 2024, better tire rotation strategies and electrical problems in 2025 that may have been caused by wiring failures, overheating or weather conditions.

Quarter 3 (Q3)

In Quarter 3 (Q3), the total cost was the highest in 2023, reaching RM 10,035,364.56, with brakes (RM 4,886,781.21) and engine repairs (RM 3,350,827.83) taking up most of the budget. In 2024,

costs dropped sharply to RM 6,190,621.95 as engine maintenance fell by -45.62% to RM 1,822,161.55 and brakes reduced by -59.85% to RM 1,961,961.91. However, in 2025, costs rebounded strongly to RM 10,949,200.96, with engine costs increasing by 132.13%, electrical repairs rising by 148.55% and tire costs going up by 65.22%. The drop in 2024 was significant because it showed that cost reduction efforts were effective, but the 2025 rebound was even more important because it erased those savings and became the highest-cost quarter in three years. These swings happened because 2023 had high breakdowns during peak travel, 2024 had temporary efficiency or fewer trips and 2025 faced long-distance demand, accident-related repairs and problems with aging fleets.

Quarter 4 (Q4)

In Quarter 4 (Q4), the total cost was RM 3,966,727.03 in 2023, which was the lowest quarter. Costs decreased further in 2024 to RM 2,665,308.31 because brake costs dropped sharply by -85.81% to only RM 239,995.80. However, tire costs increased by 403.95% to RM 272,821.56. By 2025, costs surged to RM 7,777,045.33, which was almost three times higher than the previous year. This increase was driven by engines rising 97.50% to RM 3,312,344.24, brakes jumping 850.37% to RM 2,280,838.64 and tires climbing 443.12% to RM 1,481,759.40. The sharp fall in 2024 looked like a success in cost control, but it was not sustainable. The big spike in 2025 was significant because it showed serious safety issues in brakes and tires, as well as heavy strain from high year-end travel demand. These changes happened because of reduced preventive maintenance in 2023 and 2024, followed by catch-up repairs and higher demand at the end of 2025.

Projection for 2026:

Total spare parts used for the maintenance of SWIFT COACH company in 2025 and projection for 2026 is shown below:

- 5% increase for spare parts usage

Quarter	Engine		Brake		Electrical		Tires	
	2025	2026	2025	2026	2025	2026	2025	2026
Q1	10,396	≈ 11,000	18,810	≈ 19,000	2,442	≈ 3,000	11,573	≈ 12,000
Q2	12,618	≈ 14,000	12,148	≈ 13,000	12,837	≈ 13,000	2,277	≈ 3,000

Q3	24,507	≈ 25,000	14,939	≈ 15,000	5,973	≈ 6,000	15,440	≈ 16,000
Q4	17,684	≈ 18,000	15,648	≈ 16,000	2,211	≈ 3,000	6,006	≈ 7,000

Forecast Maintenance Cost for SWIFT COACH company:

Maintenance Type	2025				Forecast in 2026 (≈)			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Engine	1,721,543	2,739,370	4,229,851	3,312,344	2,000,000	3,000,000	5,000,000	4,000,000
Brake	2,119,676	2,496,606	2,322,026	2,280,839	3,000,000	3,000,000	3,000,000	3,000,000
Electrical	493,284	1,581,095	1,256,362	702,103	600,000	2,000,000	2,000,000	800,000
Tires	2,016,180	285,540	3,140,962	3,298,010	3,000,000	300,000	2,000,000	2,000,000

The projection for 2026 indicates a 5% increase in spare parts usage and overall maintenance costs compared to 2025. Based on the table, engine parts are expected to rise from 10,396 units in Q1 2025 to 11,000 units in Q1 2026, while brake parts will increase from 18,810 to 19,000 in the same quarter. Similarly, forecasted maintenance costs for engines are projected to grow from RM1.72 million in Q1 2025 to RM2 million in Q1 2026, with comparable upward trends across brakes, electrical and tires. These figures highlight the need for SWIFT COACH to invest in preventive maintenance to manage the growing demand efficiently and optimize scheduling to reduce downtime. By using the projection above, the spare part inventories should be prepared adequately for the maintenance and the bus company can prepare for the maintenance cost for each quarter. Additionally, because of the rising long-term expenses, the planning for replacement or upgrades of aging fleet vehicles should help sustain operational efficiency and minimize future maintenance burdens.

3.1.3 Yearly Bus Ticket Sales Analysis during School Holiday

SQL Code:

```
-- Yearly Bus Ticket Sales Analysis during School Holiday
SET PAGESIZE 41
SET LINESIZE 156

TTITLE CENTER 'Bus Ticket Sales Analysis during School Holiday' -
        SKIP 1 CENTER 'Yearly Report' -
        SKIP 1 RIGHT 'Page: ' SQL.PNO

BTITLE CENTER 'End of Page ' SQL.PNO

COLUMN Holiday                FORMAT A25
COLUMN Y2023                  FORMAT 999,999,999.00
COLUMN Y2024                  FORMAT 999,999,999.00
COLUMN Y2025                  FORMAT 999,999,999.00
COLUMN "Diff_2023_2024(%)"    FORMAT 99,999,990.00
COLUMN "Diff_2024_2025(%)"    FORMAT 99,999,990.00
COLUMN Total                  FORMAT 999,999,999.00

WITH cte_festival_sales AS (
    SELECT
        dd.is_festival AS Holiday,
        dd.year,
        SUM(sf.total_sales) AS total_sales
    FROM Sales_Fact sf
    JOIN Dim_Date dd ON sf.date_key = dd.date_key
    WHERE dd.is_festival LIKE 'Sch Holi%'
    GROUP BY dd.is_festival, dd.year
),
cte_pivot_table AS (
    SELECT
        Holiday,
        MAX(CASE WHEN year = 2023 THEN total_sales END) AS y2023,
        MAX(CASE WHEN year = 2024 THEN total_sales END) AS y2024,
        MAX(CASE WHEN year = 2025 THEN total_sales END) AS y2025
    FROM cte_festival_sales
    GROUP BY Holiday
    HAVING
        (MAX(CASE WHEN year = 2023 THEN total_sales END) > 100000 OR
         MAX(CASE WHEN year = 2024 THEN total_sales END) > 100000 OR
         MAX(CASE WHEN year = 2025 THEN total_sales END) > 100000)
),
cte_with_diff AS (
    SELECT
        Holiday,
        y2023,
        y2024,
        y2025,
```

```

CASE
    WHEN y2023 IS NOT NULL AND y2024 IS NOT NULL AND y2023 != 0
    THEN ROUND(((y2024 - y2023) / y2023) * 100, 2)
END AS Diff_2023_2024,
CASE
    WHEN y2024 IS NOT NULL AND y2025 IS NOT NULL AND y2024 != 0
    THEN ROUND(((y2025 - y2024) / y2024) * 100, 2)
END AS Diff_2024_2025,
(NVL(y2023,0) + NVL(y2024,0) + NVL(y2025,0)) AS Total
FROM cte_pivot_table
)
SELECT
    COALESCE(Holiday, 'Total') AS Holiday,
    SUM(y2023) AS Y2023,
    SUM(y2024) AS Y2024,
    SUM(y2025) AS Y2025,
    CASE WHEN GROUPING(Holiday) = 0 THEN MAX(Diff_2023_2024) END AS
"Diff_2023_2024(%)",
    CASE WHEN GROUPING(Holiday) = 0 THEN MAX(Diff_2024_2025) END AS
"Diff_2024_2025(%)",
    SUM(Total) AS Total
FROM cte_with_diff
GROUP BY ROLLUP(Holiday)
ORDER BY
    CASE WHEN Holiday = 'Total' THEN 1 ELSE 0 END,
    Total DESC;

tttitle off;
bttitle off;

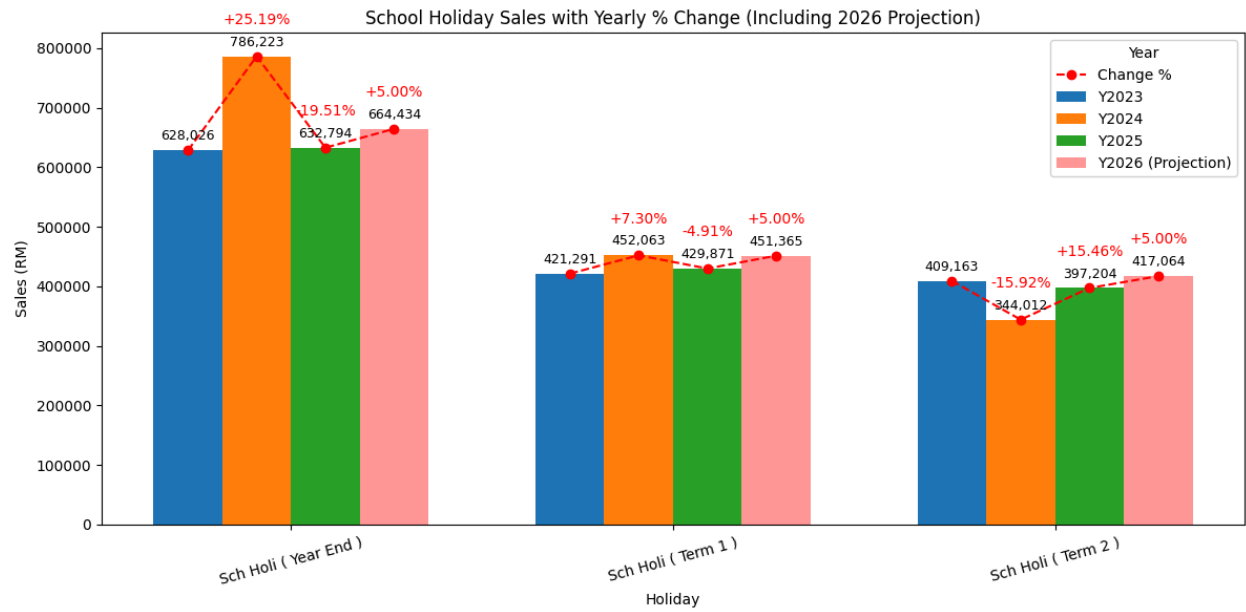
```

Output:

Bus Ticket Sales Analysis during School Holiday							Page: 1	
Yearly Report								
HOLIDAY	Y2023	Y2024	Y2025	Diff_2023_2024(%)	Diff_2024_2025(%)	Total		
Sch Holi (Year End)	628,025.63	786,223.12	632,793.94	25.19	-19.51	2,047,042.69		
Sch Holi (Term 1)	421,290.63	452,063.20	429,871.05	7.30	-4.91	1,303,224.88		
Sch Holi (Term 2)	409,162.79	344,011.69	397,203.84	-15.92	15.46	1,150,378.32		
Sch Holi (Mid Term)		374,076.17				374,076.17		
Sch Holi (Mid Term 1)	352,787.67					352,787.67		
Sch Holi (Raya)			272,800.30			272,800.30		
Sch Holi (Deepavali)			142,307.09			142,307.09		
Total	1,811,266.72	1,956,374.18	1,874,976.22			5,642,617.12		

End of Page 1

Visualization:



Analysis:

The analysis of bus ticket sales during school holidays between 2023 and 2025 shows clear patterns of demand influenced by the type of holiday. Overall sales across the three years amounted to RM 5,642,617.12, with the year-end school holiday contributing the highest total at RM 2,047,042.69, followed by Term 1 holidays with RM 1,303,224.88 and Term 2 with RM 1,150,378.32. These three periods consistently exceeded RM 100,000 each year, showing that they are the main drivers of revenue. In comparison, newly recorded periods such as Raya (RM 272,800.30) and Deepavali (RM 142,307.09) in 2025 contributed less, but they still show the importance of festive-linked travel demand.

In Year-End holidays, sales grew from RM 628,025.63 in 2023 to RM 786,223.12 in 2024, an increase of 25.19%, before falling back to RM 632,793.94 in 2025, a decline of -19.51%. This increase was significant because year-end is traditionally the busiest travel season, with longer breaks encouraging long-distance family trips and strong tourism demand. The decline in 2025 reflected the impact of economic slowdown, reduced disposable income and competition from alternative travel modes such as private cars and flights.

For Term 1 holidays, sales increased slightly from RM 421,290.63 in 2023 to RM 452,063.20 in 2024, a growth of 7.30%, but then dropped to RM 429,871.05 in 2025, a decline of -4.91%. This modest growth showed the importance of early-year travel demand and effective marketing, but the later decline was significant because it reflected shorter holiday periods and weaker demand compared to year-end seasons.

The Term 2 holiday displayed the most volatile trend. Sales dropped from RM 409,162.79 in 2023 to RM 344,011.69 in 2024, a decrease of -15.92%, before rebounding strongly in 2025 to RM 397,203.84, a growth of 15.46%. The sharp decline in 2024 reflected lower travel interest or shorter school breaks during this period, while the recovery in 2025 indicated that the timing of holidays and stronger promotions restored demand.

For mid-term breaks, sales were recorded at RM 352,787.67 in 2023 and RM 374,076.17 in 2024, showing a consistent contribution above RM 300,000. However, no mid-term figures were reported in 2025, suggesting either shorter schedules or changes in travel behavior.

In festive-linked holidays, new categories appeared in 2025, with Raya generating RM 272,800.30 and Deepavali contributing RM 142,307.09. Although these were lower than the main

school holidays, they are significant as they reflect the importance of cultural and religious holidays in shaping travel patterns.

Overall, total yearly ticket sales rose from RM 1,811,266.72 in 2023 to RM 1,956,374.18 in 2024, a growth of 8.02%, but then fell to RM 1,874,976.22 in 2025, a decline of -4.16%. The increase in 2024 was significant because it showed strong recovery momentum after pandemic restrictions, boosted by effective marketing and longer breaks. The decline in 2025 was equally important as it signaled reduced demand caused by economic challenges, higher competition and shorter holiday periods.

The reasons behind these patterns are clear. Year-end holidays consistently perform best because they attract long-distance family trips. Term holidays vary in demand because of their shorter duration and timing within the school calendar. The economic downturn in 2025 reduced disposable income, while the rise of competition from flights and private vehicles further affected demand. At the same time, festive holidays like Raya and Deepavali highlight new growth opportunities in cultural travel demand.

Projection for 2026:

Holiday	Total Bus Assigned in 2025	Forecast of Bus Assigned in 2026
School Holiday (Year End)	1,078	≈ 1,200
School Holiday (Term 1)	776	≈ 900
School Holiday (Term 2)	707	≈ 800
School Holiday (Raya)	465	≈ 500
School Holiday (Deepavali)	272	≈ 300

The projection for 2026 shows a 5% increase in bus assignments across all school holiday periods compared to 2025, reflecting the expected rise in passenger demand. Based on the table above, during the year-end school holiday, the number of buses is forecasted to increase from 1,078 in 2025 to 1,200 in 2026, while Term 1 is expected to rise from 776 to 900 buses. The growth trends can be observed for Term 2 (707 to 800 buses), Raya (465 to 500 buses) and Deepavali (272 to 300 buses). Based on the projections, the actions should be taken in 2026 to increase the sales. The company should adjust its pricing strategy to balance demand and profitability, while allocating more buses to high-demand holiday periods such as year-end and Term 1. Additional trips should be introduced

where necessary to meet rising passenger needs, supported by an expansion of online booking capacity to ensure a seamless customer experience. At the same time, offering targeted promotions and strengthening customer engagement initiatives will help attract and retain passengers, maximizing revenue during these peak periods.

3.2 Gan Khai Li

3.2.1 Annual Sales and Ticket Trend by Origin and Time Period (2023-2025)

SQL Code:

```
SET PAGESIZE 35
SET LINESIZE 130

COLUMN origin FORMAT A20
COLUMN time_period FORMAT A15
COLUMN year FORMAT 9999
COLUMN ticket_quantity FORMAT 999,999,999
COLUMN total_sales FORMAT 999,999,999.99
COLUMN avg_sales_per_ticket FORMAT 999,999.99
COLUMN diff_from_prev_year FORMAT 999,999,999.99

BREAK ON origin SKIP 1 ON time_period
COMPUTE SUM LABEL 'Subtotal' OF total_sales ON origin;

TTITLE LEFT '                          Annual Sales and Ticket Trend by Origin and Time
Period (2023-2025)' SKIP 2

WITH trip_time_classification AS (
    SELECT
        T.origin,
        D.year,
        CASE
            WHEN TO_NUMBER(TO_CHAR(T.departure_time, 'HH24')) BETWEEN 5 AND 11 THEN
'Morning'
            WHEN TO_NUMBER(TO_CHAR(T.departure_time, 'HH24')) BETWEEN 12 AND 16
THEN 'Afternoon'
            ELSE 'Night'
        END AS time_period,
        SF.total_sales,
        SF.bookingID
    FROM Sales_Fact SF
    JOIN Dim_Trip T ON SF.trip_key = T.trip_key
    JOIN Dim_Date D ON SF.date_key = D.date_key
    WHERE D.year BETWEEN 2023 AND 2025
),
sales_agg AS (
    SELECT
        origin,
        time_period,
        year,
        SUM(total_sales) AS total_sales,
        COUNT(*) AS ticket_quantity,
        CASE WHEN COUNT(*) > 0 THEN SUM(total_sales) / COUNT(*) ELSE 0 END AS
avg_sales_per_ticket
```

```

        FROM trip_time_classification
        GROUP BY origin, time_period, year
    ),
    origin_totals AS (
        SELECT origin, SUM(total_sales) AS total_origin_sales
        FROM sales_agg
        GROUP BY origin
    ),
    max_min_origins AS (
        SELECT origin FROM origin_totals
        WHERE total_origin_sales = (SELECT MAX(total_origin_sales) FROM origin_totals)
        OR total_origin_sales = (SELECT MIN(total_origin_sales) FROM origin_totals)
    ),
    filtered_sales AS (
        SELECT s.*
        FROM sales_agg s
        JOIN max_min_origins m ON s.origin = m.origin
    ),
    sales_with_diff AS (
        SELECT
            f.origin,
            f.time_period,
            f.year,
            f.ticket_quantity,
            f.total_sales,
            f.avg_sales_per_ticket,
            LAG(f.total_sales) OVER (PARTITION BY f.origin, f.time_period ORDER BY
f.year) AS prev_year_sales
        FROM filtered_sales f
    )
    SELECT
        origin,
        time_period,
        year,
        ticket_quantity,
        total_sales,
        avg_sales_per_ticket,
        CASE
            WHEN prev_year_sales IS NULL THEN NULL
            ELSE total_sales - prev_year_sales
        END AS diff_from_prev_year
    FROM sales_with_diff
    ORDER BY origin,
        CASE time_period
            WHEN 'Morning' THEN 1
            WHEN 'Afternoon' THEN 2
            WHEN 'Night' THEN 3
            ELSE 4
        END,

```

year;

CLEAR BREAKS
CLEAR COLUMNS
TTITLE OFF
CLEAR COMPUTES

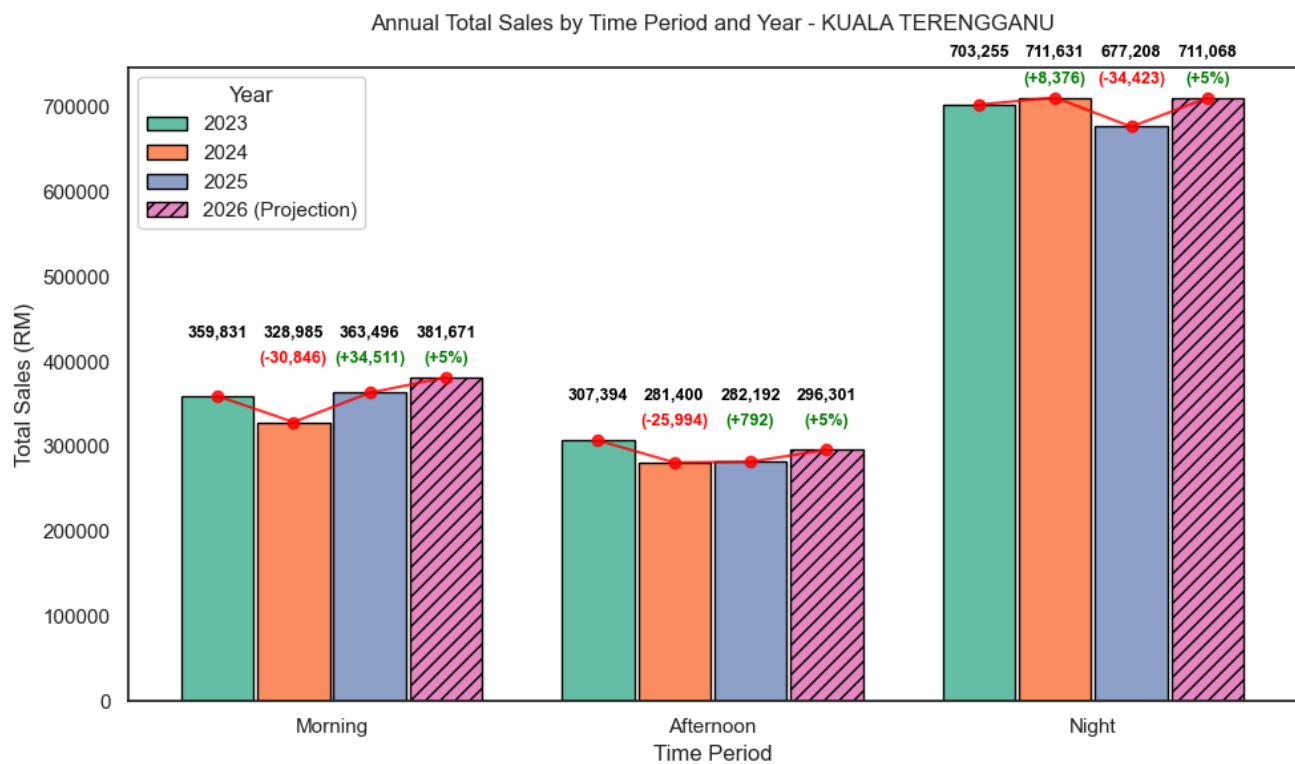
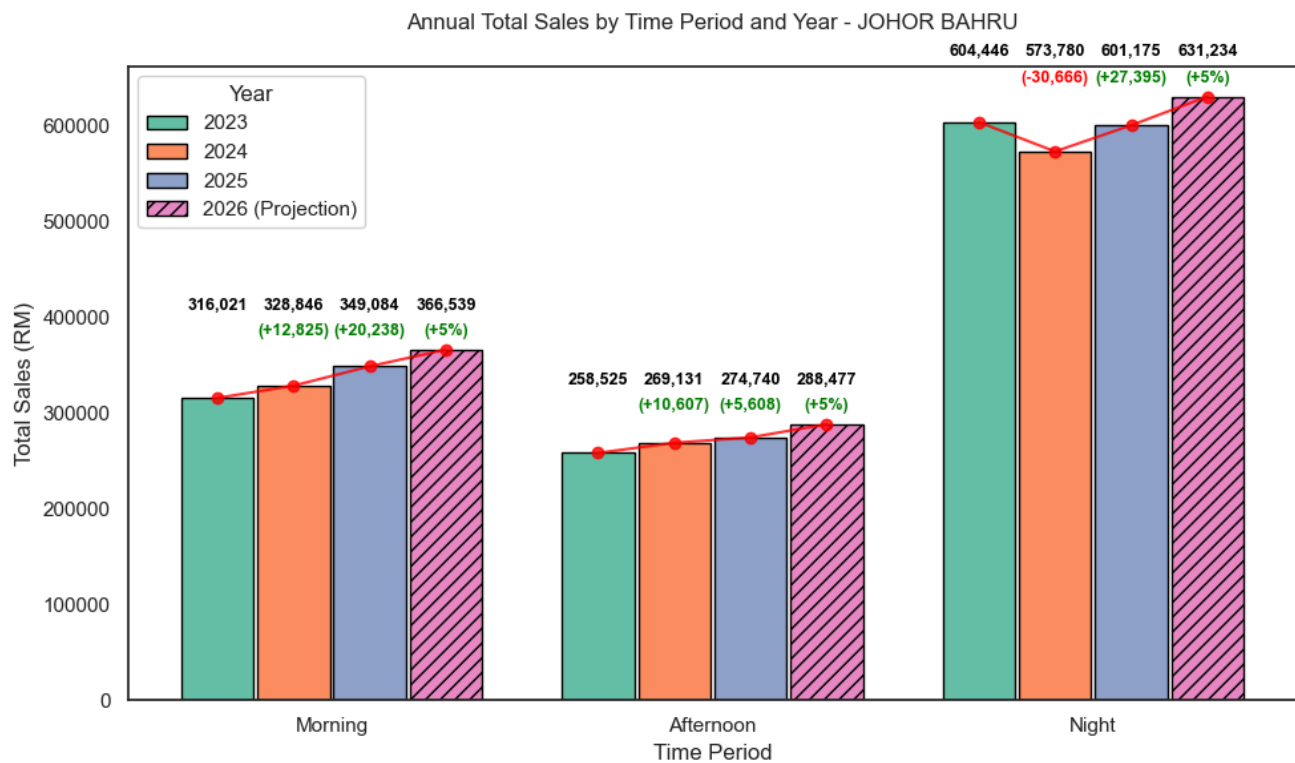
Output:

Annual Sales and Ticket Trend by Origin and Time Period (2023-2025)

ORIGIN	TIME_PERIOD	YEAR	TICKET_QUANTITY	TOTAL_SALES	AVG_SALES_PER_TICKET	DIFF_FROM_PREV_YEAR

JOHOR BAHRU	Morning	2023	742	316,020.90	425.90	
		2024	791	328,846.06	415.73	12,825.16
		2025	821	349,084.38	425.19	20,238.32
	Afternoon	2023	612	258,524.53	422.43	
		2024	631	269,131.15	426.52	10,606.62
		2025	622	274,739.63	441.70	5,608.48
	Night	2023	1,391	604,445.86	434.54	
		2024	1,313	573,780.01	437.00	-30,665.85
		2025	1,364	601,175.13	440.74	27,395.12
*****				-----		
Subtotal				3,575,747.65		
KUALA TERENGGANU	Morning	2023	834	359,830.74	431.45	
		2024	792	328,985.20	415.39	-30,845.54
		2025	823	363,495.81	441.67	34,510.61
	Afternoon	2023	672	307,394.41	457.43	
		2024	692	281,400.15	406.65	-25,994.26
		2025	667	282,191.88	423.08	791.73
	Night	2023	1,624	703,254.78	433.04	
		2024	1,644	711,630.53	432.87	8,375.75
		2025	1,531	677,207.55	442.33	-34,422.98
*****				-----		
Subtotal				4,015,391.05		

Visualization:



Analysis

From 2023 to 2025, bus sales in Johor Bahru and Kuala Terengganu showed different patterns depending on the time of day. In Johor Bahru, the number of tickets sold and the total money earned kept going up slowly during the day. This means that more people are taking the bus during daytime hours. In Kuala Terengganu, the numbers changed more often, with some times showing fewer sales than before. This means that fewer people were taking the bus during some parts of the day and the sales were not always going up each year. Based on the 2025 results, a 5% increase in total sales is expected in 2026 for all routes and times. This means that the company thinks sales will go up a little more next year for every place and time.

In Johor Bahru, morning and afternoon sales increased every year. Morning sales rose from about RM316,000 in 2023 to RM349,000 in 2025. This shows that more people are using the bus in the morning, maybe to go to work or school. Afternoon sales also went up, from RM258,000 to RM275,000. This shows that more people are also taking the bus in the afternoon, maybe after work, school or for other activities. This shows that more people are using the bus during the day. This could be because more workers are taking the bus or because the service has improved. For example, buses may now come more often or arrive on time. Night sales made the most money overall. This means that even though other times are growing, night buses still bring in the highest amount of sales. However, they were not stable. In 2024, the sales went down, which means fewer people took the bus at night that year. Sales dropped in 2024 and went up again in 2025. This shows that there were changes in how many people were using the bus at night. Night sales are expected to grow in 2026. This means the company thinks more people will take the night bus next year. But the changes in sales show that the company should keep watching this time and look for ways to make night services better. For example, they could check if there are safety issues, not enough buses or if people feel it is not comfortable or easy to take the bus at night.

In Kuala Terengganu, the trend was less stable. Morning sales dropped in 2024 but came back up in 2025. This means fewer people took the bus in the morning in 2024, but then more people used it again in 2025. They ended up close to the 2023 level. So even though sales dropped and rose again, they are still not much better than they were in 2023. Afternoon sales kept going down from 2023 to 2025. This may mean that fewer people want to travel at that time. It could also mean there are problems with the bus schedule or the service itself. For example, maybe buses are late, full or there

are not enough buses running in the afternoon. Night sales also fell over the three years. This means that every year, fewer people are taking the bus at night. This suggests that fewer people are taking the bus at night. This could be because they find other ways to travel or because they do not feel safe or comfortable using night buses. A 5% increase is still expected in 2026. The company still hopes that sales will go up a little next year. But the fall in Afternoon and Night sales shows that changes may be needed to reach this goal. The company might have to fix the problems and find new ways to bring back passengers.

In summary, Johor Bahru is growing well, especially in the morning and afternoon. The company should keep investing in these times to keep the growth going. This means they should continue to add more buses, improve the service and make it easier for people to travel during the day. Night services still make a lot of money, but they need attention to avoid future problems. This means the company should not ignore night buses, even if they earn the most now, because the ups and downs in sales could become worse. In Kuala Terengganu, the company should work on fixing the drop in afternoon and night sales. They need to look closely at what is wrong and find ways to bring people back. It may need to improve the service, ask for customer feedback or offer special deals. For example, they could ask people why they stopped using the bus, make the buses come on time or offer cheaper tickets. These actions could help sales go up and make the 5% growth in 2026 possible. If the company works on these ideas, it will be more likely that they reach their goal next year.

3.2.2 Ticket Group Sales Analysis by Year: Weekend vs Weekday Trends (2023–2025)

SQL Code:

```
SET PAGESIZE 35
SET LINESIZE 200

COLUMN YEAR FORMAT 9999
COLUMN TICKET_GROUP FORMAT A18
COLUMN WEEKEND_SALES FORMAT 999,999,999.99
COLUMN WEEKDAY_SALES FORMAT 999,999,999.99
COLUMN DIFF_SALES FORMAT 999,999,999.99
COLUMN WEEKEND_QUANTITY FORMAT 999,999
COLUMN WEEKDAY_QUANTITY FORMAT 999,999
COLUMN WEEKEND_DIFF_PREV_YEAR FORMAT 999,999,999.99
COLUMN WEEKDAY_DIFF_PREV_YEAR FORMAT 999,999,999.99

TTITLE LEFT '                                Ticket Group Sales
Analysis by Year: Weekend vs Weekday Trends (2023-2025)' SKIP 2

BREAK ON TICKET_GROUP SKIP 1
COMPUTE SUM LABEL 'Subtotal' OF weekend_sales weekday_sales ON TICKET_GROUP

WITH sales_agg AS (
  SELECT
    CASE
      WHEN sf.ticketType IN ('CHILD_SINGLE', 'ADULT_SINGLE') THEN 'Single Tickets'
      WHEN sf.ticketType IN ('CHILD_RETURN', 'ADULT_RETURN') THEN 'Return Tickets'
      WHEN sf.ticketType IN ('WEEKLY_PASS', 'MONTHLY_PASS') THEN 'Periodic Passes'
      WHEN sf.ticketType IN ('SENIOR_PASS', 'STUDENT_PASS', 'NIGHT_PASS',
'EXPRESS_PASS') THEN 'Special Passes'
      ELSE 'Other'
    END AS ticket_group,
    d.year,
    SUM(CASE WHEN d.weekend_flag = 'Y' THEN sf.total_sales ELSE 0 END) AS
weekend_sales,
    SUM(CASE WHEN d.weekend_flag = 'N' THEN sf.total_sales ELSE 0 END) AS
weekday_sales,
    COUNT(CASE WHEN d.weekend_flag = 'Y' THEN 1 END) AS weekend_quantity,
    COUNT(CASE WHEN d.weekend_flag = 'N' THEN 1 END) AS weekday_quantity
  FROM Sales_Fact sf
  JOIN Dim_Date d ON sf.date_key = d.date_key
  WHERE d.year BETWEEN 2023 AND 2025
  GROUP BY
    CASE
      WHEN sf.ticketType IN ('CHILD_SINGLE', 'ADULT_SINGLE') THEN 'Single Tickets'
      WHEN sf.ticketType IN ('CHILD_RETURN', 'ADULT_RETURN') THEN 'Return Tickets'
      WHEN sf.ticketType IN ('WEEKLY_PASS', 'MONTHLY_PASS') THEN 'Periodic Passes'
      WHEN sf.ticketType IN ('SENIOR_PASS', 'STUDENT_PASS', 'NIGHT_PASS',
'EXPRESS_PASS') THEN 'Special Passes'
      ELSE 'Other'
```

```

        END,
        d.year
    ),
    sales_with_diff AS (
        SELECT
            ticket_group,
            year,
            weekend_sales,
            weekday_sales,
            weekend_sales - weekday_sales AS diff_sales,
            weekend_quantity,
            weekday_quantity,
            weekend_sales - LAG(weekend_sales) OVER (PARTITION BY ticket_group ORDER BY
year) AS weekend_diff_prev_year,
            weekday_sales - LAG(weekday_sales) OVER (PARTITION BY ticket_group ORDER BY
year) AS weekday_diff_prev_year
        FROM sales_agg
    )
    SELECT
        ticket_group,
        year,
        NULLIF(weekend_sales, 0) AS weekend_sales,
        NULLIF(weekday_sales, 0) AS weekday_sales,
        NULLIF(diff_sales, 0) AS diff_sales,
        NULLIF(weekend_quantity, 0) AS weekend_quantity,
        NULLIF(weekday_quantity, 0) AS weekday_quantity,
        NULLIF(weekend_diff_prev_year, 0) AS weekend_diff_prev_year,
        NULLIF(weekday_diff_prev_year, 0) AS weekday_diff_prev_year
    FROM sales_with_diff
    ORDER BY ticket_group, year;

CLEAR BREAKS
CLEAR COLUMNS
TTITLE OFF
CLEAR COMPUTE

```

Output:

Ticket Group Sales Analysis by Year: Weekend vs Weekday Trends (2023-2025)								
TICKET_GROUP	YEAR	WEEKEND_SALES	WEEKDAY_SALES	DIFF_SALES	WEEKEND_QUANTITY	WEEKDAY_QUANTITY	WEEKEND_DIFF_PREV_YEAR	WEEKDAY_DIFF_PREV_YEAR
Periodic Passes	2023	645,019.43	1,636,209.08	-991,189.65	1,491	3,773		
	2024	651,314.72	1,688,004.51	-1,036,689.79	1,492	3,859	6,295.29	51,795.43
	2025	705,708.91	1,686,322.01	-980,613.10	1,628	3,834	54,394.19	-1,682.50

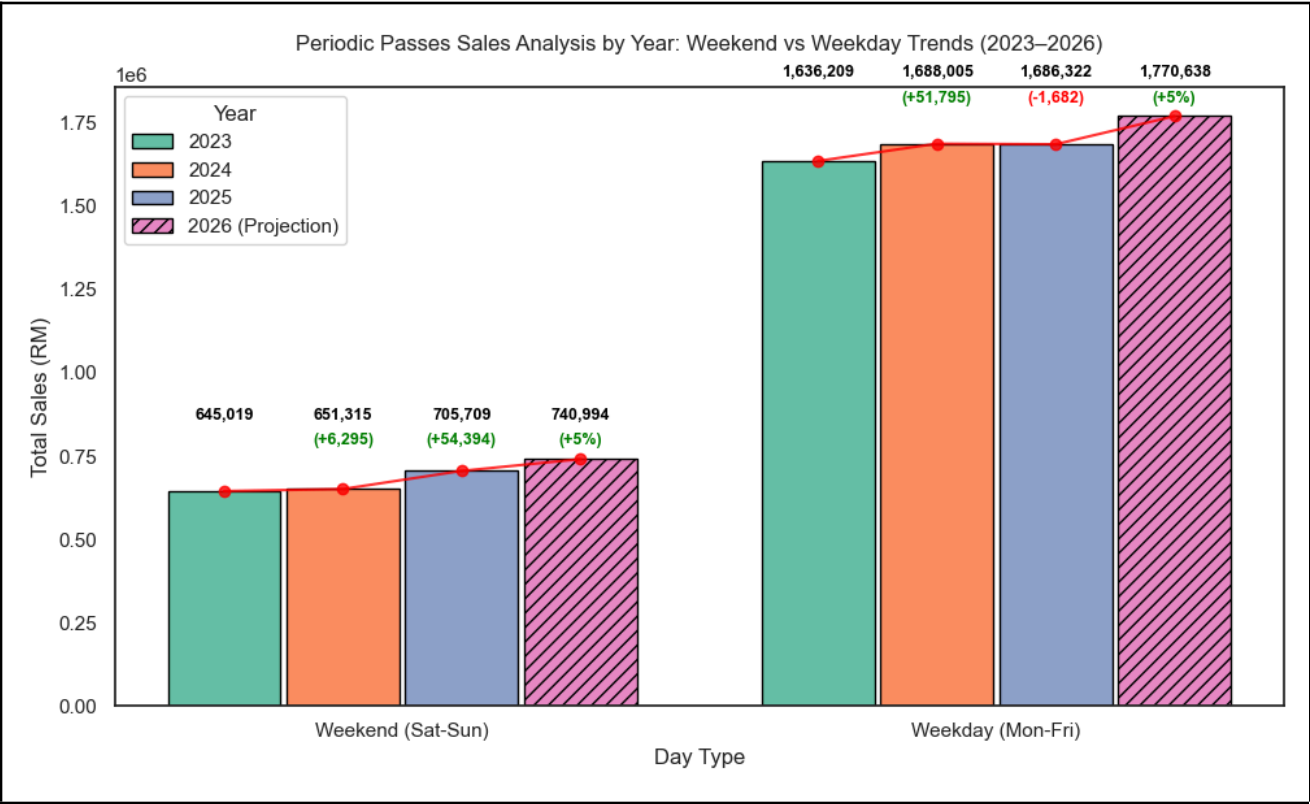
Subtotal		2,002,043.06	5,010,535.60					
Return Tickets	2023	650,566.06	1,627,481.73	-976,915.67	1,515	3,782		
	2024	645,762.70	1,620,017.91	-974,255.21	1,481	3,802	-4,803.36	-7,463.82
	2025	662,945.32	1,565,882.58	-902,937.26	1,531	3,614	17,182.62	-54,135.33

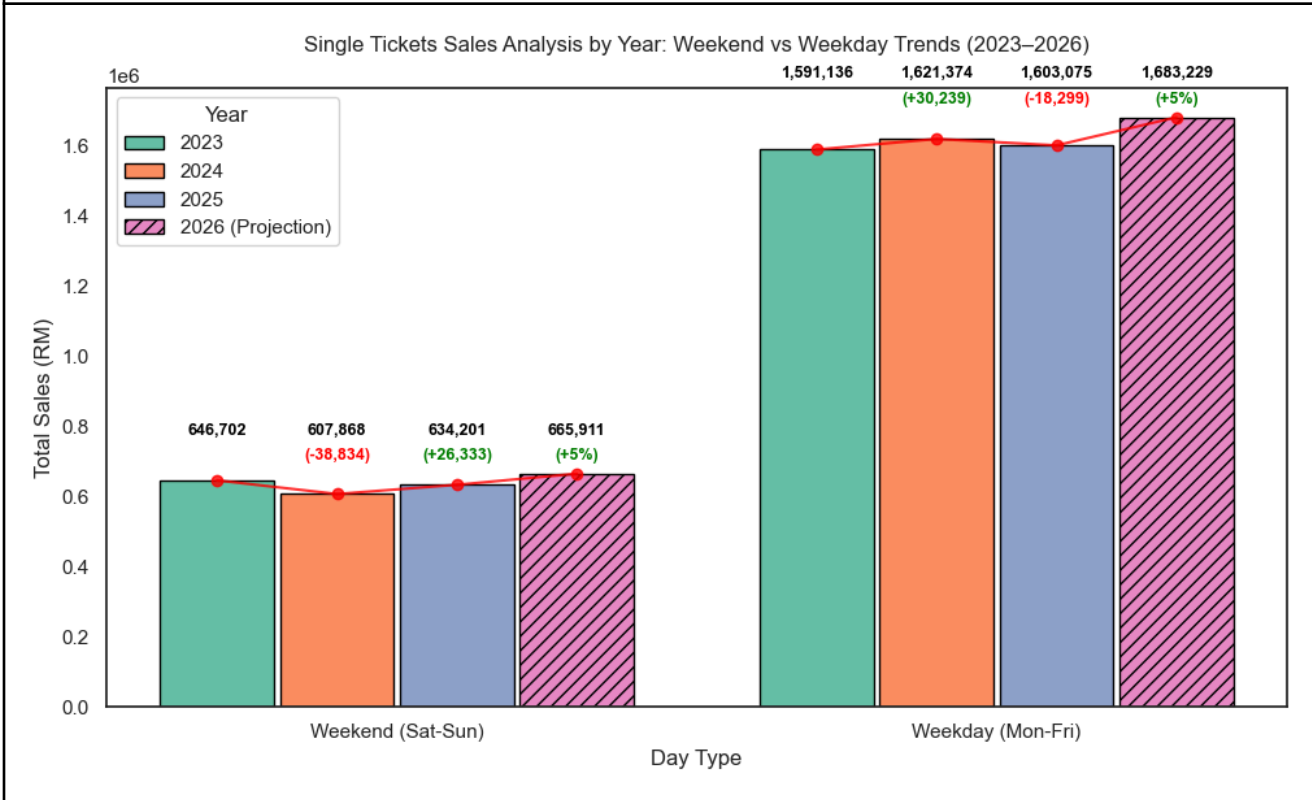
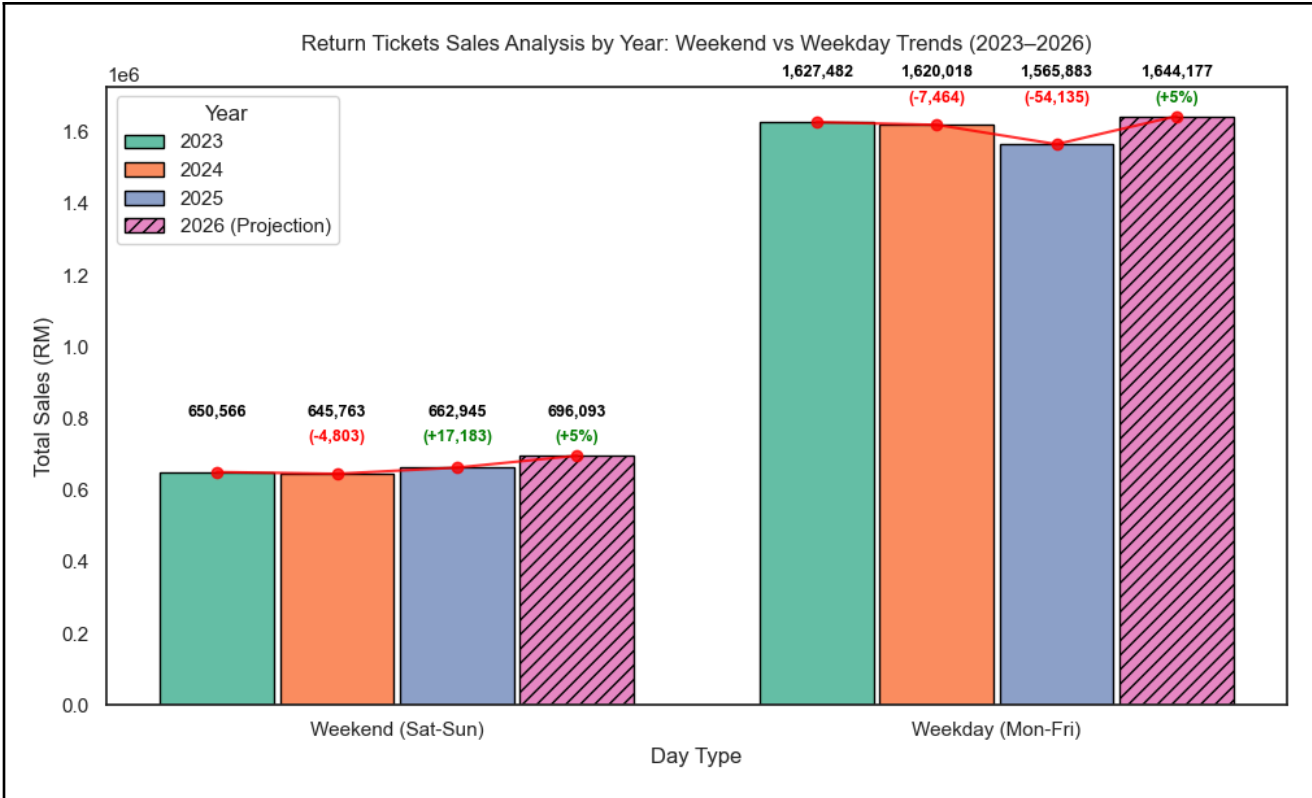
Subtotal		1,959,274.08	4,813,382.22					
Single Tickets	2023	646,701.81	1,591,135.83	-944,434.02	1,426	3,730		
	2024	607,868.19	1,621,374.34	-1,013,506.15	1,415	3,822	-38,833.62	30,238.51
	2025	634,200.77	1,603,075.18	-968,874.41	1,463	3,695	26,332.58	-18,299.16

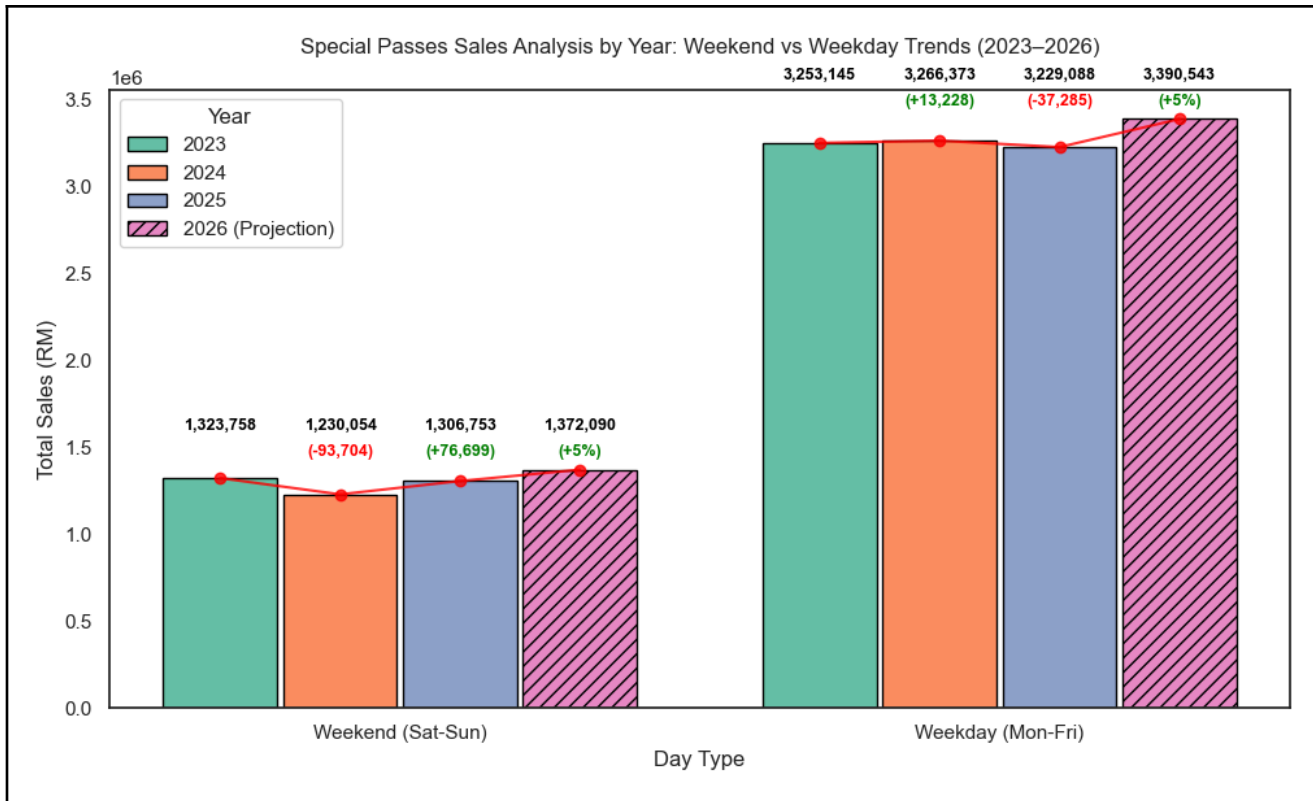
Subtotal		1,888,770.77	4,815,585.35					
Special Passes	2023	1,323,758.10	3,253,144.63	-1,929,386.53	3,050	7,534		
	2024	1,230,053.70	3,266,373.00	-2,036,319.30	2,901	7,504	-93,704.40	13,228.37
	2025	1,306,752.72	3,229,088.46	-1,922,335.74	2,992	7,472	76,699.02	-37,284.54

Subtotal		3,860,564.52	9,748,606.09					

Visualization:







Analysis:

From 2023 to 2025, ticket sales for all types showed clear patterns. Sales on weekdays were always higher than on weekends for every ticket type. Even though there were small changes each year, most ticket types either stayed steady or grew, especially on weekdays. Based on how things went in 2025, total sales are expected to grow by 5% in 2026. This is shown clearly in the chart with hatched bars.

For Periodic Passes, both weekend and weekday sales went up steadily over the three years. Weekday sales grew from about RM1.64 million in 2023 to RM1.69 million in 2025 and weekend sales went up from RM645,000 to RM706,000. This steady rise shows more people are using Periodic Passes during the week, maybe because they trust the service more or find good value in the passes. The 2026 forecast expects a 5% increase, with weekday sales reaching RM1.77 million and weekend sales about RM741,000, showing this product is doing well.

Return Tickets showed mixed results. Weekend sales stayed about the same, between RM650,000 and RM663,000 over the years, but weekday sales went up in 2024 to RM1.62 million

before dropping to RM1.57 million in 2025. This drop might be due to short-term reasons like route changes or seasonal travel. Still, the 2026 forecast expects weekday sales to go back up to RM1.64 million, showing hope for this ticket type.

Single Tickets saw a drop in weekend sales in 2024 but went back up in 2025, from RM647,000 in 2023 to RM634,000 in 2025. Weekday sales stayed steady, between RM1.59 million and RM1.62 million. This shows daily use is reliable, likely from people who travel occasionally or only once in a while. The 2026 forecast shows both weekday and weekend sales going up, with weekday sales expected to reach RM1.68 million and weekend sales RM666,000.

Special Passes, which are for groups like students and seniors, showed the most ups and downs. Weekend sales dropped a lot from RM1.32 million in 2023 to RM1.23 million in 2024 but went back up to RM1.31 million in 2025. Weekday sales also fell a little in 2025 after doing well in the past years. Despite these changes, the 2026 forecast shows a recovery, with weekday sales expected to be RM3.39 million and weekend sales RM1.37 million. This suggests the sales will grow again, possibly because of special policies or marketing campaigns.

In summary, the chart shows a strong increase in weekday ticket sales, especially for Periodic and Single Tickets. These types should be the main focus for operations and marketing. Weekend sales are lower for all ticket types and could be improved with better services or promotions. Return Tickets and Special Passes need extra attention where sales dropped, to help make sure the 2026 growth goals are met and kept steady.

3.2.3 2025 Quarterly Sales Report for State Over RM 1.0 Million

SQL Code:

```
SET LINESIZE 130
SET PAGESIZE 35

BREAK ON STATE SKIP 1 ON REPORT
COMPUTE SUM LABEL 'Total Sales' OF TOTAL_SALES ON STATE

TTITLE LEFT '                2025 Quarterly Sales Report for State Over RM 1.0
Million'

COLUMN STATE FORMAT A15
COLUMN QUARTER FORMAT A8
COLUMN TOTAL_TICKETS FORMAT 999,999
COLUMN AVG_TICKET_PRICE FORMAT 999,999.99
COLUMN TOTAL_SALES FORMAT 999,999,999.99 HEADING 'Total Sales'
COLUMN QOQ_DIFF FORMAT A15

WITH Quarterly_Sales_Raw AS (
    SELECT
        p.state AS STATE,
        d.quarter AS QUARTER,
        COUNT(*) AS TOTAL_TICKETS,
        SUM(sf.total_sales) AS TOTAL_SALES,
        CASE WHEN COUNT(sf.bookingID) = 0 THEN 0
             ELSE SUM(sf.total_sales) / COUNT(sf.bookingID)
        END AS AVG_TICKET_PRICE
    FROM Sales_Fact sf
    JOIN Dim_Passenger p ON sf.passenger_key = p.passenger_key
    JOIN Dim_Date d ON sf.date_key = d.date_key
    WHERE d.year = 2025
    GROUP BY p.state, d.quarter
),
State_Total_Sales AS (
    SELECT
        STATE,
        SUM(TOTAL_SALES) AS STATE_TOTAL_SALES
    FROM Quarterly_Sales_Raw
    GROUP BY STATE
),
Sales_With_Diff AS (
    SELECT
        q.STATE,
        q.QUARTER,
        q.TOTAL_TICKETS,
        q.AVG_TICKET_PRICE,
        q.TOTAL_SALES,
        q.TOTAL_SALES - LAG(q.TOTAL_SALES) OVER (
            PARTITION BY q.STATE
```

```

        ORDER BY CASE q.QUARTER
                WHEN 'Q1' THEN 1
                WHEN 'Q2' THEN 2
                WHEN 'Q3' THEN 3
                WHEN 'Q4' THEN 4
        END
    ) AS QOQ_DIFF
FROM Quarterly_Sales_Raw q
JOIN State_Total_Sales s ON q.STATE = s.STATE
WHERE s.STATE_TOTAL_SALES > 1000000
)
SELECT
    STATE,
    QUARTER,
    TOTAL_TICKETS,
    AVG_TICKET_PRICE,
    TOTAL_SALES,
    CASE
        WHEN QUARTER = 'Q1' THEN ''
        ELSE TO_CHAR(NVL(QOQ_DIFF, 0), '999,999.99')
    END AS QOQ_DIFF
FROM Sales_With_Diff
ORDER BY STATE,
        CASE QUARTER
            WHEN 'Q1' THEN 1
            WHEN 'Q2' THEN 2
            WHEN 'Q3' THEN 3
            WHEN 'Q4' THEN 4
        END;

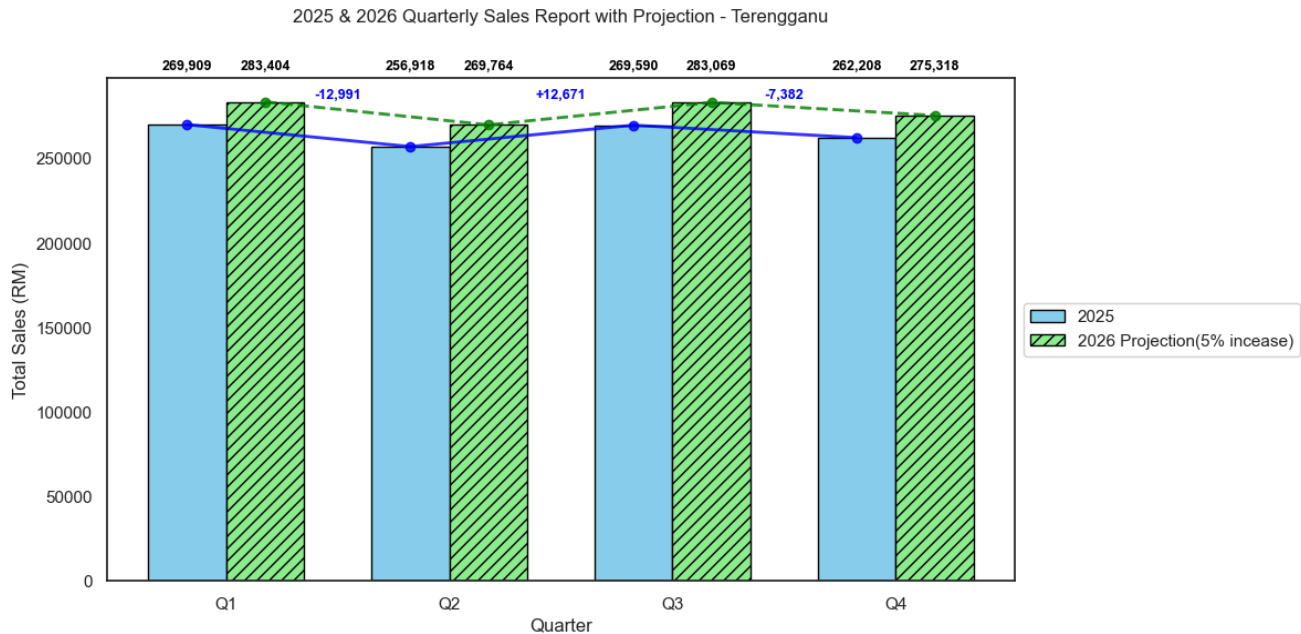
CLEAR BREAKS
CLEAR COLUMNS
TTITLE OFF
CLEAR COMPUTES

```

Output:

2025 Quarterly Sales Report for State Over RM 1.0 Million					
STATE	QUARTER	TOTAL_TICKETS	AVG_TICKET_PRICE	Total Sales	QOQ_DIFF
TERENGGANU	Q1	605	446.13	269,908.87	
	Q2	576	446.04	256,918.28	-12,990.59
	Q3	605	445.60	269,589.62	12,671.34
	Q4	611	429.15	262,207.62	-7,382.00
*****				-----	
Total Sales				1,058,624.39	

Visualization:



Analysis:

In 2025, Terengganu's ticket sales showed some ups and downs but mostly stayed steady throughout the year. The total sales were about RM1.06 million, which is quite good compared to other states. At the start of the year, sales were RM269,909 in the first quarter (Q1). Then, sales dropped a little to RM256,918 in the second quarter (Q2). After that, sales went back up to RM269,590 in the third quarter (Q3), before falling slightly again to RM262,208 in the fourth quarter (Q4).

The biggest drop in sales happened between Q1 and Q2, with a loss of almost RM13,000. However, this loss was mostly recovered in Q3 when sales increased by about RM12,600. Sales dropped again a bit in Q4, but not as much. These changes show that people's travel habits change during the year, with some quarters having more travelers than others. The prices of tickets stayed about the same throughout the year, so the sales changes were mostly because of how many people bought tickets, not because of price changes.

Looking forward to 2026, sales are expected to grow by about 5% each quarter. This means that in Q1, sales could reach RM283,404, then drop slightly in Q2 to RM269,764, rise again in Q3 to RM283,070 and fall a bit in Q4 to RM275,318. These numbers show slow but steady growth based on last year's steady sales.

The lower sales in Q2 and Q4 show a chance to do better. The company could try special promotions, marketing or change travel schedules to attract more customers during those slower times. On the other hand, the stronger sales in Q3 suggest that the efforts made in that quarter worked well. These could be copied in other quarters to improve sales.

To handle the expected 5% growth, the company will need to plan carefully. This means making sure there are enough vehicles, staff and good service to meet higher demand. Watching how sales change, listening to customer feedback and checking how full trips are will help make smart decisions to improve sales.

In conclusion, Terengganu's ticket sales stayed strong in 2025 despite small drops during the year. The outlook for 2026 looks good with steady growth expected. By focusing on improving sales during slower quarters and continuing successful efforts from busier times, Terengganu is well positioned to meet its sales goals and help the region grow.

3.3 Kit Chin Jie Ying

3.3.1: Quarterly Sales Trend Analysis by Bus Type (2024-2026)

SQL Code:

```
SET DEFINE OFF
SET LINESIZE 150
SET PAGESIZE 35
SET SERVEROUTPUT ON

COLUMN busType FORMAT A15
COLUMN metric FORMAT A15
COLUMN "Q1" FORMAT 9,999,999.99
COLUMN "Q2" FORMAT 9,999,999.99
COLUMN "Q3" FORMAT 9,999,999.99
COLUMN "Q4" FORMAT 9,999,999.99
COLUMN "TOTAL" FORMAT 9,999,999.99

TTITLE COL 25 'Report:' -
        COL 35 'Quarterly Sales Trend Analysis (2023-2025)' -
        SKIP 1 ' ' -
        LEFT 'Page:' SQL.PNO

BREAK ON year SKIP PAGE ON busType SKIP 1 ON metric

WITH monthly_sales AS (
    SELECT d.year,
           d.month,
           b.busType,
           SUM(sf.ticket_qty) AS ticket_qty,
           SUM(sf.total_sales - sf.discount_amount) AS sales_revenue
    FROM Sales_Fact sf
    JOIN Dim_Date d ON sf.date_key = d.date_key
    JOIN Dim_Bus b ON sf.bus_key = b.bus_key
    WHERE d.year BETWEEN 2023 AND 2025
    GROUP BY d.year, d.month, b.busType
),
quarterly_base AS (
    SELECT year,
           CEIL(month/3) AS quarter,
           busType,
           SUM(ticket_qty) AS ticket_qty,
           SUM(sales_revenue) AS sales_revenue,
           SUM(sales_revenue)
             - LAG(SUM(sales_revenue)) OVER (
                 PARTITION BY busType, CEIL(month/3) ORDER BY year
             ) AS yoy_diff
    FROM monthly_sales
    GROUP BY year, CEIL(month/3), busType
```

```

),
yearly_sales AS (
    SELECT year, busType,
           SUM(sales_revenue) AS total_sales,
           SUM(ticket_qty) AS total_qty
    FROM quarterly_base
    GROUP BY year, busType
),
yearly_yoy AS (
    SELECT year,
           busType,
           total_sales - LAG(total_sales) OVER (PARTITION BY busType ORDER BY year)
AS yoy_value,
           CASE WHEN LAG(total_sales) OVER (PARTITION BY busType ORDER BY year) = 0
                THEN NULL
                ELSE ROUND(
                    (total_sales - LAG(total_sales) OVER (PARTITION BY busType
ORDER BY year))
                    / LAG(total_sales) OVER (PARTITION BY busType ORDER BY year) *
100, 2
                )
           END AS yoy_pct
    FROM yearly_sales
),
unioned AS (
    SELECT year, quarter, busType, sales_revenue AS value, 'SALES' AS metric FROM
quarterly_base
    UNION ALL
    SELECT year, 0, busType, SUM(sales_revenue), 'SALES'
    FROM quarterly_base GROUP BY year, busType
    UNION ALL
    SELECT year, quarter, busType, ticket_qty, 'QTY' FROM quarterly_base
    UNION ALL
    SELECT year, 0, busType, SUM(ticket_qty), 'QTY'
    FROM quarterly_base GROUP BY year, busType
    UNION ALL
    SELECT year, quarter, busType,
           CASE WHEN ticket_qty=0 THEN NULL ELSE sales_revenue/ticket_qty END,
           'AVG_PRICE' FROM quarterly_base
    UNION ALL
    SELECT year, 0, busType,
           CASE WHEN SUM(ticket_qty)=0 THEN NULL ELSE
SUM(sales_revenue)/SUM(ticket_qty) END,
           'AVG_PRICE' FROM quarterly_base GROUP BY year, busType
    UNION ALL
    SELECT year, quarter, busType, yoy_diff, 'YOY_DIFF' FROM quarterly_base
    UNION ALL
    SELECT year, 0, busType, SUM(yoy_diff), 'YOY_DIFF'
    FROM quarterly_base GROUP BY year, busType

```

```

UNION ALL
SELECT year, 0, busType, yoy_pct, 'YOY_PCT' FROM yearly_yoy
)
SELECT year, busType, metric,
        "Q1","Q2","Q3","Q4","TOTAL"
FROM unioned
PIVOT (
        SUM(value)
        FOR quarter IN (
                1 AS "Q1", 2 AS "Q2", 3 AS "Q3", 4 AS "Q4", 0 AS "TOTAL"
        )
)
ORDER BY year, busType, metric;

```

Output:

Page: 1 Report: Quarterly Sales Trend Analysis (2023-2025)

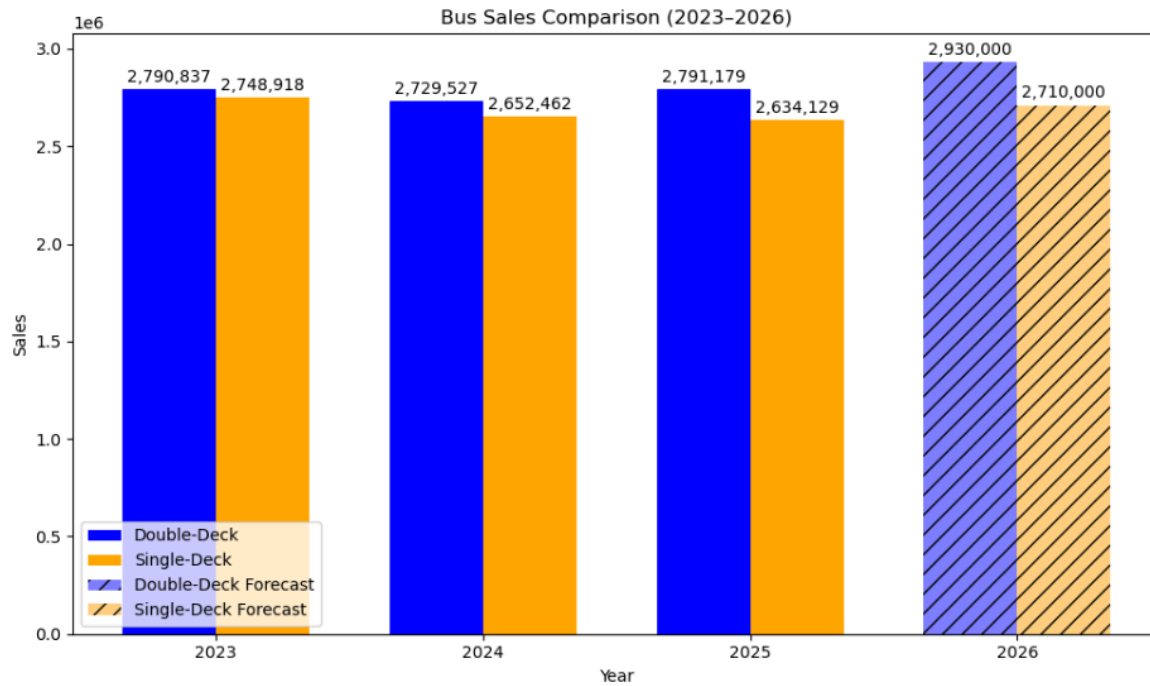
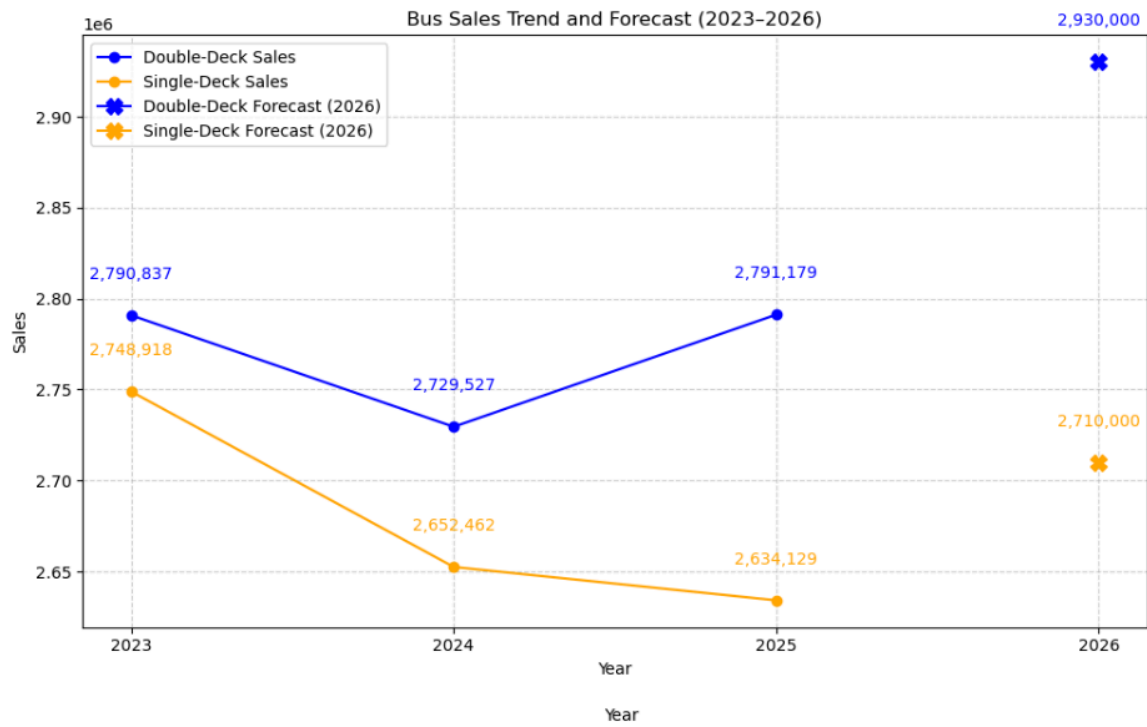
Year	BUSTYPE	Metric	Q1	Q2	Q3	Q4	TOTAL
2023	DOUBLE-DECK	AVG_PRICE	125.56	126.81	124.37	126.52	125.83
		QTY	5,451.00	5,625.00	5,437.00	5,666.00	22,179.00
		SALES	684,438.46	713,328.38	676,196.56	716,873.73	2,790,837.13
		YOY_DIFF					
		YOY_PCT					
	SINGLE-DECK	AVG_PRICE	126.16	124.59	126.37	124.94	125.52
		QTY	5,405.00	5,473.00	5,570.00	5,453.00	21,901.00
		SALES	681,887.92	681,871.51	703,881.34	681,277.50	2,748,918.27
		YOY_DIFF					
		YOY_PCT					

Page: 2 Report: Quarterly Sales Trend Analysis (2023-2025)

Year	BUSTYPE	Metric	Q1	Q2	Q3	Q4	TOTAL
2024	DOUBLE-DECK	AVG_PRICE	124.52	125.27	124.09	125.28	124.79
		QTY	5,432.00	5,493.00	5,469.00	5,479.00	21,873.00
		SALES	676,376.29	688,096.09	678,666.57	686,388.49	2,729,527.44
		YOY_DIFF	-8,062.17	-25,232.29	2,470.01	-30,485.24	-61,309.69
		YOY_PCT					-2.20
	SINGLE-DECK	AVG_PRICE	125.06	125.07	125.57	124.57	125.07
		QTY	5,212.00	5,425.00	5,346.00	5,225.00	21,208.00
		SALES	651,811.26	678,478.33	671,317.38	650,854.60	2,652,461.57
		YOY_DIFF	-30,076.66	-3,393.18	-32,563.96	-30,422.90	-96,456.70
		YOY_PCT					-3.51

Page: 3 Report: Quarterly Sales Trend Analysis (2023-2025)

Year	BUSTYPE	Metric	Q1	Q2	Q3	Q4	TOTAL
2025	DOUBLE-DECK	AVG_PRICE	127.01	125.18	124.44	125.69	125.57
		QTY	5,411.00	5,547.00	5,583.00	5,687.00	22,228.00
		SALES	687,235.70	694,362.71	694,773.41	714,807.13	2,791,178.95
		YOY_DIFF	10,859.41	6,266.62	16,106.84	28,418.64	61,651.51
		YOY_PCT					2.26
	SINGLE-DECK	AVG_PRICE	125.00	125.23	125.81	124.79	125.21
		QTY	5,282.00	5,079.00	5,323.00	5,354.00	21,038.00
		SALES	660,254.48	636,029.96	669,702.88	668,141.35	2,634,128.67
		YOY_DIFF	8,443.22	-42,448.37	-1,614.50	17,286.75	-18,332.90
		YOY_PCT					

Visualization:

Analysis:

From 2023 to 2025, the company's bus sales showed two very different trends between the Double-Deck and Single-Deck services. For Double-Deck buses, the quarterly data shows that in 2023 sales were RM 684,438 in Q1, RM 713,328 in Q2, RM 676,197 in Q3 and RM 716,874 in Q4, giving a total of about RM 2.79 million. Sales were quite balanced throughout the year, with Q2 and Q4 slightly stronger, showing that demand tends to peak in the middle and end of the year. In 2024, quarterly sales dropped slightly to RM 676,376 in Q1, RM 688,096 in Q2, RM 678,667 in Q3 and RM 686,388 in Q4, giving a total of RM 2.73 million, about -2.2% lower than 2023. This small decline was seen across all quarters, indicating a slight softening in demand. In 2025, the Double-Deck service rebounded, with Q1 sales at RM 687,236, Q2 at RM 694,363, Q3 at RM 694,773 and Q4 at RM 714,807, for a total of RM 2.79 million, almost the same as 2023 and about +2.26% higher than 2024. The strongest improvement came in Q3 and Q4, suggesting a healthy recovery in the second half of the year.

Overall, Double-Deck showed a pattern of slight decline followed by recovery, proving to be relatively stable and resilient. Based on the three-year trend, the calculated average annual growth rate is about +0.03% and after applying an additional 5% growth target, the 2026 forecast is expected to reach around RM 2.93 million, slightly higher than 2025. The projected quarterly results for 2026 are approximately RM 722,388 in Q1, RM 739,225 in Q2, RM 722,947 in Q3 and RM 747,085 in Q4, which shows growth in every quarter and suggests that Q4 will continue to be the seasonal peak.

This data tells us that Double-Deck services have a strong and loyal customer base who returned after the slight drop in 2024. The recovery in 2025 is significant because it shows that the decline was only temporary, indicating that customers responded positively to service quality, marketing efforts or external factors such as improved travel demand. The likely causes of this

recovery could be better scheduling, improved customer experience or effective promotions that helped to bring passengers back. Moving forward, the company should capitalize on this momentum by continuing to monitor customer satisfaction closely, invest in keeping buses well-maintained and comfortable and perhaps explore modest service expansions to capture more market share.

For Single-Deck buses, the quarterly data shows that in 2023 sales were RM 681,888 in Q1, RM 681,872 in Q2, RM 703,881 in Q3 and RM 681,278 in Q4, for a total of about RM 2.75 million. In 2024, quarterly sales fell to RM 651,811 in Q1, RM 678,478 in Q2, RM 671,317 in Q3 and RM 650,855 in Q4, totaling RM 2.65 million, a fall of around -3.5%. The largest drops occurred in Q1 and Q4, indicating weaker demand during those periods. In 2025, Q1 sales were RM 660,254, Q2 dropped further to RM 636,030, Q3 slightly improved to RM 669,703 and Q4 was RM 668,141, totaling RM 2.63 million, another small fall of about -0.7%.

Overall, from 2023 to 2025 Single-Deck lost about RM 114,789, which is a total decline of roughly -4.2% over two years. Using the two-year average decline rate of -2.10% and then adding a 5% adjustment to account for possible recovery initiatives, the 2026 forecast is projected to reach around RM 2.71 million. The projected quarterly breakdown is RM 671,909 in Q1, RM 672,727 in Q2, RM 689,077 in Q3 and RM 674,039 in Q4. While this slightly improves on a purely declining trend, it remains below 2023 levels, which is a clear signal that changes must be made urgently to reverse the downward trend.

The data is telling us that Single-Deck is steadily losing market share and is underperforming in almost every quarter. The decrease is significant because it is not a one-time dip but a consistent pattern over two years. The likely causes could be an outdated fleet, lower comfort compared to Double-Deck or ticket pricing that does not match what customers are willing to pay. There could

also be competition from other transportation options that offer better convenience or value. Moving forward, urgent actions should include upgrading Single-Deck buses to improve comfort, reviewing ticket prices, improving on-time performance and launching loyalty programs or bundle deals to attract passengers back. Gathering customer feedback through surveys or focus groups will be key to identifying the exact pain points so that targeted improvements can be made.

Looking at these two opposite trends, the company must take proactive steps. For Double-Deck, since sales are recovering and may continue to grow, the company should keep ticket prices stable, maintain strong service quality in terms of comfort, timing and reliability and strengthen marketing and promotions to build customer trust and ensure consistent growth. For Single-Deck, because sales are declining, urgent changes are needed. The company should review ticket prices to better fit customer demand, improve comfort, punctuality and reliability and introduce loyalty rewards or bundle deals to attract more passengers. It is also important to gather customer feedback to understand why more people prefer Double-Deck buses and then apply those insights to improve Single-Deck services.

In summary, Double-Deck is on a stable and positive path and is forecasted to grow to around RM 2.93 million in 2026, while Single-Deck is expected to remain below 2023 levels at around RM 2.71 million unless major improvements are made. If the company manages these two services carefully, overall bus sales can not only be stabilized but also grow stronger in the coming years.

3.3.2: Top Destination Quarterly Revenue (2023-2025)

SQL Code:

```
SET LINESIZE 180;
SET PAGESIZE 50;
SET SERVEROUTPUT ON;

COLUMN quarter FORMAT A7
COLUMN destination FORMAT A12
COLUMN time_of_day FORMAT A10
COLUMN ticket_group FORMAT A15
COLUMN "2023" FORMAT 9,999,999.99
COLUMN "2024" FORMAT 9,999,999.99
COLUMN "2025" FORMAT 9,999,999.99
COLUMN total_sales FORMAT 9,999,999.99
COLUMN yoy_2024_pct FORMAT 999.99
COLUMN yoy_2025_pct FORMAT 999.99

TTITLE COL 35 'Report:' -
        COL 45 'Top Destination Quarterly Revenue (2023-2025)' -
        SKIP 1 ' ' -
        LEFT 'Page:' SQL.PNO -
        SKIP 1 ' ';

BREAK ON year SKIP 1 ON quarter SKIP 1 ON destination SKIP 1 ON time_of_day SKIP 1

WITH time_category AS (
    SELECT t.trip_key,
           t.destination,
           CASE
               WHEN TO_NUMBER(TO_CHAR(t.departure_time,'HH24')) BETWEEN 6 AND 11
            THEN 'Morning'
               WHEN TO_NUMBER(TO_CHAR(t.departure_time,'HH24')) BETWEEN 12 AND 17
            THEN 'Afternoon'
               ELSE 'Night'
           END AS time_of_day
    FROM Dim_Trip t
),
ticket_grouped AS (
    SELECT tickettype,
           CASE
               WHEN tickettype IN
('ADULT_RETURN','CHILD_RETURN','ADULT_SINGLE','CHILD_SINGLE') THEN 'SINGLE/RETURN'
               WHEN tickettype IN ('STUDENT_PASS','SENIOR_PASS') THEN 'CONCESSION'
               WHEN tickettype IN ('WEEKLY_PASS','MONTHLY_PASS') THEN 'SEASON/PASS'
               WHEN tickettype IN ('EXPRESS_PASS','NIGHT_PASS') THEN 'SPECIAL'
               ELSE 'OTHER'
           END AS ticket_group
    FROM (SELECT DISTINCT tickettype FROM Sales_Fact)
),
```

```

main_sales AS (
    SELECT
        d.year,
        d.quarter,
        tc.destination,
        tc.time_of_day,
        tg.ticket_group,
        SUM(sf.ticket_revenue - sf.discount_amount) AS sales
    FROM Sales_Fact sf
    JOIN time_category tc ON sf.trip_key = tc.trip_key
    JOIN Dim_Date d ON sf.date_key = d.date_key
    JOIN ticket_grouped tg ON sf.tickettype = tg.tickettype
    WHERE d.year BETWEEN 2023 AND 2025
    GROUP BY d.year, d.quarter, tc.destination, tc.time_of_day, tg.ticket_group
),
quarter_dest_total AS (
    SELECT quarter, destination, SUM(sales) AS quarter_sales
    FROM main_sales
    GROUP BY quarter, destination
    HAVING SUM(sales) > 450000
),
filtered_sales AS (
    SELECT ms.*
    FROM main_sales ms
    JOIN quarter_dest_total qd
        ON ms.quarter = qd.quarter
        AND ms.destination = qd.destination
),
top_destination AS (
    SELECT destination
    FROM (
        SELECT destination, SUM(sales) AS total_sales
        FROM filtered_sales
        GROUP BY destination
        ORDER BY SUM(sales) DESC
    )
    WHERE ROWNUM = 1
),
filtered_top_sales AS (
    SELECT fs.*
    FROM filtered_sales fs
    JOIN top_destination td
        ON fs.destination = td.destination
),
pivot_sales AS (
    SELECT
        quarter,
        destination,
        time_of_day,

```

```

        ticket_group,
        SUM(CASE WHEN year = 2023 THEN sales END) AS "2023",
        SUM(CASE WHEN year = 2024 THEN sales END) AS "2024",
        SUM(CASE WHEN year = 2025 THEN sales END) AS "2025",
        SUM(sales) AS total_sales
    FROM filtered_top_sales
    GROUP BY quarter, destination, time_of_day, ticket_group
),
quarter_totals AS (
    SELECT
        quarter,
        destination,
        NULL AS time_of_day,
        'TOTAL' AS ticket_group,
        SUM("2023") AS "2023",
        SUM("2024") AS "2024",
        SUM("2025") AS "2025",
        SUM(total_sales) AS total_sales
    FROM pivot_sales
    GROUP BY quarter, destination
),
year_totals AS (
    SELECT
        NULL AS quarter,
        destination,
        NULL AS time_of_day,
        'YEAR TOTAL' AS ticket_group,
        SUM("2023") AS "2023",
        SUM("2024") AS "2024",
        SUM("2025") AS "2025",
        (SUM("2023") + SUM("2024") + SUM("2025")) AS total_sales
    FROM pivot_sales
    GROUP BY destination
),
combined AS (
    SELECT * FROM pivot_sales
    UNION ALL
    SELECT * FROM quarter_totals
    UNION ALL
    SELECT * FROM year_totals
)
SELECT
    quarter,
    destination,
    time_of_day,
    ticket_group,
    "2023",
    "2024",
    "2025",

```

```

total_sales,
ROUND(( NVL("2024",0) - NVL("2023",0) ) / NULLIF(NVL("2023",0),0) * 100, 2) AS
yoy_2024_pct,
ROUND(( NVL("2025",0) - NVL("2024",0) ) / NULLIF(NVL("2024",0),0) * 100, 2) AS
yoy_2025_pct
FROM combined
ORDER BY quarter,
destination,
CASE WHEN time_of_day IS NULL THEN 2 ELSE 1 END,
time_of_day,
CASE
    WHEN ticket_group = 'SINGLE/RETURN' THEN 1
    WHEN ticket_group = 'CONCESSION' THEN 2
    WHEN ticket_group = 'SEASON/PASS' THEN 3
    WHEN ticket_group = 'SPECIAL' THEN 4
    WHEN ticket_group = 'TOTAL' THEN 5
    ELSE 6
END,
ticket_group;

```

Output:

Report: Top Destination Quarterly Revenue (2023-2025)

Page: 1

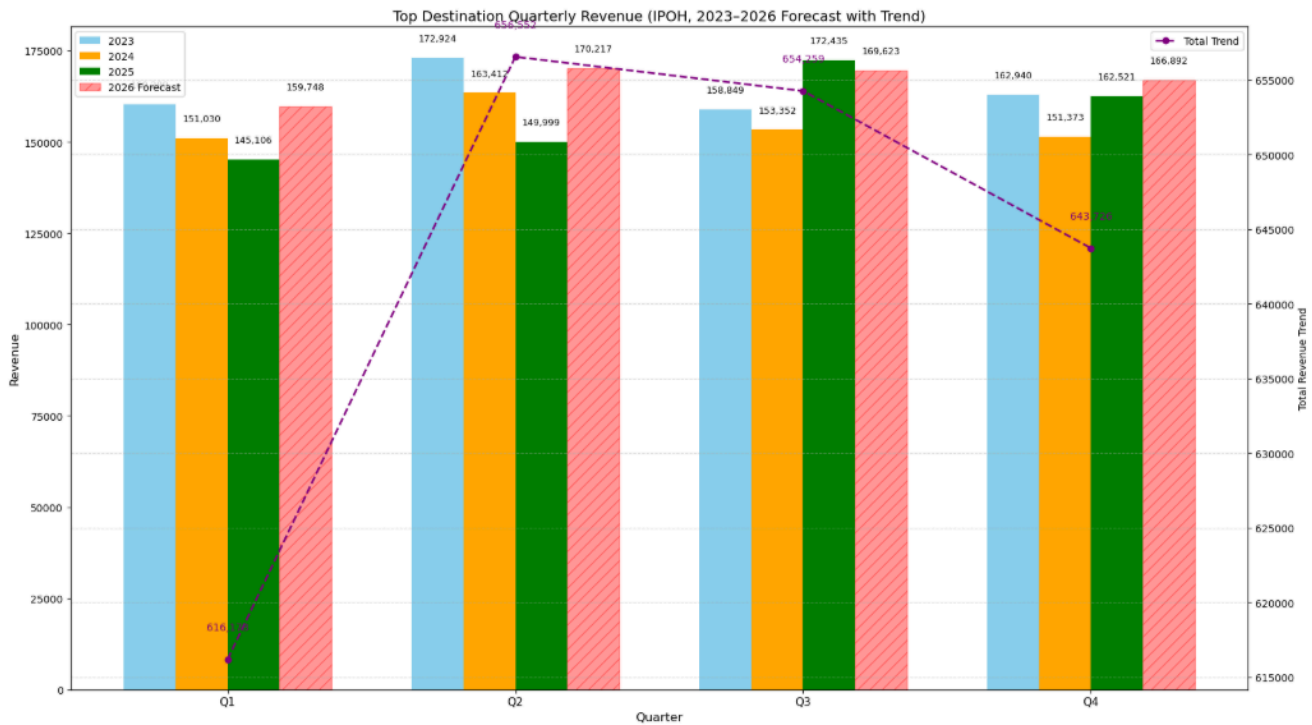
QUARTER	DESTINATION	Time	TICKET_GROUP	2023	2024	2025	TOTAL_SALES	YOY_2024_PCT	YOY_2025_PCT
Q1	IPOH	Afternoon	SINGLE/RETURN	19,346.32	13,247.12	15,958.31	48,551.75	-31.53	20.47
			CONCESSION	10,746.62	7,711.70	7,370.64	25,828.96	-28.24	-4.42
			SEASON/PASS	9,953.43	7,746.85	7,509.95	25,210.23	-22.17	-3.06
			SPECIAL	11,333.96	9,208.89	8,185.14	28,727.99	-18.75	-11.12
		Morning	SINGLE/RETURN	11,639.55	16,802.90	11,869.29	40,311.74	44.36	-29.36
			CONCESSION	6,685.26	6,027.99	5,039.71	17,752.96	-9.83	-16.39
			SEASON/PASS	8,712.88	7,340.58	8,907.47	24,960.93	-15.75	21.35
			SPECIAL	5,788.64	9,052.01	4,962.13	19,802.78	56.38	-45.18
		Night	SINGLE/RETURN	33,303.93	28,063.71	30,744.55	92,112.19	-15.73	9.55
			CONCESSION	12,142.58	12,547.77	13,480.04	38,170.39	3.34	7.43
			SEASON/PASS	12,805.46	16,945.04	15,817.10	45,567.60	32.33	-6.66
			SPECIAL	17,830.84	16,335.76	15,261.62	49,428.22	-8.38	-6.58
			TOTAL	160,289.47	151,030.32	145,105.95	456,425.74	-5.78	-3.92
	IPOH	Afternoon	SINGLE/RETURN	16,897.93	17,145.34	15,548.19	49,591.46	1.46	-9.32
			CONCESSION	11,165.69	9,963.06	6,976.51	28,105.26	-10.77	-29.98
			SEASON/PASS	8,575.38	7,088.76	10,871.12	26,535.26	-17.34	53.36
			SPECIAL	8,598.65	10,609.31	7,068.78	26,276.74	23.38	-33.37
		Morning	SINGLE/RETURN	15,384.29	11,819.72	15,144.12	42,348.13	-23.17	28.13
			CONCESSION	8,513.34	6,126.36	4,413.51	19,053.21	-28.04	-27.96
			SEASON/PASS	7,878.27	8,604.55	7,403.33	23,886.15	9.22	-13.96
			SPECIAL	6,642.90	7,604.78	7,346.13	21,593.81	14.48	-3.40
		Night	SINGLE/RETURN	34,201.99	35,146.44	33,655.00	103,003.43	2.76	-4.24
			CONCESSION	18,797.76	18,914.79	15,127.07	52,839.62	.62	-20.03
			SEASON/PASS	17,597.99	17,165.13	15,041.35	49,804.47	-2.46	-12.37
			SPECIAL	18,669.50	13,223.56	11,403.74	43,296.80	-29.17	-13.76

TOTAL	172,923.69	163,411.80	149,998.85	486,334.34	-5.50	-8.21
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Report: Top Destination Quarterly Revenue (2023-2025)

QUARTER	DESTINATION	Time	TICKET_GROUP	2023	2024	2025	TOTAL_SALES	YOY_2024_PCT	YOY_2025_PCT		
Q3	IPOH	Morning	CONCESSION	6,024.42	6,787.48	5,890.26	18,702.16	12.67	-13.22		
			SEASON/PASS	6,585.75	7,729.05	7,473.44	21,788.24	17.36	-3.31		
			SPECIAL	7,037.68	4,960.35	9,088.42	21,086.45	-29.52	83.22		
		Night	SINGLE/RETURN	29,403.99	28,315.34	29,254.20	86,973.53	-3.70	3.32		
			CONCESSION	13,157.24	15,552.49	22,043.12	50,752.85	18.20	41.73		
			SEASON/PASS	16,157.33	15,553.76	16,481.39	48,192.48	-3.74	5.96		
			SPECIAL	14,311.14	14,619.76	17,127.89	46,058.79	2.16	17.16		
		TOTAL	158,848.53	153,351.77	172,435.48	484,635.78	-3.46	12.44			
		Q4	IPOH	Afternoon	SINGLE/RETURN	18,825.06	19,361.00	15,236.30	53,422.36	2.85	-21.30
					CONCESSION	10,797.94	6,986.93	8,757.61	26,542.48	-35.29	25.34
SEASON/PASS	11,937.36				9,287.92	8,574.31	29,799.59	-22.19	-7.68		
SPECIAL	9,077.40				8,166.64	9,851.44	27,095.48	-10.03	20.63		
Morning	SINGLE/RETURN			13,761.36	11,858.73	14,087.68	39,707.77	-13.83	18.80		
	CONCESSION			5,382.51	6,240.55	7,792.38	19,415.44	15.94	24.87		
	SEASON/PASS			8,774.38	5,378.39	7,729.89	21,882.66	-38.70	43.72		
	SPECIAL			7,966.47	7,074.77	10,367.07	25,408.31	-11.19	46.54		
Night	SINGLE/RETURN			32,289.87	28,451.51	31,866.35	92,607.73	-11.89	12.00		
	CONCESSION			16,457.31	14,276.17	14,681.45	45,414.93	-13.25	2.84		
	SEASON/PASS			14,626.53	16,414.51	17,118.18	48,159.22	12.22	4.29		
	SPECIAL			13,043.77	17,875.80	16,458.70	47,378.27	37.04	-7.93		
TOTAL	162,939.96	151,372.92	162,521.36	476,834.24	-7.10	7.36					
	IPOH	YEAR TOTAL	655,001.65	619,166.81	630,061.64	1,904,230.10	-5.47	1.76			

Visualization:



Analysis:

From 2023 to 2025, Ipoh's revenue shows two different movements between the first half and the second half of the year. In the first half (Q1 and Q2), revenue kept falling. Q1 went down from RM160,289 in 2023 to RM151,030 in 2024 (-5.8%) and then to RM145,106 in 2025 (-3.9%). Q2 dropped from RM172,924 in 2023 to RM163,412 in 2024 (-5.5%) and further down to RM149,999 in 2025 (-8.2%). This data tells us that the first half of the year is consistently weaker, which drags down the overall yearly performance. This drop is significant because it shows a steady decline of almost 10% in two years, meaning fewer passengers are traveling early in the year. This may be caused by lower travel demand after the holiday season, fewer school holidays in Q1 and Q2 and possibly passengers postponing travel to the second half of the year.

In the second half (Q3 and Q4), the situation was more positive. Q3 slipped slightly from RM158,849 in 2023 to RM153,352 in 2024 (-3.5%), but then grew to RM172,435 in 2025 (+12.4%). Q4 also fell from RM162,940 in 2023 to RM151,373 in 2024 (-7.1%), but recovered to RM162,521 in 2025 (+7.4%), almost back to 2023 levels. This shows that demand comes back strongly in the second half, likely due to school holidays, festivals and year-end travel. The increase is important

because it offsets the weak first half and prevents a large yearly revenue drop. It suggests that passengers are more willing to travel when there are holidays or special events.

Looking ahead to 2026, instead of simply projecting based on the trend, we take the average of the last three years and add a 5% growth to reflect moderate recovery. Based on this approach, Q1 is forecasted at around RM159,749, Q2 at RM170,217, Q3 at RM169,623 and Q4 at RM166,892, giving a total yearly projection of about RM666,481. This approach shows a healthier picture, with growth spread across all quarters, but it still highlights that Q1 and Q2 are lower than Q3 and Q4.

To support growth, the company should work on improving the weak first half, for example by adding seasonal promotions during Q1 and Q2, adjusting schedules to match travel demand and giving discounts for advance bookings to encourage early travel. Marketing campaigns could focus on local events, long-weekend packages or collaboration with tourist attractions to boost interest. For the second half, the company should keep service quality high and add targeted marketing, especially for Q3 and Q4 when travel demand is stronger. By taking these actions, the company can balance out the weaker months, reduce yearly revenue volatility and build a more stable growth path for 2026 and beyond.

3.3.3: Yearly Student vs Worker Comparison Report (2023-2025)

SQL Code:

```
SET LINESIZE 150
SET PAGESIZE 65
SET SERVEROUTPUT ON

TTITLE COL 15 'Report:' -
        COL 25 'Yearly Student vs Worker Comparison Report (2023-2025)' -
        SKIP 1 ' ' -
        LEFT 'Page:' SQL.PNO;

COLUMN year                FORMAT 9999          HEADING 'Year'
COLUMN distance_type       FORMAT A8             HEADING 'Distance'
COLUMN time_of_day         FORMAT A10            HEADING 'Time'
COLUMN metric              FORMAT A20            HEADING 'Metric'
COLUMN student_val         FORMAT 999,999,999    HEADING 'Student'
COLUMN worker_val          FORMAT 999,999,999    HEADING 'Worker'
COLUMN r_diff_val          FORMAT 999,999,999    HEADING 'Rev_Diff'
COLUMN diff_val            FORMAT 999,999,999    HEADING 'Diff'
COLUMN pct_val             FORMAT 999,999.99     HEADING 'Percent %'

BREAK ON year SKIP PAGE ON distance_type SKIP 1 ON time_of_day SKIP 1
COMPUTE SUM OF student_val worker_val r_diff_val ON distance_type
COMPUTE SUM OF student_val worker_val r_diff_val ON year
COMPUTE SUM OF student_val worker_val r_diff_val ON REPORT

WITH Base AS (
    SELECT
        dp.occupation AS passenger_occupation,
        CASE
            WHEN t.distance <= 50 THEN 'Short'
            WHEN t.distance BETWEEN 51 AND 200 THEN 'Medium'
            ELSE 'Long'
        END AS distance_type,
        CASE
            WHEN TO_NUMBER(TO_CHAR(t.departure_time,'HH24')) BETWEEN 6 AND 11 THEN
'Morning'
            WHEN TO_NUMBER(TO_CHAR(t.departure_time,'HH24')) BETWEEN 12 AND 17 THEN
'Afternoon'
            WHEN TO_NUMBER(TO_CHAR(t.departure_time,'HH24')) BETWEEN 18 AND 22 THEN
'Evening'
            ELSE 'Night'
        END AS time_of_day,
        sf.total_sales,
        d.year
    FROM Sales_Fact sf
    JOIN Dim_Passenger dp ON sf.passenger_key = dp.passenger_key
    JOIN Dim_Trip t ON sf.trip_key = t.trip_key
    JOIN Dim_Date d ON sf.date_key = d.date_key
```

```

WHERE d.year BETWEEN 2022 AND 2025
),
Agg AS (
    SELECT
        passenger_occupation,
        distance_type,
        time_of_day,
        year,
        SUM(total_sales) AS revenue,
        COUNT(*) AS tickets
    FROM Base
    GROUP BY passenger_occupation, distance_type, time_of_day, year
),
Pivoted AS (
    SELECT
        year,
        distance_type,
        time_of_day,
        SUM(CASE WHEN passenger_occupation='STUDENT' THEN revenue END) AS
student_revenue,
        SUM(CASE WHEN passenger_occupation='WORKER' THEN revenue END) AS
worker_revenue,
        SUM(CASE WHEN passenger_occupation='STUDENT' THEN tickets END) AS
student_tickets,
        SUM(CASE WHEN passenger_occupation='WORKER' THEN tickets END) AS
worker_tickets
    FROM Agg
    GROUP BY year, distance_type, time_of_day
),
WithLag AS (
    SELECT
        p.*,
        (student_revenue + worker_revenue) AS total_revenue,
        (student_tickets + worker_tickets) AS total_tickets,
        ROUND(student_revenue / NULLIF(student_tickets,0),2) AS student_avg,
        ROUND(worker_revenue / NULLIF(worker_tickets,0),2) AS worker_avg,
        LAG(student_revenue + worker_revenue)
            OVER (PARTITION BY distance_type, time_of_day ORDER BY year) AS
prev_total_revenue,
        LAG(student_tickets + worker_tickets)
            OVER (PARTITION BY distance_type, time_of_day ORDER BY year) AS
prev_total_tickets
    FROM Pivoted p
),
Final AS (
    -- Revenue
    SELECT
        year,
        distance_type,

```

```

        time_of_day,
        'Revenue' AS metric,
        student_revenue AS student_val,
        worker_revenue AS worker_val,
        (student_revenue - worker_revenue) AS r_diff_val,
        NULL AS diff_val,
        ROUND((student_revenue - worker_revenue)/NULLIF(worker_revenue,0)*100,2) AS
pct_val
    FROM WithLag
    UNION ALL
    -- Tickets
    SELECT
        year,
        distance_type,
        time_of_day,
        'Tickets' AS metric,
        student_tickets AS student_val,
        worker_tickets AS worker_val,
        NULL AS r_diff_val,
        (student_tickets - worker_tickets) AS diff_val,
        ROUND((student_tickets - worker_tickets)/NULLIF(worker_tickets,0)*100,2) AS
pct_val
    FROM WithLag
    UNION ALL
    -- Avg Ticket
    SELECT
        year,
        distance_type,
        time_of_day,
        'Avg Ticket' AS metric,
        ROUND(student_avg,2) AS student_val,
        ROUND(worker_avg,2) AS worker_val,
        NULL AS r_diff_val,
        (ROUND(student_avg,2) - ROUND(worker_avg,2)) AS diff_val,
        NULL AS pct_val
    FROM WithLag
    UNION ALL
    -- Revenue Growth
    SELECT
        year,
        distance_type,
        time_of_day,
        'Revenue Growth' AS metric,
        NULL AS student_val,
        NULL AS worker_val,
        NULL AS r_diff_val,
        (total_revenue - prev_total_revenue) AS diff_val,
        ROUND((total_revenue -
prev_total_revenue)/NULLIF(prev_total_revenue,0)*100,2) AS pct_val

```

```

FROM WithLag
UNION ALL
-- Ticket Growth
SELECT
    year,
    distance_type,
    time_of_day,
    'Ticket Growth' AS metric,
    NULL AS student_val,
    NULL AS worker_val,
    NULL AS r_diff_val,
    (total_tickets - prev_total_tickets) AS diff_val,
    ROUND((total_tickets -
prev_total_tickets)/NULLIF(prev_total_tickets,0)*100,2) AS pct_val
    FROM WithLag
)
SELECT *
FROM Final
WHERE year >= 2023
ORDER BY year, distance_type, time_of_day,
    CASE metric
        WHEN 'Revenue' THEN 1
        WHEN 'Tickets' THEN 2
        WHEN 'Avg Ticket' THEN 3
        WHEN 'Revenue Growth' THEN 4
        WHEN 'Ticket Growth' THEN 5
    END;

```

Output:

Report: Yearly Student vs Worker Comparison Report (2023-2025)

Page: 1

Year	Distance	Time	Metric	Student	Worker	Rev_Diff	Diff	Percent %
2023	Long	Afternoon	Revenue	388,325	334,557	53,768		16.07
			Tickets	3,108	2,685		423	15.75
			Avg Ticket	125	125		0	
			Revenue Growth				559,124	341.43
			Ticket Growth				4,469	337.54
		Evening	Revenue	299,757	273,616	26,141		9.55
			Tickets	2,414	2,150		264	12.28
			Avg Ticket	124	127		-3	
			Revenue Growth				428,702	296.33
			Ticket Growth				3,413	296.52
		Morning	Revenue	358,854	308,815	50,040		16.20
			Tickets	2,890	2,440		450	18.44
			Avg Ticket	124	127		-2	
			Revenue Growth				499,166	296.24
			Ticket Growth				3,995	299.25
		Night	Revenue	422,257	356,987	65,270		18.28
			Tickets	3,359	2,837		522	18.40
			Avg Ticket	126	126		-0	
			Revenue Growth				583,563	298.22
			Ticket Growth				4,634	296.67
*****				-----	-----	-----	-----	
sum				1,481,463	1,284,591	195,218	2,088,720	
Medium	Afternoon	Revenue	229,923	189,022	40,901			21.64
		Tickets	1,788	1,477			311	21.06
		Avg Ticket	129	128			1	
		Revenue Growth					303,833	263.95
		Ticket Growth					2,370	264.80
	Evening	Revenue	183,265	154,099	29,166			18.93
		Tickets	1,464	1,223			241	19.71
		Avg Ticket	125	126			-1	
		Revenue Growth					250,767	289.58
		Ticket Growth					2,012	298.07
	Morning	Revenue	216,851	175,055	41,796			23.88
		Tickets	1,705	1,378			327	23.73
		Avg Ticket	127	127			0	
		Revenue Growth					294,756	303.40
		Ticket Growth					2,315	301.43
	Night	Revenue	237,819	198,577	39,242			19.76
		Tickets	1,936	1,584			352	22.22
		Avg Ticket	123	125			-3	
		Revenue Growth					343,802	371.30
		Ticket Growth					2,761	363.77
*****				-----	-----	-----	-----	
sum				875,254	722,921	151,105	1,203,845	
*****				-----	-----	-----	-----	
sum				2,356,717	2,007,512	346,323	3,292,565	

Report: Yearly Student vs Worker Comparison Report (2023-2025)

Page: 2

Year	Distance	Time	Metric	Student	Worker	Rev_Diff	Diff	Percent %
2024	Long	Afternoon	Revenue	376,404	321,159	55,245		17.20
			Tickets	3,013	2,612		401	15.35
			Avg Ticket	125	123		2	

		Revenue Growth				-25,318	-3.50
		Ticket Growth				-168	-2.90
Evening	Revenue	311,288	267,107	44,181			16.54
	Tickets	2,479	2,129		350		16.44
	Avg Ticket	126	125		0		
	Revenue Growth				5,022		.88
	Ticket Growth				44		.96
Morning	Revenue	356,042	290,299	65,743			22.65
	Tickets	2,821	2,309		512		22.17
	Avg Ticket	126	126		0		
	Revenue Growth				-21,327		-3.19
	Ticket Growth				-200		-3.75
Night	Revenue	423,200	341,299	81,901			24.00
	Tickets	3,405	2,719		686		25.23
	Avg Ticket	124	126		-1		
	Revenue Growth				-14,745		-1.89
	Ticket Growth				-72		-1.16

sum			1,479,153	1,230,133	247,070	-54,815	
Medium	Afternoon	Revenue	222,763	188,505	34,258		18.17
		Tickets	1,777	1,520		257	16.91
		Avg Ticket	125	124		1	
		Revenue Growth				-7,677	-1.83
		Ticket Growth				32	.98
	Evening	Revenue	180,412	146,187	34,225		23.41
		Tickets	1,468	1,172		296	25.26
		Avg Ticket	123	125		-2	
		Revenue Growth				-10,765	-3.19
		Ticket Growth				-47	-1.75
	Morning	Revenue	208,489	165,802	42,686		25.75
		Tickets	1,666	1,337		329	24.61
		Avg Ticket	125	124		1	
		Revenue Growth				-17,615	-4.49
		Ticket Growth				-80	-2.59
	Night	Revenue	232,423	198,239	34,184		17.24
		Tickets	1,884	1,588		296	18.64
		Avg Ticket	123	125		-1	
		Revenue Growth				-5,733	-1.31
		Ticket Growth				-48	-1.36

sum			851,378	704,847	145,353	-40,756	
-----	--	--	---------	---------	---------	---------	--

sum			2,330,531	1,934,980	392,423	-95,571	
-----	--	--	-----------	-----------	---------	---------	--

Report: Yearly Student vs Worker Comparison Report (2023-2025)

Page: 3

Year	Distance	Time	Metric	Student	Worker	Rev_Diff	Diff	Percent %
2025	Long	Afternoon	Revenue	385,051	296,353	88,698		29.93
			Tickets	3,096	2,407		689	28.62
			Avg Ticket	124	123		1	
			Revenue Growth				-16,160	-2.32
			Ticket Growth				-122	-2.17
		Evening	Revenue	313,384	265,483	47,901		18.04
			Tickets	2,504	2,120		384	18.11
			Avg Ticket	125	125		-0	
			Revenue Growth				472	.08
			Ticket Growth				16	.35
		Morning	Revenue	354,121	289,683	64,439		22.24

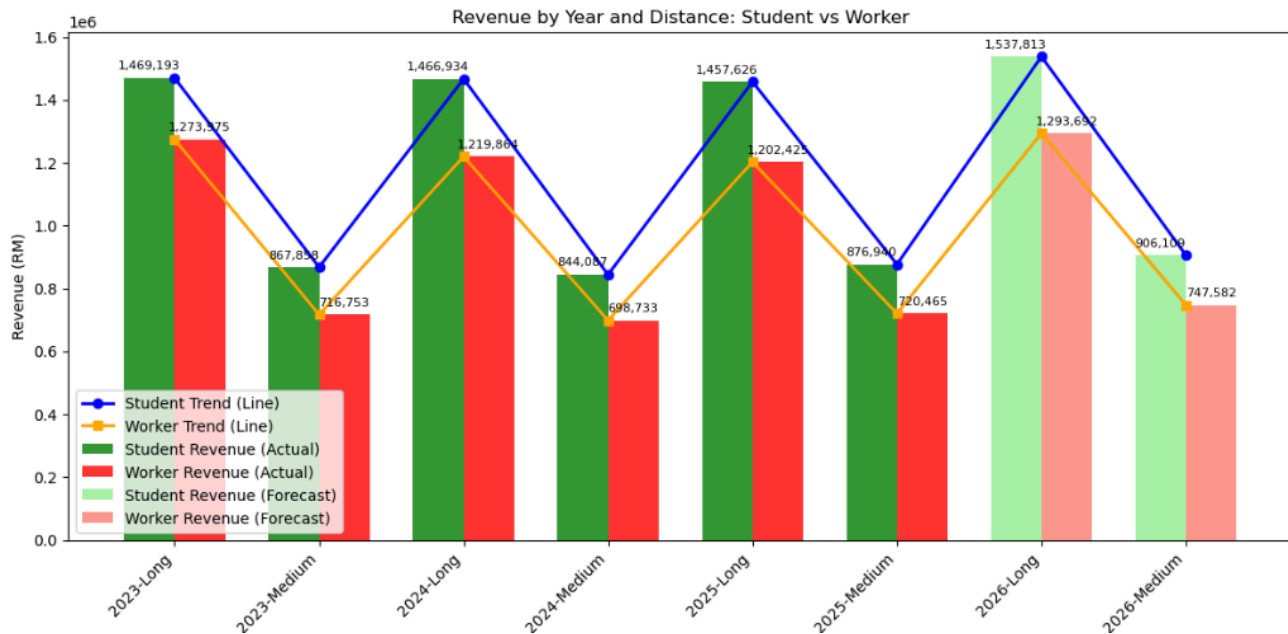
		Tickets	2,841	2,338	503	21.51
		Avg Ticket	125	124	1	
		Revenue Growth			-2,538	-.39
		Ticket Growth			49	.96
	Night	Revenue	405,070	350,906	54,163	15.44
		Tickets	3,221	2,796	425	15.20
		Avg Ticket	126	126	0	
		Revenue Growth			-8,523	-1.11
		Ticket Growth			-107	-1.75

		sum	1,469,787	1,212,583	255,201	-24,909
Medium	Afternoon	Revenue	222,295	192,561	29,734	15.44
		Tickets	1,742	1,520	222	14.61
		Avg Ticket	128	127	1	
		Revenue Growth			3,589	.87
		Ticket Growth			-35	-1.06
	Evening	Revenue	193,492	156,738	36,754	23.45
		Tickets	1,539	1,210	329	27.19
		Avg Ticket	126	130	-4	
		Revenue Growth			23,631	7.24
		Ticket Growth			109	4.13
	Morning	Revenue	229,726	178,726	51,000	28.54
		Tickets	1,827	1,450	377	26.00
		Avg Ticket	126	123	2	
		Revenue Growth			34,162	9.13
		Ticket Growth			274	9.12
	Night	Revenue	231,427	192,440	38,988	20.26
		Tickets	1,845	1,538	307	19.96
		Avg Ticket	125	125	0	
		Revenue Growth			-6,795	-1.58
		Ticket Growth			-89	-2.56

		sum	884,398	726,688	156,475	56,080

		sum	2,354,185	1,939,271	411,677	31,171

Visualization:



Analysis:

From 2023 to 2025, the company's revenue and ticket sales showed both ups and downs. In 2023, performance was strong, with total student revenue reaching about RM 2.36 million, compared to RM 2.01 million for workers. Students consistently generated higher revenue and ticket volumes than workers, leaving a revenue gap of roughly RM 0.35 million. Looking at distances, long-distance routes brought in about RM 1.48 million from students and RM 1.28 million from workers, while medium-distance routes contributed RM 0.88 million and RM 0.72 million respectively. Both distances showed large revenue differences between students and workers, with long-distance routes contributing the majority share. Students were clearly the main drivers of revenue, contributing the most to the company's financial health.

In 2024, performance declined. Student revenue fell slightly to RM 2.33 million, while worker revenue dropped to RM 1.93 million. Many long-distance time slots, especially afternoon and morning trips, recorded negative revenue and ticket growth. Total long-distance revenue declined to RM 1.48 million for students and RM 1.23 million for workers. Medium-distance routes also fell to RM 0.85 million (students) and RM 0.70 million (workers). This decline shows that overall demand weakened, meaning fewer passengers were traveling and the company was earning less than before.

Even so, the revenue gap widened to about RM 0.39 million, showing that students remained the stronger customer group and were less affected by the slowdown.

In 2025, the results were mixed. Student revenue stayed at RM 2.35 million, almost the same as in 2023, while worker revenue remained at RM 1.94 million, similar to 2024. For long-distance routes, revenue was about RM 1.47 million (students) and RM 1.21 million (workers), still showing weakness with negative growth in some time slots, especially afternoon and night. In contrast, medium-distance routes improved to RM 0.88 million (students) and RM 0.73 million (workers), showing recovery in morning and evening periods with revenue growth between 7% and 9%. This shift suggests that medium-distance routes were becoming more important, while long-distance continued to drag performance. The overall revenue gap between students and workers remained large at about RM 0.41 million, reinforcing students as the company's main market segment.

Looking ahead, the forecast for 2026 is made by taking the three-year average revenue (2023–2025) and then applying an additional 5% growth to reflect moderate recovery. Based on this approach, total student revenue is expected to reach about RM 2.47 million, while worker revenue is projected at around RM 2.05 million, giving a combined total of roughly RM 4.52 million.

For long-distance routes, the forecast suggests student revenue of about RM 1.54 million and worker revenue of RM 1.3 million, showing steady but limited improvement compared to 2025. Meanwhile, medium-distance routes are expected to generate about RM 0.91 million from students and RM 0.75 million from workers, continuing the stronger recovery trend already seen in 2025.

This forecast shows that medium-distance routes will remain the key growth driver, while long-distance routes will recover more slowly. Students will continue to contribute more than workers, with the revenue gap expected to stay wide at over RM 0.4 million.

The company must take proactive measures to ensure that revenue continues to grow in 2026 and beyond. Since students represent the largest source of income, the company should strengthen its focus on them through loyalty programs, student travel cards and more frequent services during peak hours in the morning and evening. This will keep the most valuable customer group engaged and

spending regularly. At the same time, comfort and punctuality must be consistently improved to encourage repeat usage and word-of-mouth promotion. To further secure revenue growth, the company should also expand long-distance travel demand with return-ticket discounts, group promotions and partnerships with schools, colleges and companies. In parallel, allocating more resources to medium-distance services, which are increasingly popular, will provide a stable balance and reinforce overall revenue growth.

In addition, investing in digital services such as mobile apps for easy rescheduling, real-time tracking and dynamic price alerts will improve customer convenience and loyalty. At the same time, developing dedicated routes in partnership with universities and companies will secure long-term bulk demand and guarantee stable revenue streams. By combining these actions with careful monitoring of route performance each quarter, the company can reduce losses on weak routes and build stronger revenue across the network.

In summary, from 2023 to 2025, the company first declined but then recovered. Students consistently spent more than workers, making them the main target group. In 2026, the company is expected to reach about RM 2.84 million in student revenue and RM 1.66 million in worker revenue, giving a combined total of roughly RM 4.5 million. The key to success will be focusing on student customers, improving medium-distance services and controlling losses in long-distance travel. If these strategies are carried out well, the company can not only stabilize its revenue but also lay a strong foundation for continuous growth in the coming years.

3.4 Ethel Ng Yi Yan

3.4.1 Passenger Revenue by Age Group

SQL Code:

```
SET PAGESIZE 35
SET LINESIZE 150
SET DEFINE OFF
SET SERVEROUTPUT ON

-- Title and page number
TTITLE CENTER 'Passenger Revenue by Age Group (2023-2025)' SKIP 1 -
        RIGHT 'Page: ' FORMAT 999 SQL.PNO SKIP 1 -

COLUMN AGEGROUP FORMAT A15
COLUMN "Revenue 2023" FORMAT 9,999,999.99
COLUMN "Revenue 2024" FORMAT 9,999,999.99
COLUMN "Revenue 2025" FORMAT 9,999,999.99
COLUMN Total FORMAT 9,999,999.99
COLUMN "Pct to Max Revenue" FORMAT A15
COLUMN "Ticket Sold" FORMAT 9,999,999
COLUMN "Avg Ticket Price" FORMAT 9,999,999.99

WITH AgeGroupSummary AS (
    SELECT
        CASE
            WHEN p.age BETWEEN 0 AND 17 THEN 'Age 0-17'
            WHEN p.age BETWEEN 18 AND 25 THEN 'Age 18-25'
            WHEN p.age BETWEEN 26 AND 35 THEN 'Age 26-35'
            WHEN p.age BETWEEN 36 AND 45 THEN 'Age 36-45'
            WHEN p.age BETWEEN 46 AND 60 THEN 'Age 46-60'
            WHEN p.age >= 61 THEN 'Age 61+'
        END AS age_group,
        d.year,
        s.total_sales - s.discount_amount AS revenue,
        1 AS ticket_sold
    FROM Sales_Fact s
    JOIN Dim_Passenger p ON s.passenger_key = p.passenger_key
    JOIN Dim_Date d ON s.date_key = d.date_key
    WHERE d.year BETWEEN 2023 AND 2025
        AND p.age IS NOT NULL
),
Pivoted AS (
    SELECT *
    FROM AgeGroupSummary
    PIVOT (
        SUM(revenue) AS rev, COUNT(ticket_sold) AS tix FOR year IN (
            2023 AS "2023",
            2024 AS "2024",
```

```

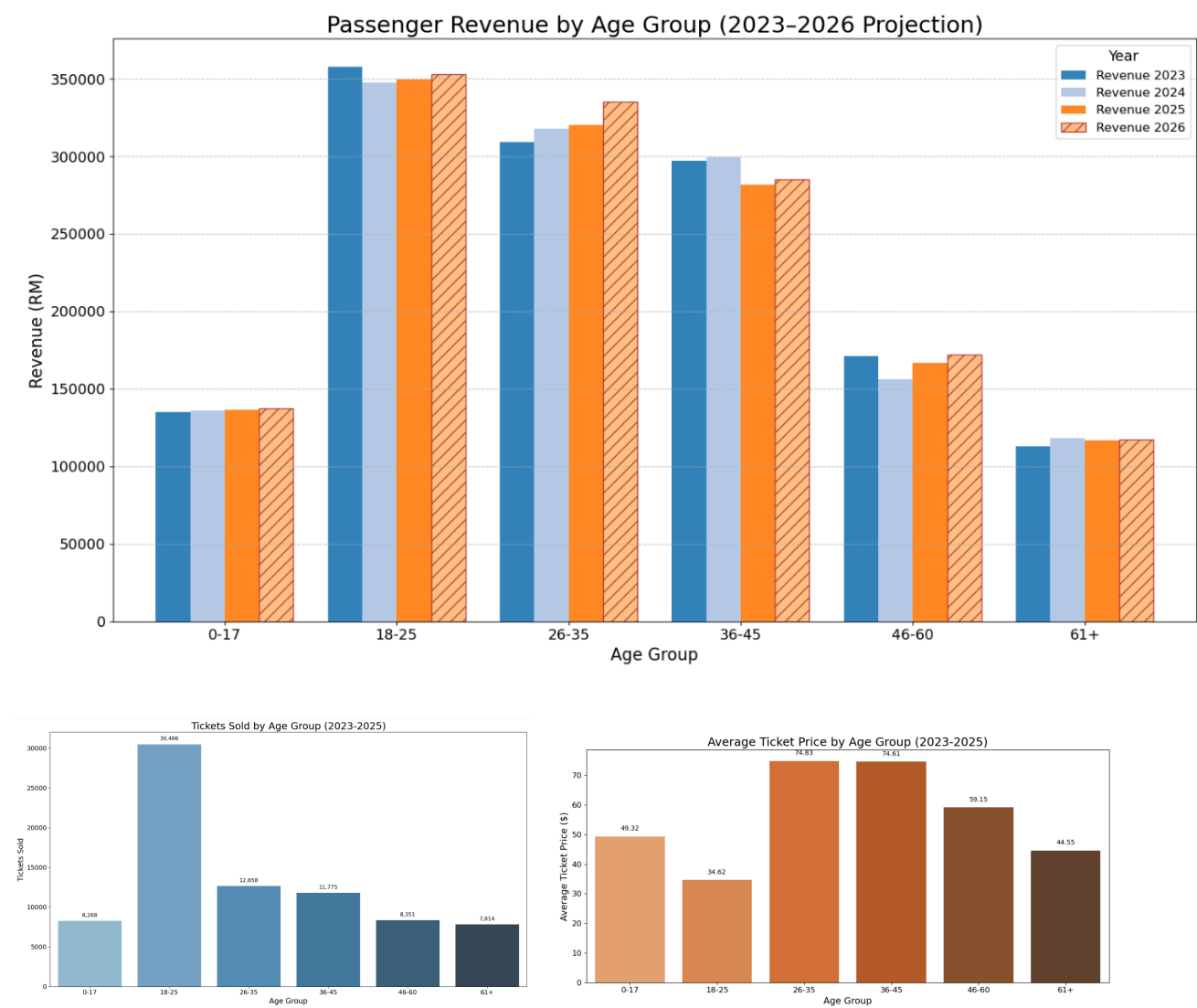
        2025 AS "2025"
    )
)
),
Final AS (
    SELECT age_group,
           "2023_REV" AS "Revenue 2023",
           "2024_REV" AS "Revenue 2024",
           "2025_REV" AS "Revenue 2025",
           NVL("2023_REV",0) + NVL("2024_REV",0) + NVL("2025_REV",0) AS
TotalRevenue,
           NVL("2023_TIX",0) + NVL("2024_TIX",0) + NVL("2025_TIX",0) AS
TotalTickets
    FROM Pivoted
),
MaxRev AS (
    SELECT MAX(TotalRevenue) AS MaxRevenue FROM Final
)
SELECT f.age_group AS AGEGROUP,
       f."Revenue 2023",
       f."Revenue 2024",
       f."Revenue 2025",
       f.TotalRevenue AS Total,
       TO_CHAR(ROUND((f.TotalRevenue / m.MaxRevenue) * 100, 2)) || '%' AS "Pct to
Max Revenue",
       f.TotalTickets AS "Ticket Sold",
       CASE WHEN f.TotalTickets > 0 THEN ROUND(f.TotalRevenue / f.TotalTickets,2)
END AS "Avg Ticket Price"
FROM Final f
CROSS JOIN MaxRev m
ORDER BY CASE f.age_group
           WHEN 'Age 0-17' THEN 1
           WHEN 'Age 18-25' THEN 2
           WHEN 'Age 26-35' THEN 3
           WHEN 'Age 36-45' THEN 4
           WHEN 'Age 46-60' THEN 5
           WHEN 'Age 61+' THEN 6
END;

```

Output:

Passenger Revenue by Age Group (2023-2025)							Page: 1
AGEGROUP	Revenue 2023	Revenue 2024	Revenue 2025	Total Revenue	Pct to Max Revenue	Ticket Sold	Avg Ticket Price
Age 0-17	135,059.56	136,059.16	136,641.05	407,759.77	38.64%	8,268	49.32
Age 18-25	357,962.92	347,878.39	349,493.87	1,055,335.18	100%	30,486	34.62
Age 26-35	309,165.67	317,881.27	320,148.02	947,194.96	89.75%	12,658	74.83
Age 36-45	296,919.87	299,650.74	281,941.36	878,511.97	83.24%	11,775	74.61
Age 46-60	171,012.63	156,255.59	166,664.39	493,932.61	46.8%	8,351	59.15
Age 61+	113,065.53	118,185.55	116,837.36	348,088.44	32.98%	7,814	44.55

Visualization:



Analysis:

The graph shows passenger revenue from 2023 to 2025 across different age groups. The 0–17 age group brings in the second-lowest total revenue compared to others. Their revenue increased only slightly, from RM135,059.56 in 2023 to RM136,641.05 in 2025. This slow growth is expected, as children and teenagers typically travel with parents, with demand mostly concentrated during school holidays or family trips. Although the increase is modest, offering promotions for family or student group travel could stimulate more demand in 2026, with projected revenue of approximately RM137,200.

The 18–25 age group is the top revenue contributor, generating over RM1,055,000 across the three years. Despite a slight drop in 2024, revenue rebounded in 2025. This group also recorded the highest number of tickets sold (30,486), although they paid the lowest average ticket price (RM34.62), suggesting they prefer affordable options like buses over more expensive transport. The 2024 dip could be due to rising competition from low-cost transport alternatives or shifts in student and work travel patterns. To maintain and grow revenue, loyalty programs or student discounts could be introduced. Revenue for this group is expected to rise steadily to around RM353,000 in 2026.

The 26–35 age group shows strong and consistent growth, with revenue rising from RM309,165.67 in 2023 to RM320,148.02 in 2025. They pay the highest average ticket price at RM74.83, indicating that young professionals are willing to pay for convenience or premium options. As the second-highest revenue group, they represent a key market segment and targeted premium travel packages or early-bird offers could further increase revenue. With such strategies, revenue in 2026 could reach RM335,000.

The 36–45 age group closely mirrors the 26–35 age group in terms of spending behavior, with an average ticket price of RM74.61. However, their revenue declined by 5.9% in 2025, falling to RM281,941.36. This drop may indicate changing travel habits due to family obligations, budget constraints or lack of targeted promotions. To reverse this trend, family-friendly travel deals or more flexible scheduling could be introduced. If implemented, revenue could recover to around RM285,000 in 2026.

The 46–60 age group displays more fluctuation. Revenue declined in 2024 and rebounded in 2025, suggesting sensitivity to external factors like economic conditions, pricing or health-related concerns. With an average ticket price of RM59.15, their spending is moderate but inconsistent. The recovery in 2025 is promising and continued growth could be supported through loyalty programs or group travel incentives. Projected revenue for 2026 is around RM172,000.

Lastly, the 61+ age group remains stable, with revenue staying between RM113,000 and RM117,000 over the three years. They had fewer tickets sold (7,814) and a moderate average ticket price of RM44.55. Their travel habits appear consistent but conservative. To encourage increased travel, off-peak discounts or health-conscious travel packages could be offered. Revenue is expected to remain stable at around RM117,000 in 2026.

3.4.2 Passenger Revenue and Travel Trends by Quarter & Holiday

SQL Code:

```
SET PAGESIZE 35
SET LINESIZE 170
SET DEFINE OFF
SET SERVEROUTPUT ON

-- Title and page number
TTITLE CENTER 'Passenger Revenue and Travel Trends by Quarter & Holiday
(2023-2025)' SKIP 1 -
          RIGHT 'Page: ' FORMAT 999 SQL.PNO

-- Column formatting
COLUMN destination          FORMAT A20          HEADING 'Destination'
COLUMN year_quarter         FORMAT A10          HEADING 'Year-Quarter'
COLUMN holiday_passenger    FORMAT 999,999      HEADING 'Ticket Sold Holi'
COLUMN holiday_revenue      FORMAT 999,999.99   HEADING 'Holiday Revenue'
COLUMN nonholiday_passenger FORMAT 999,999      HEADING 'Non-Hol Ticket Sold'
COLUMN nonholiday_revenue   FORMAT 999,999.99   HEADING 'Non-Hol Revenue'
COLUMN diff_revenue         FORMAT 999,999.99   HEADING 'Diff Revenue'
COLUMN pct_holiday          FORMAT 999.99       HEADING 'Holiday %'
COLUMN pct_nonholiday        FORMAT 999.99       HEADING 'Non-Hol %'
COLUMN pct_annual            FORMAT 999.99       HEADING 'Pct to Year'

-- Break & Compute
BREAK ON destination SKIP 1
COMPUTE SUM OF holiday_passenger holiday_revenue nonholiday_passenger
nonholiday_revenue ON destination

-- CTE for destination stats
WITH DestinationStats AS (
  SELECT
    t.destination AS destination,
    d.year || '-' || d.quarter AS year_quarter,
    d.year AS year_num,
    d.holiday_flag,
    COUNT(DISTINCT sf.passenger_key) AS total_passenger,
    SUM(NVL(sf.total_sales,0) - NVL(sf.discount_amount,0)) AS sales_revenue
  FROM Sales_Fact sf
  JOIN Dim_Trip t ON sf.trip_key = t.trip_key
  JOIN Dim_Date d ON sf.date_key = d.date_key
  WHERE t.destination IN ('KUALA LUMPUR', 'GEORGE TOWN') -- only destinations we
want
    AND d.year BETWEEN 2023 AND 2025
  GROUP BY t.destination, d.year, d.quarter, d.holiday_flag
),
PivotStats AS (
  SELECT
    d.destination,
```

```

        d.year_quarter,
        d.year_num,
        MAX(CASE WHEN d.holiday_flag='Y' THEN d.total_passenger END) AS
holiday_passenger,
        MAX(CASE WHEN d.holiday_flag='Y' THEN d.sales_revenue END) AS
holiday_revenue,
        MAX(CASE WHEN d.holiday_flag='N' THEN d.total_passenger END) AS
nonholiday_passenger,
        MAX(CASE WHEN d.holiday_flag='N' THEN d.sales_revenue END) AS
nonholiday_revenue,
        MAX(CASE WHEN d.holiday_flag='Y' THEN d.sales_revenue END) -
        MAX(CASE WHEN d.holiday_flag='N' THEN d.sales_revenue END) AS diff_revenue
FROM DestinationStats d
GROUP BY d.destination, d.year_quarter, d.year_num
)
SELECT
    p.destination,
    p.year_quarter,
    p.holiday_passenger,
    p.holiday_revenue,
    p.nonholiday_passenger,
    p.nonholiday_revenue,
    p.diff_revenue,
    -- Holiday % contribution to quarter
    ROUND((p.holiday_revenue / NULLIF(p.holiday_revenue + p.nonholiday_revenue,0))
* 100, 2) AS pct_holiday,
    -- Non-Holiday % contribution to quarter
    ROUND((p.nonholiday_revenue / NULLIF(p.holiday_revenue +
p.nonholiday_revenue,0)) * 100, 2) AS pct_nonholiday,
    -- % contribution to the year
    ROUND(((p.holiday_revenue + p.nonholiday_revenue) /
        NULLIF(SUM(p.holiday_revenue + p.nonholiday_revenue) OVER (PARTITION BY
p.destination, p.year_num),0)) * 100, 2) AS pct_annual
FROM PivotStats p
ORDER BY p.destination, p.year_quarter;

-- Clear breaks & computes
CLEAR BREAKS
CLEAR COMPUTES;

```

Output:

Passenger Revenue and Travel Trends by Quarter (2023-2025)

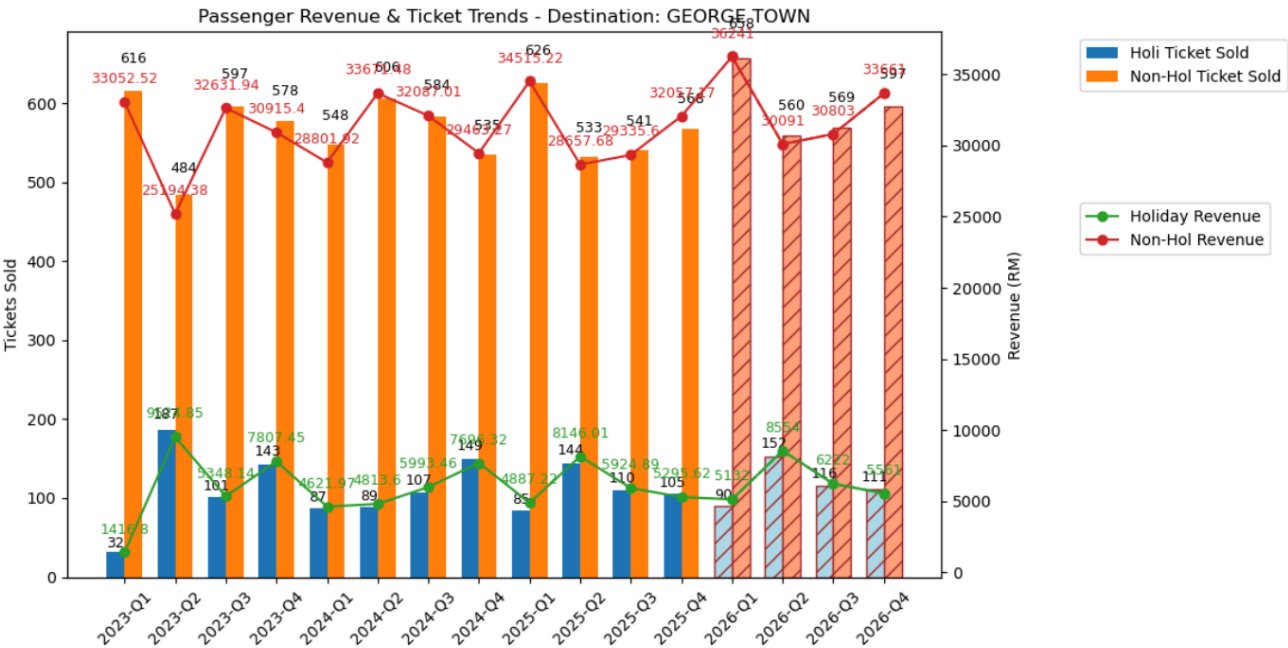
Page: 1

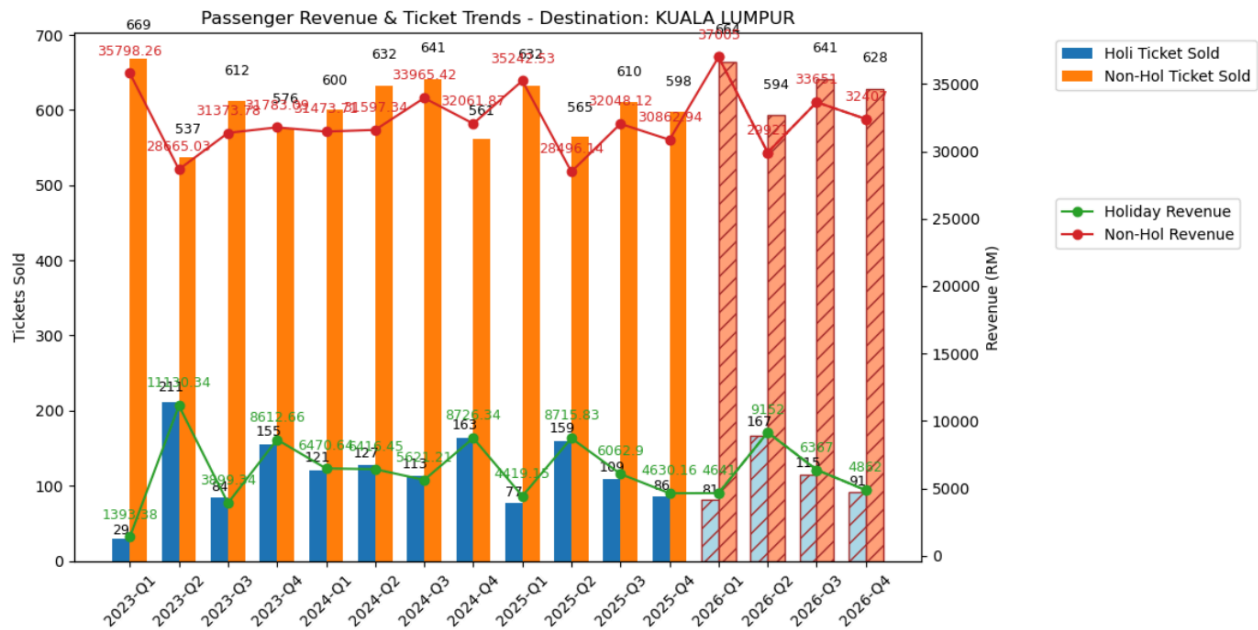
Destination	Year-Quart	Ticket Sold	Holi	Holiday Revenue	Non-Hol	Ticket Sold	Non-Hol	Revenue	Diff	Revenue	Holiday %	Non-Hol %	Pct to Year
GEORGE TOWN	2023-Q1	32		1,416.80	616		33,052.52		-31,635.72		4.11	95.89	23.63
	2023-Q2	187		9,524.85	484		25,194.38		-15,669.53		27.43	72.57	23.80
	2023-Q3	101		5,348.14	597		32,631.94		-27,283.80		14.08	85.92	26.03
	2023-Q4	143		7,807.45	578		30,915.40		-23,107.95		20.16	79.84	26.54
	2024-Q1	87		4,621.97	548		28,801.92		-24,179.95		13.83	86.17	22.71
	2024-Q2	89		4,813.60	606		33,671.48		-28,857.88		12.51	87.49	26.15
	2024-Q3	107		5,993.46	584		32,087.01		-26,093.55		15.74	84.26	25.88
	2024-Q4	149		7,696.32	535		29,463.27		-21,766.95		20.71	79.29	25.25
	2025-Q1	85		4,887.22	626		34,515.22		-29,628.00		12.40	87.60	26.48
	2025-Q2	144		8,146.01	533		28,657.68		-20,511.67		22.13	77.87	24.73
	2025-Q3	110		5,924.89	541		29,335.60		-23,410.71		16.80	83.20	23.69
	2025-Q4	105		5,295.62	568		32,057.17		-26,761.55		14.18	85.82	25.10

sum			1,339	71,476.33		6,816		370,383.59					
KUALA LUMPUR	2023-Q1	29		1,393.38	669		35,798.26		-34,404.88		3.75	96.25	24.36
	2023-Q2	211		11,130.34	537		28,665.03		-17,534.69		27.97	72.03	26.07
	2023-Q3	84		3,899.34	612		31,373.78		-27,474.44		11.05	88.95	23.11
	2023-Q4	155		8,612.66	576		31,783.09		-23,170.43		21.32	78.68	26.46
	2024-Q1	121		6,470.64	600		31,473.71		-25,003.07		17.05	82.95	24.27
	2024-Q2	127		6,416.45	632		31,597.34		-25,180.89		16.88	83.12	24.32
	2024-Q3	113		5,621.21	641		33,965.42		-28,344.21		14.20	85.80	25.32
	2024-Q4	163		8,726.34	561		32,061.87		-23,335.53		21.39	78.61	26.09
	2025-Q1	77		4,419.15	632		35,242.53		-30,823.38		11.14	88.86	26.36
	2025-Q2	159		8,715.83	565		28,496.14		-19,780.31		23.42	76.58	24.73
	2025-Q3	109		6,062.90	610		32,048.12		-25,985.22		15.91	84.09	25.33
	2025-Q4	86		4,630.16	598		30,862.94		-26,232.78		13.05	86.95	23.59

sum			1,434	76,098.40		7,233		383,368.23					

Visualization:





Analysis:

The graphs above illustrate passenger revenue and ticket trends for George Town and Kuala Lumpur across each quarter from 2023 to 2025, including the projection of the year 2026. Non-holiday tickets consistently form the backbone of the business for both destinations, contributing approximately 80–85% of total ticket sales and revenue. These tickets primarily cater to daily commuters, workers and business travelers who travel regularly regardless of seasonal fluctuations. This steady demand ensures a reliable income stream, even during quarters when holiday ticket sales decline, stabilizing overall revenue throughout the year. Holiday travel, in contrast, constitutes a smaller proportion of ticket sales and revenue, roughly 15–20%. Peaks in holiday travel typically occur during the second and fourth quarters, corresponding with festive seasons and school holidays such as Hari Raya, Chinese New Year, Deepavali and Christmas. During these periods, families and groups are more likely to travel. Conversely, holiday sales tend to decline in the first and third quarters, when major holiday periods are less frequent.

For the projection of 2026, the same trends are expected to continue, with non-holiday tickets remaining the main source of revenue. Non-holiday travel may experience slight growth or modest fluctuations in each quarter, reflecting the steady behavior of regular commuters and workers. Holiday tickets are projected to continue peaking in the second and fourth quarters and with effective marketing strategies such as early-bird promotions, family discounts and festival packages, holiday ticket sales could grow by an estimated 5%. Based

on the 2025 figures with a 5% projected growth for 2026, the quarterly revenues for George Town are expected to be RM 41,373 in Q1, RM 38,643 in Q2, RM 37,023 in Q3 and RM 39,220 in Q4. For Kuala Lumpur, the projected quarterly revenues are RM 41,644 in Q1, RM 39,073 in Q2, RM 40,016 in Q3 and RM 37,268 in Q4. These projections clearly highlight the dominance of non-holiday tickets, with non-holiday revenue ranging from RM 30,090 to RM 36,241 in George Town and from RM 29,921 to RM 37,004 in Kuala Lumpur. Holiday revenue continues to peak in Q2 and Q4, with George Town reaching RM 8,553 and Kuala Lumpur RM 9,152 during these quarters, reflecting the influence of festive and school holiday periods on travel patterns.

The quarterly stability in total revenue is largely due to the consistent performance of non-holiday tickets, which buffers the fluctuations in holiday travel. This stability allows for predictable financial planning and ensures that operational adjustments can focus on optimizing efficiency during peak and off-peak periods. To capitalize on holiday peaks, additional trip schedules could be implemented in Q2 and Q4, while quieter quarters could be optimized to reduce operational costs. Moreover, targeted marketing initiatives, such as loyalty programs, corporate travel partnerships and festival packages, could enhance both holiday and non-holiday ticket sales. The projections for 2026 indicate that George Town's quarterly revenue will range around RM 37,000 to RM 41,400, while Kuala Lumpur's will range around RM 37,300 to RM 42,000, reaffirming the critical role of non-holiday tickets in sustaining business revenue while highlighting strategic opportunities for growth during peak holiday periods.

3.4.3 Yearly Sales by Origin-Destination

SQL Code:

```
SET PAGESIZE 50
SET LINESIZE 150
SET DEFINE OFF
SET SERVEROUTPUT ON
```

```
TTITLE CENTER 'Yearly Sales by Origin-Destination (2023-2025)' SKIP 1 -
        RIGHT 'Page: ' SQL.PNO
```

```
COLUMN Origin          FORMAT A15
COLUMN Destination     FORMAT A16
COLUMN "Y 2023"        FORMAT 999,999.99
COLUMN "Y 2024"        FORMAT 999,999.99
COLUMN "Y 2025"        FORMAT 999,999.99
COLUMN "Total Revenue" FORMAT 999,999.99
COLUMN "Ticket Sold"   FORMAT 999,999
COLUMN "Avg Rev"       FORMAT 999,999.99
```

```
-- Break to avoid repeating origin
BREAK ON origin SKIP 1
```

```
-- Compute sum of each year, total revenue, ticket sold and avg rev per origin
COMPUTE SUM OF "Y 2023" "Y 2024" "Y 2025" "Total Revenue" "Ticket Sold" "Avg Rev"
ON origin
```

```
SELECT
    origin,
    destination,
    "Y 2023_REV" AS "Y 2023",
    "Y 2024_REV" AS "Y 2024",
    "Y 2025_REV" AS "Y 2025",
    NVL("Y 2023_REV",0) + NVL("Y 2024_REV",0) + NVL("Y 2025_REV",0) AS "Total
Revenue",
    NVL("Y 2023_T",0) + NVL("Y 2024_T",0) + NVL("Y 2025_T",0) AS "Ticket Sold",
    ROUND((
        NVL("Y 2023_REV",0) + NVL("Y 2024_REV",0) + NVL("Y 2025_REV",0)
    ) / 3,2) AS "Avg Rev"
FROM (
    SELECT
        t.origin,
        t.destination,
        d.year,
        (sf.total_sales - sf.discount_amount) AS revenue,
        1 AS ticket_count
    FROM Sales_Fact sf
    JOIN Dim_Trip t ON sf.trip_key = t.trip_key
    JOIN Dim_Date d ON sf.date_key = d.date_key
```

```

WHERE t.origin IN ('KUALA LUMPUR','GEORGE TOWN')
AND d.year BETWEEN 2023 AND 2025
)
PIVOT (
SUM(revenue) AS REV,
SUM(ticket_count) AS T
FOR year IN (
2023 AS "Y 2023",
2024 AS "Y 2024",
2025 AS "Y 2025"
)
)
ORDER BY origin, destination;

-- Clear breaks and computes after done
CLEAR BREAKS
CLEAR COMPUTES;

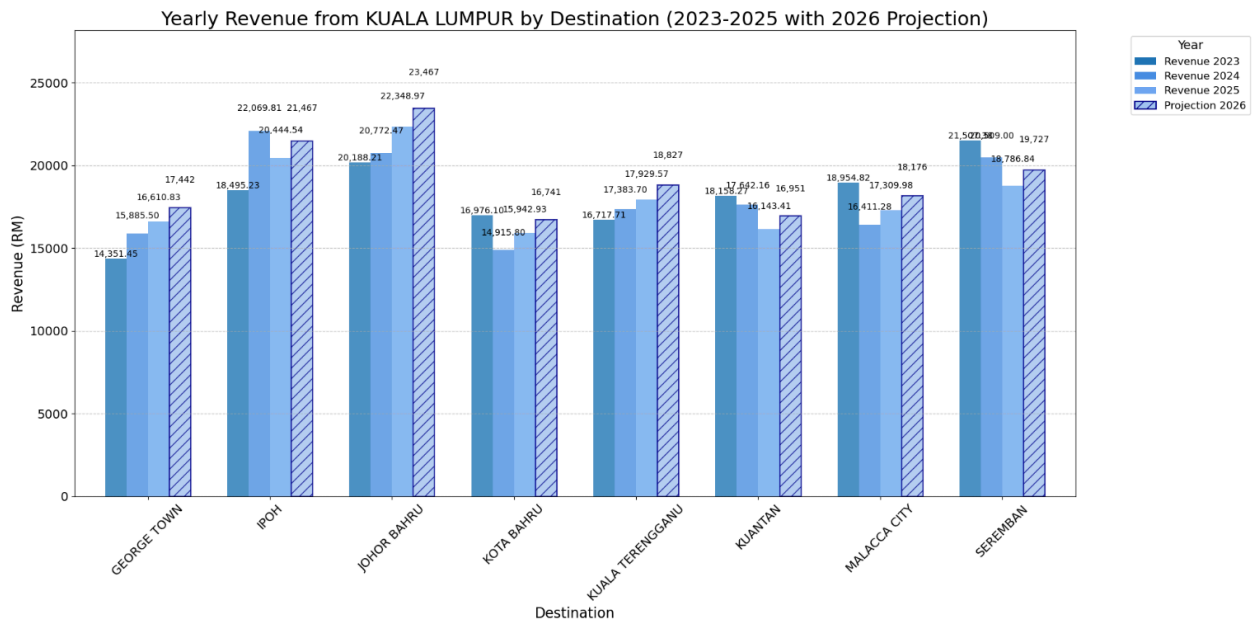
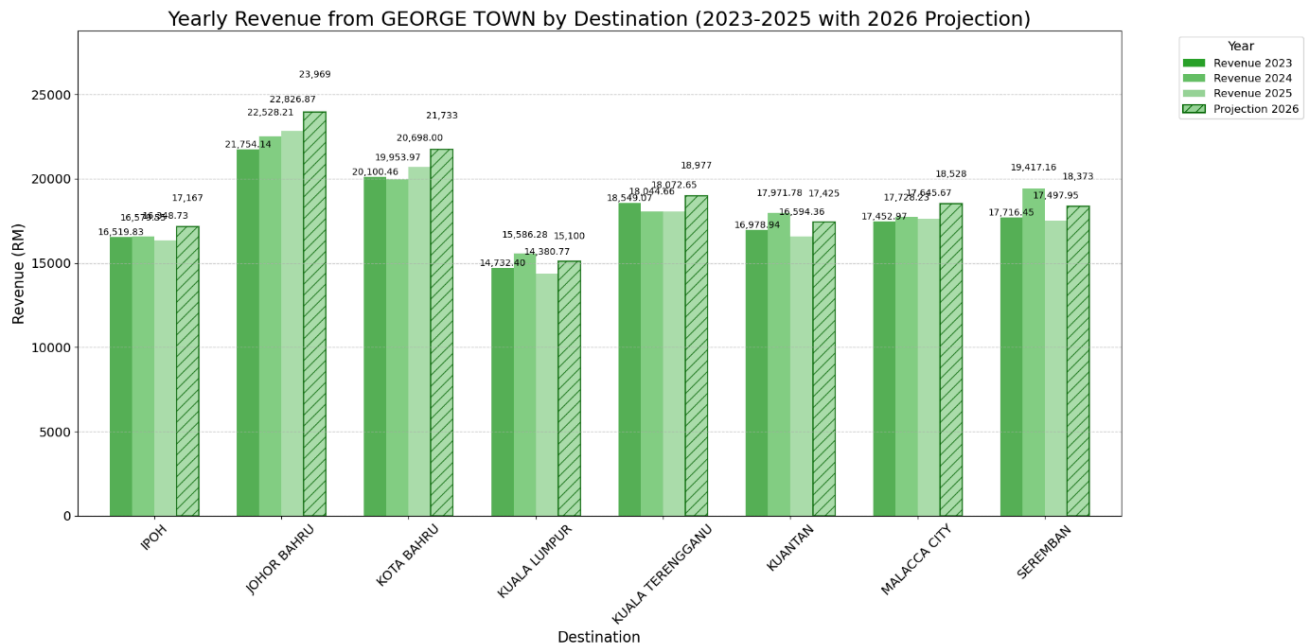
```

Output:

Page: 1

Yearly Sales by Origin-Destination (2023-2025)							
Origin	Destination	Y 2023	Y 2024	Y 2025	Total Revenue	Ticket Sold	Avg Rev
GEORGE TOWN	IPOH	16,519.83	16,579.55	16,348.73	49,448.11	935	16,482.70
	JOHOR BAHRU	21,754.14	22,528.21	22,826.87	67,109.22	1,272	22,369.74
	KOTA BAHRU	20,100.46	19,953.97	20,698.00	60,752.43	1,181	20,250.81
	KUALA LUMPUR	14,732.40	15,586.28	14,380.77	44,699.45	898	14,899.82
	KUALA TERENGGANU	18,549.07	18,044.66	18,072.65	54,666.38	1,062	18,222.13
	KUANTAN	16,978.94	17,971.78	16,594.36	51,545.08	987	17,181.69
	MALACCA CITY	17,452.97	17,728.23	17,645.67	52,826.87	990	17,608.96
*****		17,716.45	19,417.16	17,497.95	54,631.56	1,034	18,210.52
sum		143,804.26	147,809.84	144,065.00	435,679.10	8,359	145,226.37
KUALA LUMPUR	GEORGE TOWN	14,351.45	15,885.50	16,610.83	46,847.78	877	15,615.93
	IPOH	18,495.23	22,069.81	20,444.54	61,009.58	1,192	20,336.53
	JOHOR BAHRU	20,188.21	20,772.47	22,348.97	63,309.65	1,257	21,103.22
	KOTA BAHRU	16,976.10	14,915.80	15,942.93	47,834.83	940	15,944.94
	KUALA TERENGGANU	16,717.71	17,383.70	17,929.57	52,030.98	1,014	17,343.66
	KUANTAN	18,158.27	17,642.16	16,143.41	51,943.84	1,008	17,314.61
	MALACCA CITY	18,954.82	16,411.28	17,309.98	52,676.08	1,033	17,558.69
*****		21,507.38	20,509.00	18,786.84	60,803.22	1,154	20,267.74
sum		145,349.17	145,589.72	145,517.07	436,455.96	8,475	145,485.32

Visualization:



Analysis:

The two graphs above show the revenue data from the origins, Kuala Lumpur and George Town, to various destinations between 2023 and 2025, with the projection of revenue in the year 2026. These visualizations are important because they provide a clear overview of route performance, allowing the bus company to quickly identify high-performing routes, declining routes and those with fluctuating demand. Overall, the average of total yearly revenue from Kuala Lumpur remained around RM 145,000, though some routes experienced growth while others declined. For example, revenue to

Johor Bahru increased steadily from about RM 20,188.21 in 2023 to RM 22,348.97 in 2025, likely due to growing business connections or tourism. In contrast, routes such as Seremban declined from RM 21,507.38 in 2023 to RM 18,786.84 in 2025, potentially due to competition from other travel options, changing commuter habits or service issues. Other destinations, like Ipoh and Kota Bahru, showed fluctuations, possibly due to seasonal travel patterns.

From George Town, the pattern is similar, with Johor Bahru showing steady growth, while routes to Kuala Lumpur and Seremban face declines in 2025. For instance, revenue from George Town to Kuala Lumpur fell from RM 15,586.28 in 2024 to RM 14,380.77 in 2025. These trends emphasize the importance of monitoring route performance and making data-driven decisions, as the graphs clearly illustrate which services are thriving and which need attention. By understanding these patterns, the company can better allocate resources and tailor services to meet traveler demand across different destinations.

Looking ahead to 2026, revenue projections anticipate a 5% increase for each route compared to 2025, reflecting potential growth if targeted strategies are implemented. To achieve this, the company could increase service frequency on high-demand routes like Johor Bahru, run targeted marketing campaigns for routes with moderate growth potential such as Ipoh and Kota Bahru, enhance customer experience through improved punctuality, comfort and convenient booking options, adjust schedules and pricing dynamically to match fluctuating demand and review declining routes like Seremban to identify operational improvements. For example, revenue from George Town to Johor Bahru is projected to rise from RM 22,826.87 in 2025 to RM 23,969 in 2026, while Kuala Lumpur to Johor Bahru is expected to increase from RM 22,348.97 to RM 23,467. By implementing these measures, the company can not only raise overall annual revenue above RM 150,000 but also strengthen its market position and provide a higher-quality, more reliable service for passengers.